

Screening Contracts as a Barrier to Entry*

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Abstract

A seller contracts with a downstream buyer under the threat of upstream entry. The buyer privately knows both his demand and the efficiency gains from entry, so the optimal contract distorts entry to capture the buyer's rent, and with multi-unit demand this distortion shapes the tariff itself. Under market-share contracts, which condition payments on the buyer's sourcing, consumption becomes a mean-preserving spread of the monopoly schedule and entry is positively correlated with consumption. When a single non-conditional tariff must serve both market structures, capturing the gains from entry instead requires pricing below marginal cost, entry ceases to be monotone in demand, and a minimum purchase requirement emerges at a kink of the tariff. This optimum is closely approximated by an all-unit discount, so the legal distinction between incremental and retroactive rebates carries little economic content. The price-cost tests of antitrust practice track the buyer's surplus rather than total welfare, and typically sanction the contract where the seller's response to entry raises welfare.

KEYWORDS. Nonlinear pricing; barriers to entry; market-share contracts; non-conditional contracts; all-unit discounts.

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1. INTRODUCTION

The strategic role of vertical contracts in market contexts has long been a topic of debate among scholars and practitioners. The Chicago School (Posner, 1976; Bork, 1978) argued that an incumbent firm cannot profitably offer a contract to a buyer that would exclude a more efficient entrant. Such contracts would only be accepted if the buyer were fully compensated for the efficiency gains lost due to entry prevention. Consequently, any exclusion that occurs is deemed efficient. Theoretical models of vertical relations and foreclosure under complete information largely support this conclusion.¹ This reasoning has influenced U.S. antitrust authorities, where exclusive arrangements are assessed under a *rule of reason*.

However, this perspective has been challenged both in practice and theory. In the E.U., for instance, antitrust authorities often view exclusivity clauses as *per se* illegal. Theoretically, Aghion and Bolton (1987) argued that vertical contracts could create inefficient barriers to entry under asymmetric information about the benefits of entry. Although the incumbent might benefit from the efficiency gains brought by the entrant through breach penalties, informational frictions may prevent the incumbent from fully capturing these gains. As a result, entry barriers can allow the incumbent to extract additional information rent.

CAVEATS. While Aghion and Bolton's (1987) model presents a theoretical challenge to the Chicago School paradigm, it relies on assumptions that may not hold in real-world contexts. These limitations could reduce the model's practical relevance, particularly in antitrust discussions.

First, the buyer is assumed to have a unit demand for a homogeneous good, which either the incumbent or the entrant can supply. This assumption imposes exclusivity, as the buyer is restricted to purchasing from only one seller. However, in a more realistic scenario, exclusivity could emerge as an equilibrium outcome even if the buyer has the option to split purchases. Additionally, the unit demand assumption limits the analysis of pricing mechanisms such as lump-sum rebates and unit discounts, which are central to contemporary antitrust concerns. When considering a downward-sloping demand curve, exclusive dealing provisions and breach penalties could be replaced by seemingly less aggressive clauses that achieve similar objectives. All-unit discount tariffs or minimum purchase requirements, for example, are often viewed by antitrust authorities as potential tools for foreclosing upstream markets.²

Second, the unit demand assumption complicates the distinction between the fixed cost of entry and the marginal cost of production conditional on entry. Potential entrants typically evaluate entry in terms of both sunk costs and the efficiency gains they can achieve through lower marginal costs relative to the incumbent.

OUR ANALYSIS. Our primary objective is to examine the interplay between strategic and screening considerations. Specifically, we focus on the role of rebates, discounts, and minimum purchase

¹O'Brien and Shaffer (1997) and Bernheim and Whinston (1998) formalized this perspective and identified its limitations.

²Cases such as *Virgin v. British Airways* in the U.S., and *Michelin* and *Intel* in Europe illustrate this point.

requirements in response to potential entry threats.

To fully understand these contractual practices, we adopt a more realistic framework where buyers hold private information about their multi-unit demand. In this context, the nonlinear pricing literature (Maskin and Riley, 1984; Wilson, 1993) shows the role of price discounts in facilitating screening, particularly in monopolistic settings. Understanding how strategic considerations reshape screening mechanisms is relevant for regulatory policy and its implications for competition.

In the absence of strategic considerations related to entry, the incumbent's focus would be solely on screening. This would involve offering a tariff characterized by price discounts for larger quantities, sorting buyers with varying demand levels along this pricing schedule. In such a setting, clauses like minimum purchase requirements, below-cost pricing, or all-unit discounts would not arise. Using this baseline as a reference, we analyze the features of optimal nonlinear contracts when the incumbent faces the threat of entry.

MODEL. An incumbent seller contracts with a buyer who has a downward-sloping demand. The buyer's private information about preferences typically justifies the use of price discounts as a screening mechanism. However, the incumbent faces the threat of entry from a competitor, referred to as the entrant, who operates at a lower marginal cost conditional on paying a fixed cost of entry. While entry can be desirable from an allocative perspective, it may not always be socially optimal due to these fixed costs.

The buyer may sometimes find it optimal to split his purchases between the incumbent and the entrant. Due to capacity constraints, the entrant is unable to serve the entire market demand, ensuring that the incumbent retains a *non-contestable* share of the market.³

Two alternative contractual scenarios are considered. In the first one, the incumbent observes whether the buyer also purchases from the entrant and offers nonlinear tariffs, referred to as *market-share contracts*, that depend on this decision. In the second scenario, the incumbent provides a single *non-conditional* nonlinear tariff that applies regardless of whether entry occurs. This latter framework is relevant when buyers can easily conceal purchases from the entrant or when antitrust authorities prohibit contingent contracts.

Our analysis has several main features. First, in both contracting scenarios the incumbent charges lower marginal prices on the units the buyer selects when purchases are split than on those selected under exclusivity. The two scenarios differ in how far prices fall: margins remain positive under market-share contracts, whereas capturing the gains from entry with a non-conditional tariff requires pricing some of these units below marginal cost. Second, *minimum purchase requirements* emerge as contractual provisions in both scenarios. These clauses compel buyers wishing to split their purchases to commit to consuming a minimum quantity from the incumbent. Such requirements arise systematically under non-conditional tariffs. Third, both scenarios

³This assumption aligns with observations in recent antitrust cases, such as *Michelin* and *Intel*, where entrants were found capable of serving only a fraction of total demand.

address the threat of entry in the same way, by exploiting it rather than blocking it, and differ in their targeting. Market-share contracts accommodate entry for every buyer type, extracting, albeit imperfectly, the entry-induced gains. A non-conditional tariff pursues the same aim but cannot target it uniformly. Extraction remains profitable where the buyer splits his purchases, while for high-demand buyers, served exclusively at above-cost margins, entry becomes costly to the incumbent and the contract locally discourages it. Exploitative and exclusionary roles thus coexist within a single tariff, sorted across the distribution of demand.

MARKET-SHARE CONTRACTS. When the incumbent can condition tariffs on whether the buyer also purchases from the entrant, the optimal contract pairs a no-entry and a with-entry schedule, the gap between them acting as a rebate that the incumbent calibrates to influence the entry decision. Two features depart from pure monopolistic screening. First, entry and consumption become positively correlated: the incumbent uses the likelihood of entry, governed by the rebate, as a second screening instrument alongside the consumption profile, so that higher-demand buyers obtain larger rebates, lower marginal prices, and face a higher probability of entry. Second, the contract obeys a *martingale property*. With quadratic preferences, entry raises the variance of consumption but leaves its ex ante average at the monopoly level, type by type. Rent extraction through entry distortion can be strong enough to induce bunching at the bottom. Low-demand buyers who split their purchases then face the minimum-purchase requirement described above.

NON-CONDITIONAL TARIFFS. When the incumbent cannot tell a high-demand buyer who also purchases from the entrant from a low-demand buyer who does not, the same exploitative aim must be pursued through a single schedule, and this changes the channel. A market-share contract, with two tariffs at its disposal, extracts the entry gains through the *level* of the rebate, keeping marginal prices above cost. A non-conditional tariff has no such latitude. We show that whenever marginal prices stay above cost throughout, entry is necessarily costly to the incumbent, so that capturing the entry gains for some buyer types requires pricing below cost over part of the consumption profile. Below-cost pricing is thus the *marginal* counterpart of the level-based extraction a market-share contract performs directly: an accommodation of entry to share in its gains, not an exclusionary device. Relatedly, the probability of entry ceases to be monotone in demand, rising for low types and falling for high ones. Equivalently, high-demand buyers are the ones for whom splitting purchases no longer comes with lower marginal prices, a first appearance of the minimum purchase requirement. The optimal tariff splits into two segments: a high-demand branch with downward-distorted consumption, and a low-demand branch, where the buyer may also purchase from the entrant, with upward-distorted consumption. Both segments are joined at an *inward kink* where buyers bunch on a minimum purchase. Averaging over entry and no-entry, the martingale property is then replaced by a *sub-martingale* property: consumption stays above its monopoly level. This shape, a minimum purchase followed by discounts on all infra-marginal units, is that of an *all-unit discount*. By a convex-envelope argument, we show that its buyer-relevant content coincides with that of a non-conditional tariff.

WELFARE ANALYSIS AND IMPLICATIONS FOR ANTITRUST. We quantify these mechanisms numerically,

computing profit, buyer surplus, entry and welfare for both contracts over a large grid of entrant configurations.⁴ Three findings emerge. First, the non-conditional tariff dominates the market-share contract on both welfare and buyer surplus, strictly so once entry costs are moderately dispersed (the welfare ranking reverses only on a minority of high-capacity configurations at the lowest dispersion). The market-share contract extracts more rent and forecloses entry more heavily, and the extra foreclosure is a social loss, not a pure transfer. Second, the all-unit discount, the form most often litigated, comes close to the non-conditional optimum once it carries an infra-marginal instrument (a minimum-purchase clause or a fixed fee), recovering almost all of the non-conditional profit, against less than half for its purely linear version: the economic distinction between the retroactive rebate of an all-unit discount and the incremental rebate of a non-conditional tariff is thin. Third, we run the price-cost tests of antitrust practice on these contracts. Measured against a myopic incumbent who earns her full monopoly profit without responding to entry, the as-efficient-competitor test tracks the buyer’s surplus rather than welfare and reverses sign across entry regimes. It sanctions the non-conditional tariff precisely where the incumbent’s strategic response raises welfare, and clears it where the response is too small to matter. In this model the price-cost tests detect transfers, not harm.

ORGANIZATION OF THE PAPER. Section 2 reviews the relevant streams of literature. Section 3 introduces the model. Section 4 discusses several benchmarks that provide context for our analysis. Section 5 examines market-share contracts, detailing their properties and implications. Section 6 analyzes non-conditional tariffs. Section 7 draws some implications of our findings for antitrust analysis. Section 8 briefly concludes. Proofs of theoretical results are provided in the Appendix. The Online Appendix presents numerical simulations and different extensions of our basic model.

2. LITERATURE REVIEW

Our paper connects several literatures. It builds most directly on the theory of partial foreclosure, where vertical contracts serve to exploit rather than block entry. It relates to the mechanism-design analysis of rent-shifting under multidimensional private information. Finally, it also speaks to the contractual forms (market-share and all-unit discounts) at the center of recent antitrust scrutiny, including the debate over price-cost tests that we take up in Section 7.

Aghion and Bolton (1987) catalyzed a large literature on how vertical contracts facilitate entry exploitation or foster foreclosure. Subsequent research has explored how this foreclosure mechanism interacts with buyer miscoordination (e.g., Rasmusen, Ramseyer, and Wiley, 1991; Segal and Whinston, 2000a; Chen and Shaffer, 2014, 2019), downstream competition (Fumagalli

⁴An entrant configuration is a pair (k, μ) , where k is the entrant’s capacity and μ the mean of the distribution G of its fixed cost of entry, taken to be a normal distribution truncated at zero with dispersion parameter s . We compute the optimal contracts at every such pair in $[0.1, 2]^2$ at a step of 0.05 (1,521 configurations for each $s \in \{0.1, 0.5, 1, 2\}$) holding the demand side fixed ($\theta = 4$, $c_0 = 2$, and $\Delta\theta = k$). Each configuration is solved by two independent methods, a boundary-value problem built on the optimality conditions derived in the Appendix and a direct nonlinear program over discretized consumption profiles, whose results agree to within 10^{-6} on all reported aggregates. The Online Appendix details the methodology.

and Motta, 2006; Asker and Bar-Isaac, 2014), and specific investments (Spier and Whinston, 1995; Segal and Whinston, 2000b).

Closely related to our setting, Marx and Shaffer (2004) show that under complete information about entry value, nonlinear tariffs are used for rent-shifting rather than prevention, rendering exclusion inefficient only when rent-shifting is imperfect. We complement this logic by introducing asymmetric information about the entrant's fixed cost. Because the buyer, who learns this cost, then holds implicit bargaining power vis-à-vis the incumbent, rent-shifting is necessarily imperfect. The incumbent cannot extract the full entry surplus and must instead manage the entry margin through the rebate she grants the buyer. This places our analysis within the multi-dimensional mechanism design literature (Armstrong, 1996; Rochet and Choné, 1998; Rochet and Stole, 2003), but with a distinctive feature. Unlike Rochet and Stole (2002), where the outside option is an exogenous draw the principal merely screens against, here that very rebate, and hence the probability of entry, is the incumbent's choice, which turns the entry margin into a second screening instrument and gives the entry–consumption correlation and the exploit/accommodate reading their content. The same endogeneity separates us from Choné and Linnemer (2015, 2016). There the split of the entry surplus between buyer and entrant is fixed exogenously, whereas in our model it is the object the incumbent shapes through the contract.

Due to the ex ante uncertainty surrounding the fixed cost of entry, our model generates equilibria where either split or exclusive purchases arise depending on whether entry occurs. This sets our framework apart from Calzolari and Denicolò (2013), where competing sellers of equal bargaining power raise prices through the common-agency channel analyzed by Martimort and Stole (2009). Closer to our setting is the asymmetric framework of Calzolari and Denicolò (2015), where the incumbent's cost advantage is used to target low-demand buyers with exclusive clauses while split purchases occur among high-demand buyers. We instead assume that the entrant operates with a lower marginal cost than the incumbent but must incur a fixed cost to enter. This generates a different pattern of entry. Buyers with identical preferences may be served exclusively by the incumbent or split their purchases, depending on the realization of the entrant's fixed cost.

These models and ours share a primitive. Calzolari, Denicolò, and Zanchettin (2020) show that the models in this tradition, which goes back to Mathewson and Winter (1987), all rest on the same condition: contracts referencing a rival's output are worthless when marginal prices equal marginal cost, and their profitability requires a wedge between price and cost that adverse selection supplies. The wedge does similar work here, but on one dimension only. The rebate that governs entry survives without it, since the incumbent cannot extract the entry rent through a fixed fee when the cost of entry is privately known (Section 4.2); it is the margin that needs it, as the incumbent gains from expanding consumption under entry only because demand screening has depressed it. Its consequence is reversed. In Calzolari and Denicolò (2026) the wedge makes a binding market-share requirement profitable; here it makes entry costly to the incumbent, so

that sharing in its gains requires pricing part of the schedule below cost.⁵

Finally, our paper rationalizes the use of specific, heavily scrutinized contractual clauses. It contributes to the literature on market-share contracts, often studied for their effects on agency problems or intra-brand competition (Majumdar and Shaffer, 2009; Inderst and Shaffer, 2010), and all-unit discount (AUD) tariffs, which are central to recent antitrust cases (Conlon and Mortimer, 2013, 2021). While previous studies examine AUDs' roles in surplus division (Feess and Wohlschlegel, 2010; Ide, Montero, and Figueroa, 2016) or screening under uncertain demand (Kolay, Shaffer, and Ordovery, 2004; Chao, Tan, and Wong, 2018), we show that AUDs generate consumption patterns that mirror the optimal non-conditional tariff when facing both demand heterogeneity and uncertain entry gains.

3. THE MODEL

PREFERENCES. To express buyer B 's demand for multiple units while keeping the analysis tractable, we assume that B 's preferences are separable, quadratic in consumption ($q \geq 0$) and linear in payments (t):⁶

$$(3.1) \quad \theta q - \frac{q^2}{2} - t.$$

TECHNOLOGY. Incumbent I produces at constant marginal cost $c_0 > 0$. Upon paying the cost of entry c , entrant E produces at zero marginal cost up to a fixed capacity $k > 0$. Because the entrant's sale of output to the buyer will either be 0 or k , c/k can be thought of as the entrant's unit cost of producing output. The binary nature of entrant production also removes the distinction between market-share contracts and contracts conditional on exclusive dealing. If $c/k < c_0$, E 's unit cost of production for the first k units is lower than the incumbent's. It follows that the entrant would produce k units in a full-information setting with unrestricted contracts. While E may be more efficient than I at serving small quantities, we assume that k is sufficiently small that a share of B 's demand must be served by I , what we refer to as the incumbent's *non-contestable demand*, borrowing here Choné and Linnemer's (2015) terminology.

INFORMATION. B has private information on his demand parameter θ . This parameter is uniformly distributed on $\Theta = [\underline{\theta}, \bar{\theta}]$ (with $\Delta\theta = \bar{\theta} - \underline{\theta} > 0$).⁷

Both c_0 and k are common knowledge. The entrant's cost c , however, is unknown to the incumbent. c is a random variable drawn from $[0, \bar{c}]$ according to an atomless distribution function G which admits an everywhere positive density g . Following the screening literature,

⁵Their rival is active throughout and holds market power, an extension of Calzolari and Denicolò (2015) in a direction orthogonal to ours. Because the buyer holds the bargaining power here, our entrant is better read as an outside production option whose availability is stochastic and privately known.

⁶We also assume that disposal of extra units is infinitely costly. For the importance of this assumption, see Choné and Linnemer (2016).

⁷The choice of a uniform distribution is again made for tractability, as in Rochet and Stole (2002).

we assume that the Monotone Hazard Rate Property (MHRP) holds:⁸

$$\frac{d}{dc} \left(\frac{G(c) - \tau}{g(c)} \right) \geq 0 \quad \forall \tau \in [0, 1].$$

Since I remains uninformed about E 's entry cost c , she cannot assess the gains from entry. In contrast, we assume B and E both know c . This source of information rent combines with the familiar information rent related to the buyer's demand parameter θ . A primary contribution of this paper is to understand how these two sources of information rent interact.

CONTRACTS. In line with the nonlinear pricing literature, I uses a price schedule to screen buyers with different demands. With *market-share contracts*, I can also condition B 's payment on the quantity the buyer buys from the entrant. Because E produces at zero marginal cost and up to a common knowledge capacity k , that payment is thus just conditional on whether entry occurs or not. Accordingly, a market-share contract consists of a pair of tariffs $\{T_0(q), T_1(q)\}$ where index $i = 1$ (respectively, $i = 0$) refers to the case of entry (respectively, no entry). Market-share contracts thus allow some sort of third-degree price discrimination on top of the familiar second-degree price discrimination.

Market-share contracts, and more generally, contracts that “reference other contracts” in the parlance of Scott-Morton (2013), may be banned by antitrust authorities, in which case I might thus be forced to use a *non-conditional tariff* (i.e., the same price schedule applies whether or not entry occurs). In that scenario, we require

$$T_0(q) = T_1(q) \equiv T(q), \quad \forall q \geq 0.$$

The same requirement would also apply if I is unable to observe whether B buys from E or not. The contractual restriction underlying a non-conditional tariff is thus akin to adding an extra layer of asymmetric information.

TIMING. The contracting game unfolds as follows.

1. B learns his demand parameter θ .
2. I commits to a contract. In the case of market-share contracts, $\{T_0(q), T_1(q)\}$ is offered promising a payment of $T_0(q)$ paid in exchange for q units, conditional on the buyer not purchasing from the entrant, and a payment of $T_1(q)$ in exchange for q units, conditional on entry; in the case of non-conditional contracts, a payment of $T(q)$ is promised in exchange for q units.
3. B learns E 's fixed cost of entry c . B then decides whether to accept I 's contract and whether to purchase k units from E and how much to buy from I .

Last, we make the following assumptions. First, we assume that B has all relevant information and all bargaining power when dealing with E , thus capturing all gains from entry. As such, B

⁸See Bagnoli and Bergstrom (2005). This assumption ensures that the optimization problems found below remain quasi-concave and that optimal consumption profiles satisfy some monotonicity conditions.

buys k units at zero unit price from E , covering E 's fixed cost with some side-payments.^{9,10} Second, we assume that I remains an *essential* supplier, even in the case of entry. For example, contracting exclusively with E has no value because B needs to operate at some minimum scale which is not satisfied by E 's limited capacity. This simplifying assumption implies that B 's outside opportunity, had he not bought from I , is equal to zero and independent of θ .¹¹ Note that in the timing above, we also depart from Aghion and Bolton (1987) in a minor way. Even though the contract is proposed before B learns the entry cost c , the timing of the acceptance decisions ensures that B makes positive profit under all circumstances.

4. BENCHMARKS

This section outlines benchmarks that provide context for the subsequent analysis.

4.1. Efficiency

Entry is efficient whenever the fixed cost of entry is less than what can be saved by not asking I to produce the first k units, i.e.,

$$c \leq c_0 k.$$

Before the realization of the fixed cost of entry c , the efficient probability of entry is thus $G(c_0 k)$. Throughout, we will assume that $c_0 k < \bar{c}$ so that entry always arises with positive probability.

When B 's consumption is split between I and E (i.e., E supplies k units), the first-best aggregate consumption for the buyer is determined by the marginal incentives to expand B consumption above k at marginal cost c_0 :

$$Q_{fb}(\theta) \equiv \arg \max_{q \geq 0} \theta q - \frac{q^2}{2} - c_0(q - k) = \theta - c_0 \quad \forall \theta \in \Theta. \quad 12$$

Note that we use Q to refer to the buyer's total consumption; I 's production is either $Q - k$ when there is entry, or Q when the entrant is inactive. $Q_{fb}(\theta)$ is also B 's efficient consumption in the absence of entry.

To ensure that purchasing from I is efficient under all circumstances (i.e., that I has non-contestable demand), we assume that demand is sufficiently large with respect to E 's capacity k . Specifically, we assume throughout that

⁹That B might bargain with E under asymmetric information would not change the nature of our results, but only their magnitude. Not knowing the fixed cost of entry c , B would offer a constant unit price p for all k units bought from E . E would only accept that price when making a positive profit, i.e., when $p k \geq c$. That E keeps some information rent for low realizations of the fixed cost would reduce the share of the entry gains that B can capture. Yet, B would still get some extra information rent vis-à-vis I from learning c in the course of subcontracting. The reader might have recognized here an insight that is familiar from the literature on hierarchical contracting (McAfee and McMillan, 1995; Laffont and Martimort, 1998).

¹⁰Because we assume that B has all of the bargaining power vis-à-vis E , an alternative interpretation of our model is one without an entrant, but the buyer with demand type θ has the option to produce k units of output in-house at a cost of c , where (θ, c) is known only to the buyer.

¹¹This assumption is relaxed in the Online Appendix.

¹²This efficient outcome also prevails when the monopolist is informed on both θ and c ex post (i.e., when c realizes) and can engage in first-degree price discrimination so as to extract all surplus from B whether or not entry occurs. Things are different if the monopolist is uninformed about these parameters. A trade-off between efficiency and rent extraction arises and the optimal tariff balances these two objectives.

ASSUMPTION 1. $Q_{fb}(\underline{\theta}) \geq k$.

It is convenient to denote by

$$S(\theta, q) \equiv (\theta - c_0)q - \frac{q^2}{2}.$$

Absent entry, the social value of production is thus $S(\theta, Q_{fb}(\theta))$. With entry, it is given by $S(\theta, Q_{fb}(\theta)) + c_0 k - c$.

4.2. Asymmetric Information on the Cost of Entry Only

In order to build intuition for the full analysis, we begin with the simpler case where θ is publicly observed but c remains unknown to the incumbent. Two remarks are in order in this context. First, purchases are split only if the entrant's cost is sufficiently small. Second, I can extract B 's surplus by offering two-part tariffs of the form $t_i(\theta) + c_0 q$ with fixed fees $t_0(\theta)$ and $t_1(\theta)$ contingent on no entry and entry, respectively, where q denotes the quantity purchased from I . Because the marginal price equals c_0 in both cases, consumption is efficient whether or not entry occurs, and B 's payoffs are

$$\begin{aligned}\mathcal{U}_0(\theta) &= \max_{q \geq 0} \theta q - \frac{q^2}{2} - c_0 q - t_0(\theta) = S(\theta, Q_{fb}(\theta)) - t_0(\theta), \\ \mathcal{U}_1(\theta) &= \max_{q \geq 0} \theta(q + k) - \frac{(q + k)^2}{2} - c_0 q - t_1(\theta) = S(\theta, Q_{fb}(\theta)) + c_0 k - t_1(\theta),\end{aligned}$$

where the second line uses Assumption 1 and $\mathcal{U}_1(\theta)$ is computed gross of the fixed cost of entry. Upon entry, B still consumes $Q_{fb}(\theta)$ in total, but the first k units are supplied by E at a zero unit price, which saves him the payment $c_0 k$ he would otherwise make to I . Hence B purchases the first k units from the entrant if

$$\omega(\theta) \equiv \mathcal{U}_1(\theta) - \mathcal{U}_0(\theta) = c_0 k + t_0(\theta) - t_1(\theta) \geq c.$$

The term $\omega(\theta)$ measures B 's willingness to pay for entry: it combines the rebate $t_0(\theta) - t_1(\theta)$ that I grants for also purchasing from E with the saving $c_0 k$ on the units that E supplies.¹³ Because consumption is independent of entry, this setting is in the spirit of Aghion and Bolton (1987) and similar to Choné and Linnemer (2015), except that surplus is θ -dependent.

Since the marginal price equals the marginal cost, I 's profit is the fixed fee of the realized market structure, $G(\omega(\theta))t_1(\theta) + (1 - G(\omega(\theta)))t_0(\theta)$. Using $t_1(\theta) = t_0(\theta) + c_0 k - \omega(\theta)$ to treat $(t_0(\theta), \omega(\theta))$ as the incumbent's instruments, and noting that the participation constraint $\mathcal{U}_0(\theta) \geq 0$ binds while that under entry, $\mathcal{U}_0(\theta) + \omega(\theta) - c \geq 0$, is slack whenever entry occurs, I 's payoff becomes $S(\theta, Q_{fb}(\theta)) + (c_0 k - \omega)G(\omega)$. Only the second term depends on ω : when entry occurs, B retains $\omega - c$ and I appropriates the remaining $c_0 k - \omega$ of the cost saving. The optimal rebate ω_m thus maximizes

$$(4.1) \quad (c_0 k - \omega)G(\omega).$$

¹³Because $\omega_m < c_0 k$ at the optimum below, the rebate $t_0 - t_1$ is in fact negative: I charges a *higher* fixed fee upon entry, clawing back part of the saving $c_0 k$. Throughout the paper, ω is a rebate in indirect utility, not in the level of the tariff.

The trade-off identified by the earlier literature appears here in its simplest form. Raising the rebate makes entry, and hence the saving $c_0k - c$, more likely, but leaves a larger share of it to the buyer, who knows c and pockets the difference $\omega - c$. Given MHRP, this objective is quasi-concave and its maximum is determined by the following first-order condition

$$(4.2) \quad c_0k = \omega_m + \frac{G(\omega_m)}{g(\omega_m)}.$$

The optimal rebate ω_m solves this familiar rent-efficiency trade-off, and entry is inefficiently low since $\omega_m < c_0k$. Only cost considerations determine entry, so ω_m is independent of the preference parameter θ . The solution ω_m in (4.2) will play an important role in the full analysis below.

5. MARKET-SHARE CONTRACTS

We now move to a framework where I designs market-share contracts under incomplete information on the buyer's preferences. In addition to distorting entry in order to capture B 's cost-of-entry information rent, I also has incentives to capture B 's demand information rent. The interplay between these two sources of rents and the efficiency of entry leads to modifications to optimal nonlinear pricing under the threat of entry.

INCENTIVE COMPATIBILITY. The nonlinear price $T_i(q)$ induces a profile of information rents $\mathcal{U}_i(\theta)$ and optimal aggregate purchases $Q_i(\theta) = q_i(\theta) + ik$ (for each market structure $i = 0, 1$) that are defined by the familiar conditions¹⁴

$$\begin{aligned} \mathcal{U}_i(\theta) &= \max_{q \geq 0} \theta(q + ik) - \frac{(q + ik)^2}{2} - T_i(q), \\ q_i(\theta) &\in \arg \max_{q \geq 0} \theta(q + ik) - \frac{(q + ik)^2}{2} - T_i(q), \end{aligned}$$

for $i \in \{0, 1\}$. We focus on settings where $q_i(\theta)$ remains non-negative, and thus I remains an essential supplier. Under such conditions (made precise below), $Q_1(\theta)$ exceeds E 's capacity for all possible values of θ . I always serves a non-contestable share of demand. Bunching, if any, is not justified by I 's desire to exclude B if demand is too low.

Define the difference in B 's indirect utility

$$\omega(\theta) \equiv \mathcal{U}_1(\theta) - \mathcal{U}_0(\theta)$$

¹⁴The *Revelation Principle* of course applies in this multi-dimensional context. There is no loss of generality in considering direct and truthful mechanisms that specify payments t_i and consumption levels Q_i under each market structure $i = 0, 1$ together with a probability of entry P_1 that would depend on the report $(\hat{c}, \hat{\theta})$ that B makes on his whole array of private information (c, θ) . Private information now includes not only B 's preferences parameter but also the fixed cost of entry. Within this class, consider a subset of direct revelation mechanisms corresponding to payments and quantities $\{t_0(\hat{\theta}), t_1(\hat{\theta}), q_0(\hat{\theta}), q_1(\hat{\theta})\}_{\hat{\theta} \in \Theta}$ together with a probability of entry such that $P_1(\hat{c}, \hat{\theta}) = \mathbb{1}\{\hat{c} \leq \mathcal{U}_1(\hat{\theta}) - \mathcal{U}_0(\hat{\theta})\}$ (where $\mathbb{1}\{\cdot\}$ is the indicator function). Those direct mechanisms replicate what can be achieved with market-share contracts $\{T_0(q), T_1(q)\}$. With such schemes, the choice of the market structure is left to B who decides whether or not to purchase from E based on his knowledge of the cost of entry and his preference parameter. Also, consumption levels are not contingent on the cost of entry but only on whether entry arises or not. This restriction is justified when more complex stochastic mechanisms are hard to verify in practice.

when B splits his purchases. It follows that whenever $\omega(\theta) \geq c$, B will buy k units from E . The probability of entry $G(\omega(\theta))$ thus only depends on the rent profiles $\mathcal{U}_0(\theta)$ and $\mathcal{U}_1(\theta)$. Raising (respectively, decreasing) the rent profile $\mathcal{U}_1(\theta)$ (respectively, $\mathcal{U}_0(\theta)$) upon entry (respectively, with no entry) is akin to a rebate that increases the likelihood of entry.

Incentive compatibility requires that $\mathcal{U}_i(\theta)$ is convex (so that the overall quantity $Q_i(\theta) = q_i(\theta) + ik$ remains non-decreasing) and thus almost everywhere differentiable with $\dot{\mathcal{U}}_i(\theta) = \dot{Q}_i(\theta) \geq 0$. It is useful to rewrite incentive compatibility in terms of $\mathcal{U}_0(\theta)$ and the rebate $\omega(\theta)$ as

$$(5.1) \quad \dot{\mathcal{U}}_0(\theta) = \dot{Q}_0(\theta),$$

$$(5.2) \quad \dot{\omega}(\theta) = \dot{Q}_1(\theta) - \dot{Q}_0(\theta).$$

PARTICIPATION CONSTRAINTS. If B does not purchase from E , he buys from I exclusively whenever

$$(5.3) \quad \mathcal{U}_0(\theta) \geq 0 \quad \forall \theta \in \Theta.$$

Because incentive compatibility implies that $\mathcal{U}_0$ is non-decreasing, the most stringent participation constraint is obtained for the lowest-demand type $\underline{\theta}$. Condition (5.3) also implies that, once B knows c ,

$$\mathcal{U}_1(\theta) - c = \mathcal{U}_0(\theta) + \omega(\theta) - c \geq 0,$$

where the last inequality follows from the definition of $\omega(\theta)$ and (5.3). Participation constraints when entry occurs are slack; we need only to enforce the participation constraint (5.3) in the no-entry scenario. The intuition here is that entry is partially subsidized by I , and rent extraction is less of a concern in this case.¹⁵

OPTIMAL MARKET-SHARE CONTRACTS. I 's expected profit can be expressed in terms of $(\mathcal{U}_0(\theta), \omega(\theta), Q_0(\theta), Q_1(\theta))_{\theta \in \Theta}$ as follows:

$$\mathcal{V}_I(T_0, T_1) = \mathbb{E}_\theta \left[G(\omega(\theta)) \underbrace{\left(\overbrace{c_0 k}^{\text{cost saving}} + \underbrace{\left[S(\theta, q) \right]_{Q_0(\theta)}^{Q_1(\theta)}}^{\text{incremental surplus}} - \overbrace{\omega(\theta)}^{\text{rebate}} \right)}_{\text{entry gains}} + \underbrace{S(\theta, Q_0(\theta)) - \mathcal{U}_0(\theta)}_{\text{monopoly profit with no entry}} \right],$$

where $\mathbb{E}_\theta$ is the expectation operator with respect to the distribution of θ . Optimal market-share contracts maximize $\mathcal{V}_I(T_0, T_1)$ subject to the relevant incentive and participation constraints, respectively (5.1), (5.2) and (5.3).

I 's overall payoff is made of two parts: first, the monopoly profit that I gets in the absence of a competitor; second, the expected gains that I obtains from entry. Entry gains account for how much can be saved in terms of variable costs, namely $c_0 k - c$, but also for the incremental surplus obtained when choosing a consumption level $Q_1(\theta)$ closer to the first-best level than $Q_0(\theta)$.

¹⁵This participation constraint is consistent with our assumption that I is an essential supplier for B . Relying on E only is not a valuable option because E is not able to serve the whole demand anyway.

With unit demand, this latter term disappears. It will be the source of much of the difference between our results and Aghion and Bolton's (1987).

To raise the expected gains that I can share from entry, attractive deals must be offered to B . These deals are reflected both in the absolute levels of the tariffs and in their margins. Regarding levels, raising the rebate $\omega(\theta)$ enhances the likelihood that the fixed cost of entry is covered. As for margins, the incremental surplus $[S(\theta, q)]_{Q_0(\theta)}^{Q_1(\theta)}$ grows when the marginal price is reduced in scenarios involving entry.

5.1. Joint Distortions on Entry and Consumption

SEPARATING ALLOCATIONS. We solve the relaxed problem, omitting the monotonicity and positivity constraints on $Q_0(\theta)$ and $Q_1(\theta)$, and discuss below whether the omitted constraints bind, giving rise to bunching or exclusion. It is helpful to define $Q_m(\theta)$ as the solution to the classic monopoly screening problem with no entrant, which here is (assuming it is non-negative)

$$Q_m(\theta) \equiv 2\theta - c_0 - \bar{\theta}, \quad \forall \theta \in \Theta.$$

PROPOSITION 1. MARKET-SHARE CONTRACTS. *The solution of the relaxed problem has the following features.*

1. ENTRY. *The optimal rebate $\bar{\omega}(\theta)$ is increasing and concave in θ , and*

$$(5.4) \quad c_0 k > \bar{\omega}(\bar{\theta}) > \omega_m.$$

Entry is inefficiently low: $G(\bar{\omega}(\theta)) < G(c_0 k)$, $\forall \theta \in \Theta$.

2. CONSUMPTION. *Consumption is below its first-best level. Distortions are more pronounced without entry, with a consumption level lower than at the classic monopoly outcome:*

$$(5.5) \quad \bar{Q}_0(\theta) \leq Q_m(\theta) \leq \bar{Q}_1(\theta) \leq Q_{fb}(\theta), \quad \forall \theta \in \Theta,$$

and

$$(5.6) \quad Q_m(\theta) = G(\bar{\omega}(\theta))\bar{Q}_1(\theta) + (1 - G(\bar{\omega}(\theta)))\bar{Q}_0(\theta), \quad \forall \theta \in \Theta.$$

Consumption is always at its first-best level for the highest-demand type, but also for the lowest-demand type if entry occurs

$$(5.7) \quad \begin{aligned} \bar{Q}_1(\theta) &= Q_{fb}(\theta) \quad \forall \theta \in \{\underline{\theta}, \bar{\theta}\}, \\ \bar{Q}_0(\bar{\theta}) &= Q_{fb}(\bar{\theta}). \end{aligned}$$

ENTRY AND CONSUMPTION DISTORTIONS. The probability of entry and the consumption levels induced for screening are positively correlated. Lower-demand buyers, who are more captive, are less likely to engage with the entrant and consume less than higher-demand buyers. This reduced entry among lower-demand buyers serves as an additional screening tool for I , enabling greater rent extraction from higher-demand buyers. This correlation arises despite the independence of the two private information variables, θ (buyer's type) and c (entry cost), both of which contribute to the information rent available to B .

To understand this correlation, we first consider simple (but suboptimal) tariffs $T_0(Q)$ and $T_1(Q)$ using the monopoly tariff, $T_m(Q)$, that implements $Q_m(\theta)$: $T_0(Q) = T_m(Q)$ and $T_1(Q - k) = T_m(Q) - \omega_m$ ($\forall Q \geq k$). Although these two tariffs differ in both margins and levels, B always purchases the same monopoly quantity whether he contracts with one or both suppliers:¹⁶ $Q_0(\theta) = Q_1(\theta) = Q_m(\theta)$ ($\forall \theta \in \Theta$). Proceeding as in Section 4.2, it follows immediately that the optimal rebate is still ω_m . There is inefficiently low entry.

Now fix $\varepsilon > 0$ (sufficiently small) and consider the following mean-preserving spread of consumption levels around the monopoly solution

$$(5.8) \quad \begin{aligned} Q_0^\varepsilon(\theta) &= Q_m(\theta) - \varepsilon G(\omega_m), \\ Q_1^\varepsilon(\theta) &= Q_m(\theta) + \varepsilon(1 - G(\omega_m)). \end{aligned}$$

Consumption is increased beyond the monopoly solution when entry occurs and reduced otherwise. This transformation also impacts the rebate $\omega^\varepsilon(\theta)$ since now (5.2) implies $\omega^\varepsilon(\theta) = \varepsilon$. Taking $\omega^\varepsilon(\bar{\theta}) = \omega_m$ to pin down the profile of rebates, we get $\omega^\varepsilon(\theta) = \omega_m - \varepsilon(\bar{\theta} - \theta) < \omega_m$. There is thus less entry compared to the fixed rebate ω_m .

This perturbation leaves the monopoly base profit unchanged up to a second-order loss, $-\frac{1}{2}\varepsilon^2 G^2(\omega_m)$. Its effect on I 's expected gains from entry is, by contrast, first-order. To see this, consider first the incremental surplus between entry and no entry, which the spread raises by

$$\left[S(\theta, q) \right]_{Q_0^\varepsilon(\theta)}^{Q_1^\varepsilon(\theta)} = (\bar{\theta} - \theta) \varepsilon - \frac{1}{2}\varepsilon^2 (1 - 2G(\omega_m)).$$

The spread also lowers the rebate to $\omega^\varepsilon(\theta) = \omega_m - \varepsilon(\bar{\theta} - \theta)$, which both reduces the probability of entry and raises I 's payoff conditional on entry. These last two effects exactly cancel: their combined first-order contribution is proportional to $-g(\omega_m)(c_0 k - \omega_m) + G(\omega_m)$, which vanishes by the definition of ω_m in (4.2). What remains is the first-order gain in incremental surplus:

$$\begin{aligned} & \mathbb{E}_\theta \left(G(\omega^\varepsilon(\theta)) \left(c_0 k + \left[S(\theta, q) \right]_{Q_0^\varepsilon(\theta)}^{Q_1^\varepsilon(\theta)} - \omega^\varepsilon(\theta) \right) \right) - G(\omega_m)(c_0 k - \omega_m) \\ &= \varepsilon \mathbb{E}_\theta \left(\underbrace{\left(-g(\omega_m)(c_0 k - \omega_m) + G(\omega_m) \right)}_{=0} (\bar{\theta} - \theta) + G(\omega_m)(\bar{\theta} - \theta) \right) \\ &= \varepsilon \frac{\Delta\theta}{2} G(\omega_m) > 0. \end{aligned}$$

Since this first-order gain dominates the second-order loss on the base profit, the monopoly allocation is suboptimal.

The monopoly allocation is improved, then, by raising consumption when entry occurs and lowering it otherwise, while leaving B 's overall information rent approximately unchanged. In terms of tariffs, I decreases the marginal price $T_1'(Q - k)$ under entry and increases the marginal price $T_0'(Q)$ without entry, thereby linking entry and consumption distortions.

The characterization of the optimal consumption levels follows a similar logic although

¹⁶This requires that overall consumption always exceeds E 's capacity, or $Q_m(\theta) > k$.

the mean-preserving spread now depends on the taste parameter θ , a generalization of (5.8) above. Consumption levels can be expressed in terms of the optimal rebate $\bar{\omega}(\theta)$ using (5.6) in Proposition 1:

$$(5.9) \quad \begin{aligned} \bar{Q}_0(\theta) &= Q_m(\theta) - \dot{\bar{\omega}}(\theta)G(\bar{\omega}(\theta)), \\ \bar{Q}_1(\theta) &= Q_m(\theta) + \dot{\bar{\omega}}(\theta)(1 - G(\bar{\omega}(\theta))). \end{aligned}$$

Making entry less likely at lower values of θ , i.e., having $\bar{\omega}(\theta)$ non-decreasing, increases (respectively, reduces) consumption with (respectively, without) entry. Doing so thus allows I to enjoy a positive incremental surplus when entry occurs.

Yet, both entry and consumption distortions become less attractive at the margin as θ increases, i.e., $\bar{\omega}(\theta)$ is also concave. Echoing a familiar result from the nonlinear pricing literature, there is no consumption distortion at the top of the distribution, i.e., $\bar{Q}_0(\bar{\theta}) = \bar{Q}_1(\bar{\theta}) = Q_{fb}(\bar{\theta})$, and thus $\dot{\bar{\omega}}(\bar{\theta}) = 0$. To affect the probability of entry in that neighborhood, I only plays on the absolute difference between $T_0(Q)$ and $T_1(Q - k)$ around $Q_{fb}(\bar{\theta})$, but not on their margins. Marginal prices are both equal to marginal cost there, i.e., $T'_0(Q_{fb}(\bar{\theta})) = T'_1(Q_{fb}(\bar{\theta}) - k) = c_0$.

MARTINGALE PROPERTY. We refer to (5.6) in Proposition 1 as the *martingale property*. Under the threat of entry, the distribution of purchases is a mean-preserving spread of the monopoly distribution. Keeping average consumption equal to its monopoly level, I now spreads the cost of screening B 's demand across the entry and no-entry scenarios. Reducing consumption distortions under entry and exacerbating them without allows I to enjoy a positive incremental surplus when entry occurs, which makes entry more attractive than if consumption stayed at its monopoly level regardless of entry. This extra benefit arises only because I can fine-tune consumption when demand is multi-unit, and it is absent from unit-demand models.¹⁷

EFFICIENCY AT THE BOTTOM. When the optimal allocation is separating, a buyer of type $\underline{\theta}$ faces no downward distortion in consumption if entry occurs. This is reminiscent of Rochet and Stole (2002), and arises technically from *Pontryagin's Maximum Principle*, whose transversality condition on the costate variable associated with (5.2) forces that variable to vanish at both ends of the type distribution, since B 's participation constraint under entry is automatically slack. Economically, the absence of distortion at $\underline{\theta}$ is best understood by recalling that extracting B 's information rent matters less to I when entry occurs: for the marginally participating type, raising surplus to support participation is more valuable than capturing its rent. I therefore subsidizes entry, though it remains inefficiently low, to ultimately capture part of its benefits, and this motive, which runs counter to rent extraction, makes it unattractive to distort the lowest type's entry consumption $\bar{Q}_1(\underline{\theta})$ below $Q_{fb}(\underline{\theta})$.

¹⁷By the envelope theorem, the buyer's expected rent grows with his type at the rate of his average consumption, $(1 - G(\bar{\omega}(\theta)))\bar{Q}_0(\theta) + G(\bar{\omega}(\theta))\bar{Q}_1(\theta)$: the incentive cost of the contract depends on the two consumption profiles only through their mean. With quadratic preferences, so does the marginal expected surplus. The choice of the average consumption profile therefore separates from the choice of the spread and solves the standard monopolistic screening problem (whose solution is Q_m) while the entry problem loads entirely onto the spread $\bar{Q}_1 - \bar{Q}_0 = \dot{\bar{\omega}}$, which affects surplus only at the second order but screens the entry margin at the first. Absent quadratic preferences, a Jensen term would tie the two choices together and the average would drift from Q_m in proportion to the spread.

MARGINAL PRICES. Because consumption never exceeds the first best, the optimal market-share tariffs keep marginal prices above cost, and order them by entry status.

COROLLARY 1. *Under the optimal market-share contract, both tariffs are differentiable and, comparing marginal prices at a given level of total consumption, $\bar{T}'_0(Q) \geq \bar{T}'_1(Q - k) \geq c_0$, with equality to c_0 for the $\bar{\theta}$ -buyer.*

Recall that a type θ buys $\bar{Q}_0(\theta)$ units from I when entry does not occur, but only $\bar{Q}_1(\theta) - k$ units when it does, since the entrant then supplies k units. The relevant marginal prices are therefore $\bar{T}'_0(\bar{Q}_0(\theta))$ without entry and $\bar{T}'_1(\bar{Q}_1(\theta) - k)$ with entry; comparing them at a common total consumption Q is what the corollary does. Two properties hold: the marginal price is everywhere above cost, so a market-share contract never prices below cost; and, once the entrant's k units are accounted for, it is lower, and closer to cost, when entry occurs. We will see that both properties fail under non-conditional tariffs.

BUNCHING. The profiles in (5.9) are the solution only if they are non-decreasing. Differentiating $\bar{Q}_1(\theta)$, strict monotonicity of the entry profile holds when

$$(5.10) \quad -\frac{d}{d\theta} \left(\dot{\bar{\omega}}(\theta) (1 - G(\bar{\omega}(\theta))) \right) \leq 2.$$

Since $\bar{\omega}$ is concave (Proposition 1), this is the binding requirement: the monotonicity of $\bar{Q}_0$ then follows automatically.¹⁸

Two forces act on the entry profile $\bar{Q}_1$ near the bottom. On the one hand, consumption under entry is efficient at $\underline{\theta}$, $\bar{Q}_1(\underline{\theta}) = Q_{fb}(\underline{\theta})$, a level strictly above the monopoly benchmark $Q_m(\underline{\theta})$. On the other hand, the martingale property ties $\bar{Q}_1$ to the monopoly profile with a weight equal to the probability of entry: $Q_m = G(\bar{\omega})\bar{Q}_1 + (1 - G(\bar{\omega}))\bar{Q}_0$. When entry is likely, this weight is large and pulls $\bar{Q}_1$ down toward $Q_m(\theta)$ just above $\underline{\theta}$. Starting from $Q_{fb}(\underline{\theta})$ and having to descend toward $Q_m(\theta)$, the entry profile would need to decrease, which monotonicity rules out. The incumbent then optimally pools the lowest types: $\bar{Q}_1$ is flat at a floor $Q_1^b + k$ over an interval $[\underline{\theta}, \theta_0]$. In effect, I requires buyers with sufficiently low demand to purchase a minimum quantity from it whenever they also source from E .¹⁹

These forces can be made fully explicit. In the Appendix we show that the optimal rebate admits a closed-form characterization: together with endpoint conditions at $\underline{\theta}$ and $\bar{\theta}$, it solves the first-order differential equation

$$(5.11) \quad \dot{\bar{\omega}}(\theta) = \sqrt{\frac{2}{G(\bar{\omega}(\theta))(1 - G(\bar{\omega}(\theta)))} \int_{\bar{\omega}(\theta)}^{\bar{\omega}(\bar{\theta})} g(x)(c_0 k - x) dx}.$$

Equation (5.11) confirms the mechanism: as entry becomes near-certain ($G(\bar{\omega}) \rightarrow 1$), the factor $G(\bar{\omega})(1 - G(\bar{\omega}))$ in the denominator vanishes, so $\dot{\bar{\omega}}$, and with it the slope of $\bar{Q}_1$, must blow

¹⁸Differentiating $\bar{Q}_0$ gives the symmetric condition $\frac{d}{d\theta}(\dot{\bar{\omega}} G(\bar{\omega})) \leq 2$; concavity of $\bar{\omega}$ makes it slack whenever (5.10) holds.

¹⁹This tension is specific to likely entry. When entry is improbable, the weight $G(\bar{\omega})$ on $\bar{Q}_1$ in the martingale vanishes: consumption under entry barely enters average consumption, is no longer pulled toward Q_m , and can rise smoothly from $Q_{fb}(\underline{\theta})$ without violating monotonicity.

up near $\underline{\theta}$ to sustain the efficient level $Q_{fb}(\underline{\theta})$, violating (5.10). Proposition 2 characterizes the optimum with bunching.

PROPOSITION 2. BUNCHING ON AN INTERVAL $[\underline{\theta}, \theta_0]$. *The optimal ‘minimum Purchase Requirement’ Q_1^b requested by I if entry arises and the upper bound of the bunching interval θ_0 (if interior) altogether satisfy the following optimality conditions*

$$(5.12) \quad \mathbb{E}_\theta[Q_m(\theta)|\theta \leq \theta_0] = \mathbb{E}_\theta[(1 - G(\bar{\omega}(\theta)))\bar{Q}_0(\theta)|\theta \leq \theta_0] + \mathbb{E}_\theta[G(\bar{\omega}(\theta))|\theta \leq \theta_0](Q_1^b + k),$$

and

$$(5.13) \quad \bar{Q}_1(\theta_0) = Q_1^b + k.$$

Condition (5.12) generalizes the martingale property to the possibility of bunching. The average monopoly outcome is still equal to a convex combination of consumption levels with (in which case, this level is fixed) and without entry. Figure 1 represents the quantity profiles with and without bunching.

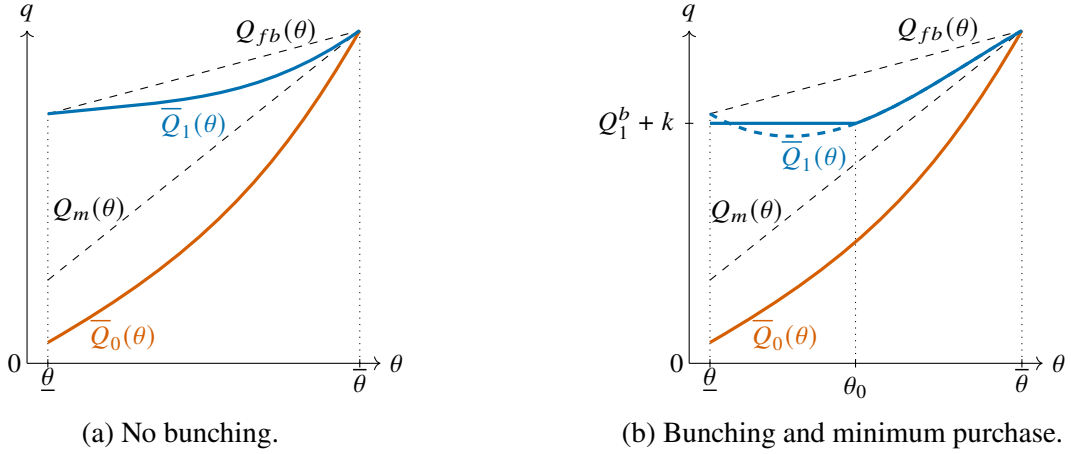


Figure 1: Optimal consumption profiles under market-share contracts. Panel (a): $\bar{Q}_0(\theta)$ and $\bar{Q}_1(\theta)$ are everywhere non-decreasing. Panel (b): $\bar{Q}_1(\theta)$ is locally decreasing around $\underline{\theta}$ and bunching occurs on $[\underline{\theta}, \theta_0]$ at the optimum.

MINIMUM PURCHASE REQUIREMENT AND EXCLUSION AS STRUCTURAL FEATURES. Bunching, the minimum purchase requirement of Proposition 2, is structural rather than distributional, since the monotone hazard-rate property holds throughout. To exploit entry, I manipulates the rebate $\bar{\omega} = \mathcal{U}_1 - \mathcal{U}_0$, whose slope $\dot{\bar{\omega}} = \bar{Q}_1 - \bar{Q}_0$ governs how sharply the two consumption profiles separate. Near the bottom, the martingale property leaves no slack: with consumption under entry at its efficient level, $\bar{Q}_1(\underline{\theta}) = Q_{fb}(\underline{\theta})$, the entire monopoly distortion $\Delta\theta$ must be borne by the no-entry profile alone, whose weight in the average is only $1 - G(\bar{\omega}(\underline{\theta}))$. Hence the leverage in $\dot{\bar{\omega}}(\underline{\theta}) = \Delta\theta/(1 - G(\bar{\omega}(\underline{\theta})))$: the more likely entry is at the bottom, the further $\bar{Q}_0$ must be pushed down, and the steeper the rebate. Such steepness, however, undermines itself. A fast-rising rebate raises the probability of entry and erodes the very weight that sustains the gap between the profiles. The separation therefore collapses just above $\underline{\theta}$ and drives each profile against its bound: $\bar{Q}_1$ against monotonicity, $\bar{Q}_0$ against positivity.

The positivity bound can be made explicit. Using this slope in (5.9), the lowest type's consumption without entry is $\bar{Q}_0(\underline{\theta}) = Q_m(\underline{\theta}) - \Delta\theta G(\bar{\omega}(\underline{\theta}))/ (1 - G(\bar{\omega}(\underline{\theta})))$. Whenever the lowest type would be served by an ordinary monopolist ($Q_m(\underline{\theta}) > 0$), I nonetheless excludes it absent entry ($\bar{Q}_0(\underline{\theta}) = 0$) precisely when

$$\frac{G(\bar{\omega}(\underline{\theta}))}{1 - G(\bar{\omega}(\underline{\theta}))} \geq \frac{Q_m(\underline{\theta})}{\Delta\theta}.$$

Exclusion and bunching are thus two faces of the same lever, and neither is an artifact of an ill-behaved distribution.

Exclusion is more likely the more probable entry is at the bottom (a larger $G(\bar{\omega}(\underline{\theta}))$) relative to the lowest type's monopoly demand, capacity acting on the former through the rebate. Our simulations confirm that bunching on $\bar{Q}_1$ and exclusion on $\bar{Q}_0$ (both near $\underline{\theta}$) are the rule rather than the exception.²⁰ Read through an antitrust lens, these distortions look like abuses (a forced minimum quantity under entry, a refusal to serve the buyer in its absence). Yet, they arise for the opposite reason: not to deter entry but to exploit it. The clauses that look exclusionary are the form taken by the incumbent's appropriation of the gains from entry. This resemblance to foreclosure is no guide to its effect on entry. The shortfall of (expected) entry below its first-best level is largest not where the contract looks most exclusionary, but at intermediate capacities, under the minimum-purchase regime with no exclusion.

6. NON-CONDITIONAL TARIFFS

ALL-UNIT DISCOUNT TARIFFS. All-unit discount tariffs are among the contractual forms most closely scrutinized in vertical antitrust cases.²¹ Under such a tariff the buyer pays a price p_1 on the first Q^b units and a lower price $p_0 < p_1$ on every unit beyond that threshold, the discount $p_1 - p_0$ applying not only to the marginal unit but to all infra-marginal ones (a feature competition authorities often read as a barrier to entry). Formally,

$$T_{AUD}(q) = \begin{cases} p_1 q & \text{if } q \leq Q^b, \\ p_0 q & \text{otherwise.} \end{cases}$$

This abrupt drop in the unit price at Q^b , a *price cliff* in the language of practitioners, is a feature of the contract's *form*. But the form alone does not determine the contract's effects, which depend on the behavior the tariff induces.

The reason is that the buyer does not respond to the whole tariff. An all-unit discount carries options that play no role in his decision. With quadratic-linear preferences as in (3.1), the buyer's

²⁰Over the grid of entrant configurations described in the Online Appendix, at the reference dispersion of entry costs, the fully separating contract is optimal on only 29% of configurations. Bunching on $\bar{Q}_1$ arises on 33%, bunching combined with exclusion on a further 20%, and on the remaining 18%, at the largest capacities, the incumbent both excludes and withdraws supply under entry altogether ($\bar{Q}_1 = k$). The regimes are ordered by the entrant's capacity: as k rises, bunching sets in first and exclusion second. The pattern is stable across the dispersion of entry costs, though the exact shares depend on the bounds of the grid.

²¹All-unit discounts and retroactive loyalty rebates have been central to a series of major cases (*Michelin II* and *British Airways/Virgin* in the E.U., *Intel* on both sides of the Atlantic, *Tomra*) in which such schemes were challenged as instruments of exclusion.

optimal consumption is found by moving down his indifference curve along $T_{AUD}(q)$ until it last meets the schedule (Figure 2); the relevant object is therefore not T_{AUD} itself but its lower convex envelope $\bar{T}_{AUD}(q)$,²² which is what the buyer effectively faces: he makes the same choice under T_{AUD} and under the lower tariff $\bar{T}_{AUD}$. This envelope is the *effects* counterpart of the price cliff. It has three segments, a linear part of slope p_1 , a strictly concave part tracing the indifference curve of a type θ_0 , and a linear part of slope p_0 :

$$\bar{T}_{AUD}(q) = \begin{cases} p_1 q & \text{if } q \leq \bar{q}(\theta_0) = \theta_0 - p_1, \\ \theta_0 q - \frac{q^2}{2} - \frac{1}{2}(\theta_0 - p_1)^2 & \text{if } q \in [\bar{q}(\theta_0), \bar{q}(\theta_1)], \\ p_0 q & \text{if } q \geq \bar{q}(\theta_1) = Q^b. \end{cases}$$

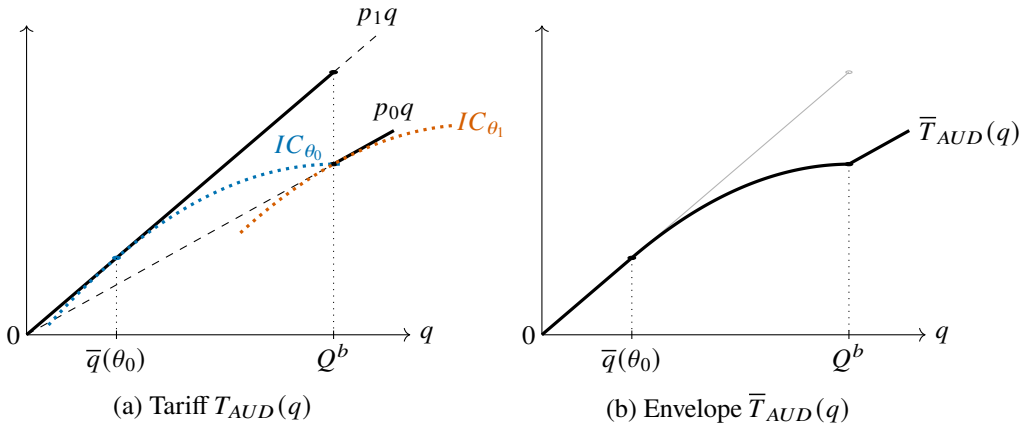


Figure 2: An all-unit discount tariff and its convex envelope. In panel (a), type θ_0 is tangent to the branch $p_1 q$ at $\bar{q}(\theta_0) = \theta_0 - p_1$ and indifferent up to the minimum purchase Q^b ; type θ_1 is tangent to the branch $p_0 q$ at Q^b . Panel (b) shows the resulting envelope.

Through its envelope, the price cliff becomes an *inward kink* at Q^b : the marginal price jumps upward there,

$$\bar{T}'_{AUD-}(Q^b) = \theta_0 - Q^b < p_0 = \bar{T}'_{AUD+}(Q^b),$$

a non-differentiability unusual among screening tariffs. Seen from the buyer's side, this kink is a *minimum purchase*: a range of types $\theta \in [\theta_0, \theta_1]$ bunch at Q^b , consuming exactly that quantity from I even though they would otherwise buy less. The two are one and the same feature, viewed on the tariff and on the allocation it induces. Read on its form, it is the clause antitrust authorities flag as foreclosing the rival, a forced minimum that crowds out the buyer's purchases from E . Whether its effect matches that reading is what the analysis below puts in question. The rest of this section shows that exactly this feature characterizes the optimal non-conditional tariff, so that the all-unit discount is not an arbitrary contractual form but the practical counterpart of the optimum we derive.

MORE GENERAL TARIFFS. Now suppose that I offers the same nonlinear price $T(q)$ regardless of

²²Formally $\bar{T}_{AUD}(q) + \frac{q^2}{2} = \text{conv}\{T_{AUD}(q) + \frac{q^2}{2}\}$. The induced consumption profile is $\bar{q}(\theta) = \theta - p_1$ on $[\theta, \theta_0]$, $\bar{q}(\theta) = Q^b$ on $[\theta_0, \theta_1]$, and $\bar{q}(\theta) = \theta - p_0$ on $[\theta_1, \bar{\theta}]$, with thresholds $\theta_1 = p_0 + Q^b$ and $(\theta_0 - p_0)Q^b - \frac{1}{2}(Q^b)^2 = \frac{1}{2}(\theta_0 - p_1)^2$.

entry. In this scenario I has a single instrument to pursue two goals: extracting the buyer's rent, as a monopolist would, and capturing the gains from entry. Serving both market structures with one tariff ties together the rent profiles B obtains under each. Indeed, B 's rent with entry (gross of the fixed cost), namely

$$\mathcal{U}_1(\theta) = \max_{q \geq 0} \theta(q + k) - \frac{(q + k)^2}{2} - T(q) \quad \forall \theta \in \Theta,$$

can be rewritten as

$$(6.1) \quad \mathcal{U}_1(\theta) = \theta k - \frac{k^2}{2} + \mathcal{U}_0(\theta - k) \quad \forall \theta \in \Theta.$$

Upon entry, a buyer of type θ gets an information rent $\mathcal{U}_1(\theta)$, which comprises the gross surplus from purchasing k units from E and the information rent $\mathcal{U}_0(\theta - k)$ associated with a *fictitious* lower $\theta - k$ in the absence of entry. From I 's perspective, a buyer of type θ already contracting with E behaves as if he had a reduced demand parameter $\theta - k$. Consequently, consumption levels across both scenarios satisfy the key identity

$$Q_1(\theta) = Q_0(\theta - k) + k.$$

The *delay* k in (6.1) makes it as though each type were split into two selves: one buying exclusively from I at parameter θ , the other, having bought k from E , dealing with I as the lower fictitious type $\theta - k$.²³ Accordingly, we define the set of such fictitious types as the *pushed-backward* types set $\Theta_{-k} = [\underline{\theta} - k, \bar{\theta} - k]$. Notice that $Q_0(\theta)$ has now to be defined over the joint set $\Theta_{-k} \cup \Theta$.

Using our previous notations, the cut-off value $\omega(\theta)$ for E 's fixed cost, above which entry does not occur, now satisfies

$$(6.2) \quad \omega(\theta) = \theta k - \frac{k^2}{2} + \mathcal{U}_0(\theta - k) - \mathcal{U}_0(\theta).$$

Although $\omega(\theta)$ could still be interpreted as an implicit rebate, the fact that the tariff is non-conditional implies that this rebate is not contractually specified. Nonetheless, the probability of entry remains expressed as $G(\omega(\theta))$. For future reference, differentiating (6.2) yields the following expression for how the cut-off $\omega(\theta)$ evolves

$$(6.3) \quad \dot{\omega}(\theta) = k + Q_0(\theta - k) - Q_0(\theta) \quad \forall \theta \in \Theta.$$

A closed-form characterization is not available in general. The probability of entry $G(\omega(\theta))$, and hence I 's payoff, depend on $\mathcal{U}_0$ and Q_0 at two distinct types, θ and $\theta - k$, so the problem is no longer pointwise and calls for techniques from optimal control with delay (Kamien and Schwartz, 1991). We therefore focus on the main properties of the solution, under the following assumption.

ASSUMPTION 2. $\Delta\theta = k$.

²³This echoes a standard mechanism in the common agency literature (e.g. Martimort and Stole, 2009), where competing suppliers of complements induce the buyer to shade his type toward each, distorting consumption downward. Here the goods are a priori substitutes, but B 's fixed purchase of k units from E creates a similar complementarity at the extensive margin.

Under Assumption 2, the real types and their pushed-backward counterparts fill a single connected interval, joined at $\underline{\theta} = \bar{\theta} - k$, and the problem becomes tractable. The assumption admits a clean decomposition of the optimal tariff into two branches: an upper portion, at higher consumption, serving high-demand types who buy exclusively from I ; and a lower portion, at lower consumption, taken either by low-demand exclusive buyers or by high-demand types who also source k units from E . Far from a knife-edge, $\Delta\theta = k$ is the point at which the contract's distinctive features (the minimum-purchase requirement and the inward kink) are present and a characterization is available. These features persist when $\Delta\theta > k$ and over much of $\Delta\theta < k$ as well, dissolving only when capacity is large relative to dispersion and entry is unlikely.²⁴ Where it does dissolve, the non-conditional tariff reproduces the market-share allocation exactly, through purely incremental discounts. When the buyer's two consumption ranges lie on well-separated parts of the schedule, a single tariff recovers the two degrees of freedom of a conditional contract. There, a ban on contracts conditioned on entry would leave the outcome unchanged: the incumbent attains the same allocation with an ordinary quantity discount.

INCENTIVE COMPATIBILITY. Any incentive compatible profile $(\mathcal{U}_0(\theta), Q_0(\theta))$ must still satisfy the standard implementability conditions (5.1) and (5.2), now imposed not only on Θ but also on the fictitious types in Θ_{-k} .

Monotonicity of Q_0 , required by incentive compatibility on $\Theta_{-k} \cup \Theta$, applies in particular at the junction point $\bar{\theta} - k = \underline{\theta}$, where Q_0 may take different left and right values. It requires

$$(6.4) \quad Q_0(\underline{\theta}^-) \leq Q_0(\underline{\theta}^+).$$

PARTICIPATION CONSTRAINTS. The relevant participation constraint remains that of the lowest-demand buyer who buys exclusively from I . Types in Θ_{-k} instead receive more than their reservation payoff: the extra surplus $\theta k - \frac{k^2}{2}$ obtained from the k units bought from E facilitates entry.

OPTIMAL NON-CONDITIONAL TARIFF. The two objectives of capturing the gains from entry and extracting the buyer's surplus appear in I 's expected payoff, derived in the Appendix:

$$\mathcal{V}_I(T) = \mathbb{E}_\theta \left(G(\omega(\theta)) \left(c_0 k + \left[S(\theta, q) \right]_{Q_0(\theta)}^{Q_0(\theta-k)+k} - \omega(\theta) \right) + S(\theta, Q_0(\theta)) - \mathcal{U}_0(\theta) \right).$$

The payoff retains the same structure as under market-share contracts: monopoly profit plus expected gains from entry. The single tariff, however, ties together the consumption profiles in the two market structures and $Q_1(\theta)$ is no longer free: $Q_1(\theta) = Q_0(\theta - k) + k$, so Q_0 enters the payoff at both θ and $\theta - k$. The rebate $\omega(\theta)$ loses its independent status as well, becoming a function of Q_0 through (6.3). The incumbent thus controls a single profile Q_0 , with all three objects of the market-share program (Q_0 , Q_1 , and ω) now tied to it.

Before characterizing the optimal non-conditional contract, we first illustrate the main forces through an example.

²⁴See the analysis of $\Delta\theta \neq k$ in the Online Appendix, which extends the characterization and finds the welfare ranking and the antitrust conclusions qualitatively unchanged.

EXAMPLE 1: MONOPOLY PRICING. Suppose I takes a myopic stance, offering the tariff $T_m(Q)$ that is optimal absent entry without adapting to the entry threat. Consumption is then at the monopoly level when entry does not occur, $Q_0(\theta) = Q_m(\theta)$, and at $Q_1(\theta) = Q_m(\theta) - k$ when it does.²⁵ the monopoly downward distortion outweighs the k units bought from E , and overall consumption falls under entry. From (6.2), the rebate is

$$\omega_m(\theta) = c_0 k + (\bar{\theta} - \theta)k + \frac{k^2}{2} \quad \forall \theta \in \Theta,$$

decreasing in θ : higher-demand types face less entry. Moreover, since $\omega_m(\theta) > c_0 k$ for every type, the myopic incumbent induces strictly more entry than is efficient. The reason is the exercise of market power: absent a competitive constraint, I screens the buyer through a tariff whose marginal price exceeds marginal cost and the buyer values the k contestable units at that marginal price rather than at c_0 . He therefore turns to the entrant even for cost realizations above the efficient threshold. The gap $\omega_m(\theta) - c_0 k$ decomposes as the sum of a screening markup, $(\bar{\theta} - \theta)k$, and an inframarginal residual, $\frac{k^2}{2}$. Over-entry is therefore tied to the exercise of market power rather than to the specific shape of T_m : only a two-part tariff priced at marginal cost, available under complete information (Section 4.1), removes the markup, so any non-reacting benchmark that exercises market power through an above-cost marginal price over-admits entry by the same mechanism.

The myopic tariff maximizes base profit but precludes any capture of entry gains. The net gain from entry is strictly negative for every type,

$$c_0 k + \left[S(\theta, q) \right]_{Q_m(\theta)}^{Q_m(\theta-k)+k} - \omega_m(\theta) = -2(\bar{\theta} - \theta)k - k^2 < 0,$$

so entry is costly to I throughout. The example thus exposes the central tension of the non-conditional contract: extracting entry gains and screening demand pull in opposite directions when only one tariff is available. ■

ABOVE-COST PRICING PRECLUDES POSITIVE ENTRY GAINS. Example 1 is a special case of a more general result.

PROPOSITION 3. *If $T'(Q) \geq c_0$ for all Q , then $c_0 k + \left[S(\theta, q) \right]_{Q_0(\theta)}^{Q_0(\theta-k)+k} - \omega(\theta) \leq 0 \quad \forall \theta \in \Theta$, where $\omega(\theta)$ satisfies (6.2).*

The mechanism of Example 1 thus extends to any tariff that prices every unit above cost: when it occurs, entry transfers more to the buyer than it generates for I . Capturing positive entry gains for some type therefore requires below-cost pricing on some quantities. Two opposing forces shape the optimal tariff. With entry, the buyer consumes less from I , and lowering the marginal price below cost makes entry more attractive to the buyer, allowing the incumbent to capture a larger share of its gains through the rebate. In the absence of entry, marginal price above cost remains the standard instrument for screening. Unlike under market-share contracts, where two distinct schedules can implement these opposite slopes directly, a single tariff must

²⁵Provided this quantity remains positive, i.e. $Q_m(\theta) > k$.

reconcile both within one continuous schedule. Doing so requires a discontinuity in slope at the junction of the two branches, which forces a range of types to bunch at that point. The three propositions that follow establish, in turn, that the optimal non-conditional contract exhibits such bunching, characterize the corresponding quantity schedules, and show that the tariff itself has an inward kink at the bunching point. We further impose that the problem with non-conditional contracts is regular in the following sense.

ASSUMPTION 3. *At the optimum,*

$$(6.5) \quad \frac{G(\omega(\theta)) - 1}{g(\omega(\theta))} < c_0 k + \left[S(\theta, q) \right]_{Q_0(\theta)}^{Q_0(\theta-k)+k} - \omega(\theta) < \frac{G(\omega(\theta))}{g(\omega(\theta))} \quad \forall \theta \in \Theta.$$

Condition (6.5) essentially requires that both rents left to the buyer, $\mathcal{U}_0(\theta)$ without-entry or $\mathcal{U}_0(\theta - k)$ with-entry, are costly to the incumbent. Each rent plays two roles. Directly, it is a cost that I must grant to the buyer in exchange for information revelation. Indirectly, it shifts the rebate and thereby the probability of entry, which can either benefit or hurt I depending on the sign of the entry gains. Assumption 3 requires that the direct effect dominates the indirect one for each rent profile. This is the natural extension of the standard screening logic to the present setting with entry and a unique tariff, and it is observed throughout our simulations.

MINIMUM PURCHASE REQUIREMENT. A feature of the optimal non-conditional tariff is the systematic emergence of a *minimum purchase requirement*. A range of types is required to consume the same quantity Q^b when buying from I .

PROPOSITION 4. *Suppose Assumptions 2 and 3 hold. At the optimum, the consumption profile $\bar{Q}_0$ defined on $\Theta_{-k} \cup \Theta$ exhibits bunching in a neighborhood of the junction $\bar{\theta} - k = \underline{\theta}$.*

The result reflects a structural tension specific to the non-conditional contract. Consider first the buyers who consume exclusively from I . Their information rent is governed by standard screening. The participation constraint $\mathcal{U}_0(\underline{\theta}) = 0$ leaves the lowest type indifferent between buying or not, and consumption levels are distorted downward to deter higher types from understating their demand. Consider now the buyers who also source from E . From (6.1), the rent of their highest type is forced to $\mathcal{U}_1(\bar{\theta}) = \bar{\theta}k - \frac{k^2}{2}$. This high anchor creates incentives for lower types of this population to *overstate* their demand, incentives that are countered by distorting consumption *upward*. With two separate tariffs T_0 and T_1 , as under market-share contracts, the incumbent could implement these opposite distortions on independent schedules. The non-conditional contract, constrained to a single tariff, cannot. The downward distortion on the exclusive branch and the upward distortion on the non-exclusive branch can only be reconciled by bunching a range of types at a common consumption level Q^b .

OPTIMAL CONSUMPTION LEVELS. The bunching can affect the exclusive buyers (on an interval $[\underline{\theta}, \theta_0]$), the non-exclusive buyers (on an interval $[\theta_1, \bar{\theta}]$), or both. We focus on the generic case where bunching affects both with $\theta_0 < \theta_1$.²⁶

²⁶The boundary configurations follow as limits by collapsing one of the bunching intervals. When $\theta_0 > \theta_1$, both quantity profiles are constant ($\bar{Q}_0(\theta) = Q^b$, $\bar{Q}_1(\theta) = Q^b + k$) on $[\theta_1, \theta_0]$.

PROPOSITION 5. Suppose Assumptions 2 and 3 hold. Then the optimal non-conditional contract has the following features.

1. DISTORTIONS ON CONSUMPTION LEVELS.

- On $[\underline{\theta}, \theta_0]$, $\bar{Q}_1(\theta)$ is increasing with $\bar{Q}_1(\underline{\theta}) = Q_{fb}(\underline{\theta})$ while $\bar{Q}_0(\theta) = Q^b$.
- On $[\theta_0, \theta_1]$, both $\bar{Q}_0(\theta)$ and $\bar{Q}_1(\theta)$ are separating and satisfy

$$(6.6) \quad \begin{aligned} \bar{Q}_0(\theta) &= Q_m(\theta) + \gamma - \dot{\bar{w}}(\theta)G(\bar{w}(\theta)), \\ \bar{Q}_1(\theta) &= Q_m(\theta) + \gamma + \dot{\bar{w}}(\theta)(1 - G(\bar{w}(\theta))). \end{aligned}$$

with $\gamma \in (0, \Delta\theta)$.

- On $[\theta_1, \bar{\theta}]$, $\bar{Q}_0(\theta)$ is increasing with $\bar{Q}_0(\bar{\theta}) = Q_{fb}(\bar{\theta})$ while $\bar{Q}_1(\theta) = Q^b + k$.

2. ENTRY DISTORTION. The optimal rebate $\bar{w}(\theta)$ decreases around $\bar{\theta}$ and increases around $\underline{\theta}$.

Distortions in Proposition 5 resemble those observed under market-share contracts, as evident when comparing (6.6) with (5.9). The lower range of consumption levels, $\bar{Q}_1(\Theta) - k$, corresponds to purchases from I when the buyer splits; the upper range, $\bar{Q}_0(\Theta)$, corresponds to exclusive purchases from I .

Unlike under market-share contracts, however, $\bar{w}(\theta)$ is no longer monotone in θ . For higher consumption levels, which correspond to high-demand types purchasing exclusively from I , marginal prices exceed marginal cost, so by Proposition 3 entry is strictly costly to I ; reducing $\bar{w}$ around $\bar{\theta}$ is the way to mitigate this cost. For lower consumption levels, where non-exclusive buyers are tempted to overstate their demand, I counters these incentives by correlating consumption with entry decisions, which forces $\bar{w}$ to increase with θ near $\underline{\theta}$.

SUB-MARTINGALE PROPERTY. Equation (6.6) implies a weakened version of the martingale property of market-share contracts. On the separating interval $[\theta_0, \theta_1]$,

$$(6.7) \quad Q_m(\theta) + \gamma = G(\bar{w}(\theta))\bar{Q}_1(\theta) + (1 - G(\bar{w}(\theta)))\bar{Q}_0(\theta),$$

and, on average on Θ ,

$$\mathbb{E}_\theta[Q_m(\theta)] + \gamma = \mathbb{E}_\theta[G(\bar{w}(\theta))\bar{Q}_1(\theta) + (1 - G(\bar{w}(\theta)))\bar{Q}_0(\theta)].$$

The wedge γ measures the cost of the single-tariff constraint: under market-share contracts $\gamma = 0$ and the weighted average of consumption reproduces the monopoly level Q_m , but under non-conditional contracts the bunching at Q^b lifts the average above Q_m .

OPTIMAL NON-CONDITIONAL TARIFF. The optimal non-conditional tariff $\bar{T}(Q)$ consists of two branches pieced together at a kink at Q^b . The upper branch ($Q > Q^b$) targets high-demand types who purchase exclusively from I . The lower branch ($Q < Q^b$) corresponds to types who also purchase k units from E . Low types in $[\underline{\theta}, \theta_0]$ are bunched to a minimum purchase requirement in the no-entry branch. High types in $[\theta_1, \bar{\theta}]$ are bunched to a quantity ceiling in the entry branch. This bunching at Q^b has a direct counterpart on the tariff:

COROLLARY 2. Under Assumptions 2 and 3, the optimal non-conditional tariff $\bar{T}$ has a kink with an upward jump at Q^b with left- and right-derivatives $\bar{T}'(Q^{b-}) = \theta_1 - Q^b - k < \bar{T}'(Q^{b+}) = \theta_0 - Q^b$.

The kink is where the two branches meet, and its orientation reflects the asymmetric distortions discussed above. On the entry branch, $\bar{T}'$ is low, possibly below marginal cost: the incumbent reduces the marginal price to capture entry gains by inducing the buyer to consume more. On the no-entry branch, $\bar{T}'$ is above marginal cost: standard downward screening distortion for exclusive purchases. Figure 3 illustrates the quantity profiles defined in Proposition 5 and the corresponding tariff.

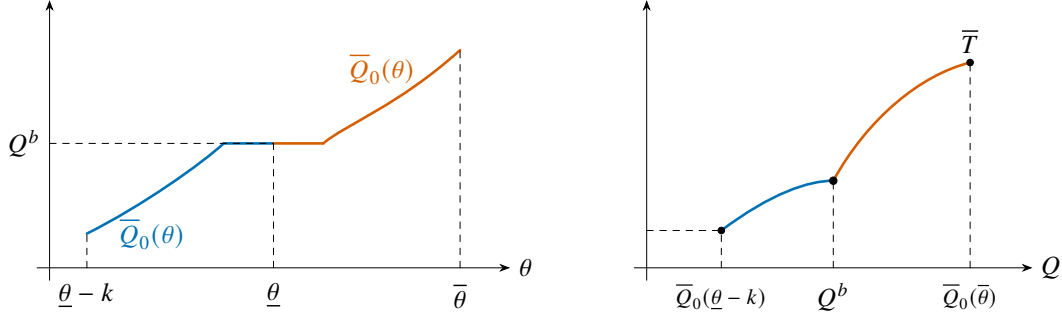


Figure 3: Optimal quantity profile (left) and optimal tariff (right) under non-conditional contracts.

Numerical simulations help document the magnitude and pervasiveness of the theoretical results. First, the below-cost pricing on the entry branch holds systematically: $\bar{T}' < c_0$ over essentially all of the entry-branch quantity support, with a median depth of -0.57 relative to marginal cost. Weighting types by the probability of effectively choosing the entry branch, a median of 83% of all buyers are served below cost. Second, the kink at Q^b traverses marginal cost across the entire grid: $\bar{T}'(Q^{b-}) < c_0 < \bar{T}'(Q^{b+})$ holds without exception, with a median jump of 0.91. Third, the wedge γ remains strictly interior to $(0, \Delta\theta)$, and increases with the entry threat.

The kinked, piecewise structure of $\bar{T}$ resembles the convex envelope of an all-unit discount tariff. Our simulations show that a two-tier AUD with linear prices, complemented with a minimum-purchase requirement, approximates $\bar{T}$ to within about 5% of profit at every grid point and 1.5% on average; a two-tier two-part tariff without the requirement is tighter, within 1% on average. Practical AUD contracts are therefore close counterparts of the theoretical optimum derived here.

7. IMPLICATIONS FOR ANTITRUST

Three contract forms recur in the antitrust scrutiny of conditional pricing, and the legal tradition treats them differently: all-unit discounts with retroactive rebates have been *per se* suspect in the E.U. (*Hoffmann-La Roche*, *Michelin II*, *British Airways*), above-cost incremental discounts have benefited from a safe harbor in the U.S. (*Brooke Group*, *Concord Boat*), and market-share contracts have been scrutinized in both regimes. Yet the previous sections show that the three perform the same economic function, extracting the buyer's rent while accommodating rather than blocking entry, and differ only in the instrument through which they do so: the level of a rebate, an incremental discount, or a minimum-purchase requirement combined with price tiers.

7.1. Two Tests

Recent antitrust practice has converged on the As-Efficient Competitor (AEC) test, typically implemented through an Effective Price (EP) proxy: the EP test in the E.U. (*Intel*, *Unilever*) and its U.S. counterpart, the Discount Attribution Test (*Cascade Health*). A large literature debates their appropriateness for conditional discounts under complete information and no uncertainty about entry, showing how such discounts may inefficiently exclude a viable competitor.²⁷ On price-cost tests specifically, Calzolari and Denicolò (2020) go further within the demand-boost framework, where loyalty discounts serve to raise the buyer's demand rather than to exclude, and find price-cost tests to be misleading or uninformative there. Our framework identifies a different failure. The incumbent may optimally accommodate entry in order to extract the buyer's informational rent. Since the AEC test asks whether the entrant can profitably match the effective price offered to the buyer rather than what entry does to welfare, the bias runs opposite to the conventional exclusionary concern. The test may protect excessive rather than insufficient entry.

Both tests have explicit expressions here. The AEC test compares the buyer's willingness to pay for the entrant's capacity to the cost an as-efficient entrant would incur in supplying k units, and fails when $\mathcal{U}_1(\theta) - \mathcal{U}_0(\theta) \leq c_0 k$.²⁸ The EP test is an observable proxy for an authority that does not observe the buyer's utility: a scheme fails it whenever $T_0(Q) - T_1(Q - k) < c_0 k$, a condition that separates a marginal-price from a level effect.²⁹ The two coincide only under perfectly inelastic demand. More generally the EP margin is a lower bound on the AEC margin, so AEC failure implies EP failure.³⁰

PROPOSITION 6 (Antitrust tests).

1. *The optimal market-share contract always fails both the AEC and the EP tests.*
2. *A non-conditional tariff satisfies the AEC test if and only if*

$$(7.1) \quad \int_{\theta-k}^{\theta} [Q_0(\tau) - Q_{fb}(\tau)] d\tau \leq 0 \quad \text{for all } \theta.$$

The first result follows directly from Proposition 1. The level instrument falls short of an as-efficient entrant's cost, so the market-share contract fails the AEC test for every type and the EP test a fortiori. The second is more surprising since a non-conditional tariff carries only incremental discounts, with no retroactive component on inframarginal units. It may nonetheless fail, because the test turns on the allocation the tariff induces, itself endogenously distorted by rent extraction, rather than on the tariff's structure. Section 7.2 locates that failure.

²⁷See, among others, Greenlee, Reitman, and Sibley (2008), Salop (2017), Fumagalli and Motta (2017), and Durkin (2017).

²⁸Throughout, a test is evaluated under the strict criterion. It fails when the relevant margin is negative for at least one served type. A less stringent criterion, aggregating the margin in expectation over types, yields the same qualitative results in the simulations; see the Online Appendix.

²⁹Namely $\int_{Q-k}^Q [T'_0(x) - c_0] dx < T_1(Q - k) - T_0(Q - k)$.

³⁰By revealed preference, $\mathcal{U}_1(\theta) \geq \theta Q_0(\theta) - \frac{1}{2} Q_0(\theta)^2 - T_1(Q_0(\theta) - k)$, hence $\omega(\theta) \geq T_0(Q_0(\theta)) - T_1(Q_0(\theta) - k)$.

7.2. The Envelope and the Cliff

Both the AEC and the EP tests are statements about average marginal prices but they differ in how those averages are taken as we show below.

Consider first the AEC test with an unconditional tariff T . Section 6 demonstrated that only the lower convex envelope $\widehat{T}$ (where the envelope is taken in the sense of Footnote 22) matters for the buyer's choice. Since the with-entry rent is the no-entry rent of the pushed-backward type, we have $\mathcal{U}_1(\theta) = \theta k - \frac{k^2}{2} + \mathcal{U}_0(\theta - k)$. Condition (5.1) then makes $\mathcal{U}_0(\theta) - \mathcal{U}_0(\theta - k)$ the integral of consumption over the *contestable window* $[\theta - k, \theta]$, the types whose no-entry choices bear on the purchase of k units from the entrant. Since the buyer's first-order condition identifies his marginal valuation with the marginal price $\widehat{T}'$ he faces almost everywhere, we get

$$(7.2) \quad \omega(\theta) = \mathcal{U}_1(\theta) - \mathcal{U}_0(\theta) = \int_{\theta-k}^{\theta} [\tau - Q_0(\tau)] d\tau = \int_{\theta-k}^{\theta} \widehat{T}'(Q_0(\tau)) d\tau.$$

The AEC test $\omega(\theta) \geq c_0 k$ is thus equivalent to

$$(7.3) \quad \frac{1}{k} \int_{\theta-k}^{\theta} \widehat{T}'(Q_0(\tau)) d\tau \geq c_0.$$

Subtracting $c_0 k$ and using $Q_{fb}(\tau) = \tau - c_0$ returns condition (7.1) of Proposition 6, the same statement which is now read in terms of the allocation. Whatever the schedule specifies over quantities that no type selects therefore cannot enter the AEC margin, by construction rather than by approximation. The effects-based test is form-blind.

The identity (7.2) admits three readings. First, since $\omega(\theta)$ is the cost cutoff below which entry occurs and $c_0 k$ the efficient cutoff, the AEC margin measures type by type the gap between induced and efficient entry. The test is an exact detector of under-entry, measured against the first best. Proposition 6(1) follows immediately, since market-share contracts induce low entry for every type (Proposition 1). This benchmark is the difficulty, and Section 7.3 returns to it.

Second, read on (7.1), the test has a disconcerting orientation on the allocation. It passes when consumption over the contestable window is on average *below* first best and fails when it is above. Hence, a tariff that restricts output tends to pass and one that expands it tends to fail. The myopic-monopoly tariff of Example 1 therefore passes everywhere, while the optimal non-conditional tariff expands consumption on its entry branch and may fail the test there.³¹

Third, if an unconditional tariff has $T' \geq c_0$ everywhere, the integrand in (7.3) is non-negative for every type and the AEC test necessarily passes. By Proposition 3 such an incumbent has nothing to gain from entry, since capturing part of the gains entry creates requires expanding consumption on the entry branch and hence pricing below cost somewhere. The U.S. above-cost safe harbor thus does more than rule out a pricing practice. It confines the incumbent to contracts under which she cannot share in the entrant's efficiency gains at all. This also disposes of the pointwise test, whose literal application to the with-entry branch would, following Areeda-Turner

³¹Below cost-pricing in the sense of $T' < c_0$ is pervasive on the entry branch as shown by our simulations. See Section .

test, find predation since the marginal price falling below c_0 over essentially the whole branch. Instead, a market-share contract pursuing the identical objective through the level of its rebate keeps both schedules above c_0 and passes. Economically identical conduct receives opposite verdicts depending on the instrument of extraction.

The EP test, by contrast, averages the marginal prices the contract *specifies*,

$$\frac{1}{k} \int_{Q_0(\theta)-k}^{Q_0(\theta)} T'(x) dx \geq c_0,$$

in the distributional sense, a downward jump of T contributing a negative Dirac. It differs from (7.3) in integrating T' rather than $\widehat{T}'$, and over quantities rather than types.³² The residual is the gap between the schedule and its envelope: the cliff. For an all-unit discount the two averages diverge on the bridge segment $[\bar{q}(\theta_0), Q^b]$ that no type ever selects, where $\widehat{T}'$ descends smoothly from p_1 to $\theta_0 - Q^b$ while T' stays at p_1 and concentrates the cliff as a downward Dirac of mass $(p_1 - p_0)Q^b$ at the right endpoint. On the purely linear all-unit discount the EP test applied to the raw schedule fails on 35.6% of configurations while the AEC test never fails;³³ applied to the convex envelope, which induces the same choices and differs only at the cliff, it never fails and coincides with the AEC verdict. The cliff alone generates the adverse verdict, the test reading the schedule at $Q_0 - k$, which for the affected types falls on that bridge segment. Each test, in short, reads a channel of extraction; none reads the conduct.

7.3. Comparing with the Monopoly Counterfactual

Before asking what the tests measure, we ask what they should measure. Competition law does not condemn a dominant firm for exercising market power, and second-degree price discrimination is a lawful means of extracting rent. The concern is whether the incumbent modifies its conduct in response to the threat of entry in a way that impedes competition. The natural counterfactual is the myopic incumbent of Example 1, who offers the monopoly tariff T_m , optimal absent any entry threat, and faces entry without adapting to it. Comparing the equilibrium contract with this scenario holds fixed the underlying exercise of market power and isolates the entry-induced adaptation.

The benchmark is not a permissive one. The myopic incumbent already over-admits entry, its threshold $\omega_m(\theta)$ lying above the efficient $c_0 k$, because monopoly screening entails an above-cost marginal price, so the buyer values the contestable units above their social opportunity cost. Being structural rather than a peculiarity of T_m , this means no non-strategic incumbent implements efficient entry, and strategic conduct must be judged against an already distorted allocation. An incumbent that reduces entry relative to the myopic monopoly may be correcting pre-existing over-entry rather than creating an exclusionary distortion. The question is not whether adaptation reduces entry but whether it does so beyond what that inefficiency justifies.

³²The change of variable $x = Q_0(\tau)$ disposes of the second difference wherever $\dot{Q}_0 \equiv 1$, as along any linear branch of a tariff.

³³Strict criterion, on the grid where the contestable quantity is admissible; see the Online Appendix.

Under the non-conditional tariff the strategic response is localized. The strategic reaction $\Delta \mathcal{V}_I = \mathcal{V}_I(\text{real}) - \mathcal{V}_I(\text{myopic})$ has a grid-wide median of 35% of first-best welfare, but differs sharply with the direction of the entry distortion. It reaches 37% where realized entry falls below its efficient level (under-entry) and drops below 6% where it remains above the social optimum (over-entry), roughly an order of magnitude smaller.³⁴ The incumbent reacts where adaptation restricts entry and stays myopic where entry is still excessive. The adaptation is welfare-productive rather than merely rent-extractive. Controlling for the entrant's capacity, the partial correlation between the strategic reaction and the welfare gain relative to the myope ranges from +0.47 at low cost dispersion to +0.77 at high. Stronger adaptation corrects, rather than exacerbates, the benchmark's distortion.

The market-share contract differs markedly. Its strategic reaction is larger, a median of 53% of first-best at $s = 1$ against 35% (2.14 versus 1.52 in model units), yet far less closely associated with welfare, the same partial correlation ranging only from +0.10 to +0.30. The level instrument permits additional extraction, much of it absorbed by a reduction in buyer surplus rather than translated into welfare gains. On about 20% of configurations when $s \geq 1$, concentrated at extreme values of entrant capacity, adaptation is welfare-destructive relative to the myope.³⁵ A stronger strategic response, but not a more efficient one.

7.4. The Diagnostic Inversion

Against the myopic benchmark, the AEC test does not track the welfare consequences of the conduct it flags. Across the grid, its failure rate is strongly associated with the loss of buyer surplus (−0.58) but essentially uncorrelated with the welfare loss (−0.05).³⁶ The near-zero correlation is a cancellation rather than indifference. Writing $\Delta \mathcal{W} = \Delta \mathcal{U} + \Delta \mathcal{V}_I$, the negative association with the buyer-surplus gap and the positive one with the incumbent-profit gap (+0.42) offset each other. The test reacts to the redistribution of surplus, not to its efficiency consequences. Configuration by configuration the disconnect is stark: adaptation is welfare-improving relative to the myope on 99% to 100% of the grid across all values of cost dispersion, yet the AEC test fails on 82% to 92% of configurations, and still on 56% to 77% with an average criterion.

The contradiction is only apparent. The test is anchored on first-best entry, our assessment of conduct on a myopic allocation that itself induces excessive entry. Hence, strategic adaptation can reduce entry relative to the myope, raise welfare, and still leave entry below first best. The same gap explains the buyer's ambivalence. With under-entry, an efficient entrant is foreclosed, hurting buyer and welfare alike, so a buyer-based diagnostic is incidentally aligned with welfare.

³⁴In model units, 1.52, 1.94 and 0.15 respectively. Across all four values of cost dispersion s , the median strategic reaction ranges from 36% to 39% of first-best under under-entry, compared with 2% to 6% under over-entry (1.91 to 1.94 versus 0.04 to 0.15 in model units).

³⁵Configuration shares are computed on the grid $(k, \mu) \in [0.10, 2.00]^2$ at step 0.05 and depend on its bounds; the qualitative patterns are robust, but the exact percentages would change with the parameter range. Precisely, $\Delta \mathcal{W} < 0$ on 19.7% of configurations at $s = 1$ and 20.0% at $s = 2$, with a median loss of 0.5% to 0.7% of first-best (0.03 to 0.04 in model units).

³⁶Partial correlations controlling for k and k^2 , $s = 1$.

With over-entry, an inefficient entrant is admitted, which lowers welfare but benefits the buyer since entry disciplines an incumbent who otherwise prices above marginal cost. The numbers show this reversal. The non-conditional contract induces under-entry on 71% of configurations and over-entry otherwise. The correlation between AEC failure and the welfare gap changes sign across the two regions, -0.60 against $+0.74$, while correlation with buyer surplus stays negative.³⁷

Read together with the strategic-reaction results, the inversion sharpens. The incumbent adapts most strongly on the under-entry configurations, where adaptation generates its largest welfare gains. These are precisely configurations on which the AEC test is most likely to fail. The test sanctions the contract where adaptation is most welfare-improving and clears it where there is little to assess, because the relevant object, the incumbent's departure from the conduct it would have chosen absent the threat of entry, cannot be recovered from a price-cost comparison alone.

These conclusions rely on a total-welfare standard. Under a consumer-welfare standard, the alignment with buyer surplus would instead read as the AEC test tracking its intended objective. The two diverge here for an unexpected reason. The welfare cost of over-entry is negligible, the duplication loss never exceeding 0.1% of first-best welfare on the grid,³⁸ since the inefficient entry that occurs is marginal in value. The divergence arises with insufficient entry, where the test fails by construction while adaptation generates its largest welfare gains. The cost of a buyer-surplus standard here is not tolerated duplication but false alarms.

8. CONCLUSION

This paper studies nonlinear screening contracts when an incumbent is threatened by entry and the buyer–entrant coalition holds private information about both demand and the gains from entry. Under market-share contracts, the entry decision and the consumption choices are both distorted away from efficiency and end up positively correlated, so as to extract the buyer's rent when he sources exclusively from the incumbent and to capture the gains from a more efficient competitor. These dual objectives are prone to bunching, in which case the optimal contract features an endogenous minimum purchase requirement.

Banning market-share contracts would not restore efficiency. The optimal non-conditional tariff pursues the same extraction, through the margin of a single schedule rather than the level of a conditional rebate, and is itself closely approximated by an all-unit discount once the latter carries an infra-marginal instrument such as a minimum purchase requirement. The legal distinction between incremental and retroactive rebates thus carries little economic content: the three forms differ in the channel through which they extract the buyer's rent, not in what they do.

The price-cost tests of antitrust practice inherit this. The conduct that scrutiny exists to assess is the incumbent's departure from a myopic benchmark, who earns her full monopoly profit

³⁷ -0.87 under under-entry and -0.66 under over-entry.

³⁸ See the Online Appendix.

without responding to entry, and measured against it these tests track the buyer's surplus rather than welfare. The as-efficient-competitor test reverses sign across the entry regime, sanctioning the non-conditional tariff where the incumbent's strategic response raises welfare and clearing it where there is no response to assess. It identifies a channel of extraction, not a conduct, and no price-cost screen recovers the departure a rule of reason would need.

Two extensions seem worth pursuing. First, situations where the playing field is more balanced between incumbent and entrant (see Calzolari and Denicolò, 2013). Second, how the incumbent's behavior might be constrained: a blanket ban on below-cost pricing would be counterproductive here, since below-cost marginal prices on the with-entry branch accommodate entry rather than deter it, so the relevant question is not whether to constrain conduct but how to target it.

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APPENDIX

PROOF OF PROPOSITION 1. I 's objective function can be written as follows

$$\mathcal{V}_I(T_0, T_1) = \mathbb{E}_\theta \left[G(\omega(\theta)) \left(c_0 k + \left[S(\theta, q) \right]_{Q_0(\theta)}^{Q_1(\theta)} - \omega(\theta) \right) + S(\theta, Q_0(\theta)) - \mathcal{U}_0(\theta) \right].$$

Since $\mathcal{U}_0(\theta)$ is non-decreasing, (5.3) holds for all θ if

$$(A.1) \quad \mathcal{U}_0(\underline{\theta}) \geq 0.$$

(A.1) must be binding at the optimum. Otherwise, it would be optimal to increase uniformly $T_0(q)$ and $T_1(q)$ by some sufficiently small ϵ . This would leave $\omega(\theta)$ and $Q_i(\theta)$ unchanged for $i \in \{0, 1\}$. At the same time, it would increase the monopoly base profit by ϵ . Hence,

$$(A.2) \quad \mathcal{U}_0(\theta) = \int_{\underline{\theta}}^{\theta} Q_0(\tilde{\theta}) d\tilde{\theta}.$$

Using (A.2) allows us to rewrite the maximand as follows

$$\mathbb{E}_\theta \left[G(\omega(\theta)) \left(c_0 k + \left[S(\theta, q) \right]_{Q_0(\theta)}^{Q_1(\theta)} - \omega(\theta) \right) + Q_m(\theta) Q_0(\theta) - \frac{Q_0^2(\theta)}{2} \right].$$

We first look for a solution without requiring that $Q_0(\theta)$ and $Q_1(\theta)$ are strictly increasing. Up to a multiplicative factor, the corresponding Hamiltonian becomes

$$\mathcal{H}(\theta, Q_0, Q_1, \omega, \lambda) = G(\omega) \left(c_0 k + \left[S(\theta, q) \right]_{Q_0}^{Q_1} - \omega \right) + Q_m(\theta) Q_0 - \frac{Q_0^2}{2} + \lambda(Q_1 - Q_0),$$

where λ is the adjoint variable for (5.2). Applying *Pontryagin's Maximum Principle* (Seierstad and Sydsaeter, 1987) yields the following necessary optimality conditions that are satisfied by a normal maximum $(\bar{\omega}(\theta), \bar{Q}_0(\theta), \bar{Q}_1(\theta))$:

1. *Condition on the adjoint variable.* $\lambda(\theta)$ is piecewise continuously differentiable with

$$(A.3) \quad -\dot{\lambda}(\theta) = -G(\bar{\omega}(\theta)) + g(\bar{\omega}(\theta)) \left(c_0 k + \left[S(\theta, q) \right]_{\bar{Q}_0(\theta)}^{\bar{Q}_1(\theta)} - \bar{\omega}(\theta) \right).$$

2. *Transversality conditions.* There are no transversality conditions for $\omega(\theta)$:

$$(A.4) \quad \lambda(\underline{\theta}) = \lambda(\bar{\theta}) = 0.$$

3. *Optimality conditions w.r.t. (Q_0, Q_1)* : The Hamiltonian is strictly concave in (Q_0, Q_1) . Hence, the optimality conditions w.r.t. Q_0 and Q_1 are, respectively,

$$(A.5) \quad Q_m(\theta) - \bar{Q}_0(\theta) - G(\bar{\omega}(\theta))(Q_{fb}(\theta) - \bar{Q}_0(\theta)) = \lambda(\theta),$$

$$(A.6) \quad G(\bar{\omega}(\theta))(Q_{fb}(\theta) - \bar{Q}_1(\theta)) = -\lambda(\theta).$$

Several implications follow. Summing (A.5) and (A.6) gives (5.6). Define now

$$(A.7) \quad \mu(\theta) \equiv \lambda(\theta) + G(\bar{\omega}(\theta))(\bar{\theta} - \theta).$$

The transversality conditions (A.4) imply

$$(A.8) \quad \mu(\bar{\theta}) = 0,$$

$$(A.9) \quad \mu(\underline{\theta}) = \Delta\theta G(\bar{\omega}(\underline{\theta})).$$

The solutions to (A.5) and (A.6) can thus be written in terms of $\mu(\theta)$ as follows

$$(A.10) \quad \bar{Q}_0(\theta) = Q_m(\theta) - \frac{\mu(\theta)}{1 - G(\bar{\omega}(\theta))},$$

$$(A.11) \quad \bar{Q}_1(\theta) = Q_m(\theta) + \frac{\mu(\theta)}{G(\bar{\omega}(\theta))}.$$

Using (A.13) below, these equations imply (5.9). Differentiating (A.7) w.r.t. θ yields

$$(A.12) \quad \dot{\mu}(\theta) = \dot{\lambda}(\theta) + g(\bar{\omega}(\theta))\dot{\bar{\omega}}(\theta)(\bar{\theta} - \theta) - G(\bar{\omega}(\theta)).$$

Using (A.10) and (A.11), we obtain

$$(A.13) \quad \dot{\bar{\omega}}(\theta) = \frac{\mu(\theta)}{G(\bar{\omega}(\theta))(1 - G(\bar{\omega}(\theta)))}.$$

We can now rewrite (A.3) as follows

$$-\dot{\lambda}(\theta) = -G(\bar{\omega}(\theta)) + g(\bar{\omega}(\theta))\left(c_0k + \frac{1}{2}\left(Q_{fb}(\theta) - \bar{Q}_0(\theta)\right)^2 - \frac{1}{2}\left(Q_{fb}(\theta) - \bar{Q}_1(\theta)\right)^2 - \bar{\omega}(\theta)\right).$$

Using again (A.5) and (A.6) to simplify the r.h.s., the latter equation becomes

$$(A.14) \quad -\dot{\lambda}(\theta) = -G(\bar{\omega}(\theta)) + g(\bar{\omega}(\theta))\left(c_0k + \frac{(\bar{\theta} - \theta + \lambda(\theta))^2}{2(1 - G(\bar{\omega}(\theta)))^2} - \frac{\lambda^2(\theta)}{2G^2(\bar{\omega}(\theta))} - \bar{\omega}(\theta)\right).$$

Inserting (A.13) and (A.14) into (A.12) and simplifying yields

$$(A.15) \quad \dot{\mu}(\theta) = -g(\bar{\omega}(\theta))\left(c_0k - \bar{\omega}(\theta) + \frac{(2G(\bar{\omega}(\theta)) - 1)\mu^2(\theta)}{2G^2(\bar{\omega}(\theta))(1 - G(\bar{\omega}(\theta)))^2}\right).$$

The system of autonomous first-order differential equations (A.13)-(A.15) and the transversality conditions (A.8)-(A.9) define $(\mu(\theta), \bar{\omega}(\theta))$. Transversality conditions are defined at both $\underline{\theta}$ and $\bar{\theta}$. To make progress, we now transform this problem.

TRAJECTORIES OF THE SYSTEM. From (A.13) and (A.15), we obtain a first-order differential equation that defines μ in terms of ω along any trajectory of the system. Denote by $\mu(\omega|\bar{\omega}(\bar{\theta}))$ such a function where we index those solutions with the upper value of $\bar{\omega}$. This function satisfies

$$(A.16) \quad \frac{\mu(\omega|\bar{\omega}(\bar{\theta}))}{G(\omega)(1-G(\omega))} \frac{d\mu}{d\omega}(\omega|\bar{\omega}(\bar{\theta})) = -g(\omega)(c_0k - \omega) + \frac{g(\omega)(1-2G(\omega))}{2G^2(\omega)(1-G(\omega))^2} \mu^2(\omega|\bar{\omega}(\bar{\theta})).$$

The transversality conditions (A.8) and (A.9) can be written as follows

$$(A.17) \quad \mu(\bar{\omega}(\bar{\theta})|\bar{\omega}(\bar{\theta})) = \Delta\theta G(\bar{\omega}(\bar{\theta})),$$

$$(A.18) \quad \mu(\bar{\omega}(\bar{\theta})|\bar{\omega}(\bar{\theta})) = 0.$$

The first-order differential equation (A.16) together with the transversality condition (A.17) fully determines $\mu(\omega|\bar{\omega}(\bar{\theta}))$ and $\bar{\omega}(\bar{\theta})$. After manipulations, (A.16) can be rewritten as follows

$$(A.19) \quad \frac{d}{d\omega} \left(\frac{\mu^2(\omega|\bar{\omega}(\bar{\theta}))}{2G(\omega)(1-G(\omega))} \right) = -g(\omega)(c_0k - \omega).$$

Using (A.18) and integrating (A.19) yields $\mu^2(\omega|\bar{\omega}(\bar{\theta})) = 2G(\omega)(1-G(\omega)) \int_{\omega}^{\bar{\omega}(\bar{\theta})} g(\tilde{\omega})(c_0k - \tilde{\omega})d\tilde{\omega}$. Taking the positive root to ensure that (A.17) holds yields

$$(A.20) \quad \mu(\omega|\bar{\omega}(\bar{\theta})) = \sqrt{2G(\omega)(1-G(\omega)) \int_{\omega}^{\bar{\omega}(\bar{\theta})} g(\tilde{\omega})(c_0k - \tilde{\omega})d\tilde{\omega}}.$$

We look for a solution such that $\bar{\omega}(\bar{\theta}) \leq c_0k$ and $\bar{\omega}(\bar{\theta})$ non-decreasing. It is straightforward to check that $\mu(\omega|\bar{\omega}(\bar{\theta}))$ has infinite derivatives at both 0 and $\bar{\omega}(\bar{\theta})$ when $\bar{\omega}(\bar{\theta}) \in [0, c_0k)$ and has derivative $-\sqrt{g(c_0k)G(c_0k)(1-G(c_0k))}$ when $\bar{\omega}(\bar{\theta}) = c_0k$. Moreover, $\mu(\omega|\bar{\omega}(\bar{\theta}))$ is an increasing function of $\bar{\omega}(\bar{\theta})$, whose upper bound we denote $M_{c_0k}(\omega)$.

Observe also that $\mu(\omega|\bar{\omega}(\bar{\theta}))$ necessarily crosses the function $\Gamma(\omega) = \Delta\theta G(\omega)$ at least once in $(0, \bar{\omega}(\bar{\theta}))$. This is so since $\dot{\mu}(0|\bar{\omega}(\bar{\theta})) = +\infty > \dot{\Gamma}(0) = \Delta\theta g(0)$, $\mu(0|\bar{\omega}(\bar{\theta})) = \Gamma(0) = 0$ and $\mu(\bar{\omega}(\bar{\theta})|\bar{\omega}(\bar{\theta})) = 0 < \Gamma(\bar{\omega}(\bar{\theta}))$. Let denote by $\omega^0(\bar{\omega}(\bar{\theta}))$ the highest value in $(0, \bar{\omega}(\bar{\theta}))$ such that $\mu(\omega^0(\bar{\omega}(\bar{\theta}))|\bar{\omega}(\bar{\theta})) = \Gamma(\omega^0(\bar{\omega}(\bar{\theta})))$. Finally, notice that $\omega^0(\bar{\omega}(\bar{\theta}))$ is one-to-one.

A FIRST-ORDER DIFFERENTIAL EQUATION WHOSE SOLUTION IS $\bar{\omega}(\bar{\theta})$. Inserting (A.20) into (A.13), we obtain that $\bar{\omega}(\bar{\theta})$ solves the following first-order differential equation

$$(A.21) \quad \dot{\bar{\omega}}(\bar{\theta}) = \sqrt{\frac{2}{G(\bar{\omega}(\bar{\theta}))(1-G(\bar{\omega}(\bar{\theta})))} \int_{\bar{\omega}(\bar{\theta})}^{\bar{\omega}(\bar{\theta})} g(\tilde{\omega})(c_0k - \tilde{\omega})d\tilde{\omega}}.$$

We now study the local behavior of $\bar{\omega}(\bar{\theta})$ in the neighborhood of $\bar{\omega}(\bar{\theta}) \leq c_0k$. Two cases must be distinguished. First, for $\bar{\omega}(\bar{\theta}) < c_0k$, a first-order Taylor expansion of the term in the square root of the right-hand side gives us in that neighborhood

$$(A.22) \quad \dot{\bar{\omega}}(\bar{\theta}) = \sqrt{\frac{2g(\bar{\omega}(\bar{\theta}))(c_0k - \bar{\omega}(\bar{\theta}))}{G(\bar{\omega}(\bar{\theta}))(1-G(\bar{\omega}(\bar{\theta})))}} (\bar{\omega}(\bar{\theta}) - \bar{\omega}(\bar{\theta})).$$

For $\bar{\omega}(\bar{\theta}) = c_0 k$, a second-order Taylor expansion of the term in the square root of the right-hand side gives us the following

$$(A.23) \quad \dot{\bar{\omega}}(\theta) = \sqrt{\frac{g(c_0 k)}{G(c_0 k)(1 - G(c_0 k))}} (c_0 k - \bar{\omega}(\theta)).$$

The Taylor approximations (A.22) and (A.23) imply that the right-hand side in (A.21) is integrable in the neighborhood of $\bar{\omega}(\bar{\theta})$ only if $\bar{\omega}(\bar{\theta}) < c_0 k$. In other words, the trajectory that solves (A.21) and corresponds to $\bar{\omega}(\bar{\theta}) = c_0 k$ never reaches that value in finite time. Necessarily, we have $\bar{\omega}(\bar{\theta}) < c_0 k$. The monotonicity of $\bar{\omega}(\theta)$ then implies $\bar{\omega}(\theta) < c_0 k \forall \theta \in \Theta$.

From (A.21), we also obtain

$$(A.24) \quad d\theta = d\omega \left(\frac{2}{G(\omega)(1 - G(\omega))} \int_{\omega}^{\bar{\omega}(\bar{\theta})} g(\tilde{\omega})(c_0 k - \tilde{\omega}) d\tilde{\omega} \right)^{-\frac{1}{2}}.$$

Integrating (A.24) yields $\Delta\theta = \int_{\bar{\omega}(\underline{\theta})}^{\bar{\omega}(\bar{\theta})} \left(\frac{2}{G(\omega)(1 - G(\omega))} \int_{\omega}^{\bar{\omega}(\bar{\theta})} g(\tilde{\omega})(c_0 k - \tilde{\omega}) d\tilde{\omega} \right)^{-1/2} d\omega$, or

$$(A.25) \quad \Delta\theta = \int_{\bar{\omega}(\underline{\theta})}^{\bar{\omega}(\bar{\theta})} \frac{G(\tilde{\omega})(1 - G(\tilde{\omega}))}{\mu(\tilde{\omega}|\bar{\omega}(\bar{\theta}))} d\tilde{\omega}.$$

The transversality condition (A.9) now reads $\mu(\bar{\omega}(\underline{\theta})|\bar{\omega}(\bar{\theta})) = \Delta\theta G(\bar{\omega}(\underline{\theta}))$, or $\bar{\omega}(\underline{\theta}) = \omega^0(\bar{\omega}(\bar{\theta}))$. Inserting into (A.25) gives $\bar{\omega}(\bar{\theta})$ as a solution to

$$(A.26) \quad \Delta\theta = \int_{\omega^0(\bar{\omega}(\bar{\theta}))}^{\bar{\omega}(\bar{\theta})} \frac{G(\tilde{\omega})(1 - G(\tilde{\omega}))}{\mu(\tilde{\omega}|\bar{\omega}(\bar{\theta}))} d\tilde{\omega}.$$

Observe now that

$$\lim_{\bar{\omega}(\bar{\theta}) \rightarrow c_0 k} \int_{\omega^0(\bar{\omega}(\bar{\theta}))}^{\bar{\omega}(\bar{\theta})} \frac{G(\tilde{\omega})(1 - G(\tilde{\omega}))}{\mu(\tilde{\omega}|\bar{\omega}(\bar{\theta}))} d\tilde{\omega} = +\infty > \Delta\theta > \lim_{\bar{\omega}(\bar{\theta}) \rightarrow 0} \int_{\omega^0(\bar{\omega}(\bar{\theta}))}^{\bar{\omega}(\bar{\theta})} \frac{G(\tilde{\omega})(1 - G(\tilde{\omega}))}{\mu(\tilde{\omega}|\bar{\omega}(\bar{\theta}))} d\tilde{\omega} = 0,$$

where the left-hand side equality follows from the fact that $1/\mu(\tilde{\omega}|c_0 k)$ is not integrable in a left-neighborhood of $c_0 k$, and the right-hand side equality follows from the fact that $1/\mu(\tilde{\omega}|0)$ is integrable in a right-neighborhood of 0 and $\lim_{\bar{\omega}(\bar{\theta}) \rightarrow 0} \omega^0(\bar{\omega}(\bar{\theta})) = 0$. (This is so since $\omega^0(\bar{\omega}(\bar{\theta})) \in (0, \bar{\omega}(\bar{\theta}))$.) By continuity, a solution to (A.26) exists.

COMPARATIVE STATICS. Since $\mu(\theta) = \mu(\bar{\omega}(\theta)|\bar{\omega}(\bar{\theta}))$, the following Lemma immediately follows from (A.20).

LEMMA A.1. $\mu(\theta) \geq 0 \quad \forall \theta \in \Theta$ (with equality only at $\bar{\theta}$).

Using Lemma A.1 with (A.10) and (A.11) gives the first two inequalities in (5.5). It follows from (A.13) that $\bar{\omega}(\theta)$ is everywhere increasing in θ . Note that $\bar{\omega}(\theta)$ has zero derivative at $\bar{\theta}$ only.

BOUNDS ON $\bar{\omega}(\bar{\theta})$. We now prove a useful lemma.

LEMMA A.2.

$$(A.27) \quad \bar{\omega}(\bar{\theta}) \in (\omega_m, c_0 k),$$

$$(A.28) \quad \lambda(\theta) \leq 0 \quad \forall \theta \text{ in a left-neighborhood of } \bar{\theta}.$$

Proof of Lemma A.2.

Fact 1. From (A.14), we obtain

$$(A.29) \quad -\dot{\lambda}(\bar{\theta}) = -G(\bar{\omega}(\bar{\theta})) + g(\bar{\omega}(\bar{\theta}))(c_0 k - \bar{\omega}(\bar{\theta})) < 0,$$

when $\bar{\omega}(\bar{\theta}) > \omega_m$. Since the transversality condition is $\lambda(\bar{\theta}) = 0$, in the neighborhood of $\bar{\theta}$, a solution to (A.14) such that $\bar{\omega}(\bar{\theta}) > \omega_m$ satisfies $\lambda(\theta) \leq 0$. Hence, (A.28) follows from (A.27).

Fact 2. Suppose now that $\bar{\omega}(\bar{\theta}) \leq \omega_m$. From (A.14), $-\dot{\lambda}(\bar{\theta}) = -G(\bar{\omega}(\bar{\theta})) + g(\bar{\omega}(\bar{\theta}))(c_0 k - \bar{\omega}(\bar{\theta})) \geq 0$. Hence $\lambda(\theta)$ is decreasing around $\bar{\theta}$, and, since $\lambda(\bar{\theta}) = 0$, it is also positive in a left-neighborhood of $\bar{\theta}$. Denote by θ_0 the highest value less than $\bar{\theta}$, if one exists, such that $\lambda(\theta_0) = 0$. From the monotonicity of $\bar{\omega}(\theta)$, we thus have $\bar{\omega}(\theta_0) < \omega_m$ and thus $-G(\bar{\omega}(\theta_0)) + g(\bar{\omega}(\theta_0))(c_0 k - \bar{\omega}(\theta_0)) > 0$. At such a point θ_0 , we have

$$-\dot{\lambda}(\theta_0) = -G(\bar{\omega}(\theta_0)) + g(\bar{\omega}(\theta_0))(c_0 k - \bar{\omega}(\theta_0)) + g(\bar{\omega}(\theta_0)) \left(\frac{(\bar{\theta} - \theta_0 + \lambda(\theta_0))^2}{2(1 - G(\bar{\omega}(\theta_0)))^2} - \frac{\lambda^2(\theta_0)}{2G^2(\bar{\omega}(\theta_0))} \right)$$

and thus

$$-\dot{\lambda}(\theta_0) > g(\bar{\omega}(\theta_0)) \left(\frac{(\bar{\theta} - \theta_0 + \lambda(\theta_0))^2}{2(1 - G(\bar{\omega}(\theta_0)))^2} - \frac{\lambda^2(\theta_0)}{2G^2(\bar{\omega}(\theta_0))} \right) = \frac{g(\bar{\omega}(\theta_0))(\bar{\theta} - \theta_0)^2}{2(1 - G(\bar{\omega}(\theta_0)))^2} > 0.$$

In a right-neighborhood of θ_0 , we would thus have $\lambda(\theta) < 0$; a contradiction with the definition of θ_0 . Hence, (A.27) necessarily holds for the optimal trajectory. $\square$

CONCAVITY OF $\bar{\omega}$. Differentiating (5.11) yields the following

$$(A.30) \quad -\ddot{\omega}(\theta) \frac{(1 - G(\bar{\omega}(\theta)))G(\bar{\omega}(\theta))}{g(\bar{\omega}(\theta))} = c_0 k - \bar{\omega}(\theta) + \frac{1 - 2G(\bar{\omega}(\theta))}{G(\bar{\omega}(\theta))(1 - G(\bar{\omega}(\theta)))} \int_{\bar{\omega}(\theta)}^{\bar{\omega}(\bar{\theta})} g(x)(c_0 k - x) dx.$$

Define $\Psi(\omega) = (c_0 k - \omega)G(\omega)(1 - G(\omega)) + (1 - 2G(\omega)) \int_{\omega}^{\bar{\omega}(\bar{\theta})} g(x)(c_0 k - x) dx$. Observe that $\omega \leq \bar{\omega}(\bar{\theta}) < c_0 k$ implies $\dot{\Psi}(\omega) = -G(\omega)(1 - G(\omega)) - 2g(\omega) \int_{\omega}^{\bar{\omega}(\bar{\theta})} g(x)(c_0 k - x) dx < 0$. The monotonicity of $\bar{\omega}$ then implies $\Psi(\bar{\omega}(\theta)) \geq \Psi(\bar{\omega}(\bar{\theta})) > 0$. Using (A.30) then shows that $\bar{\omega}(\theta)$ is concave.

CONSUMPTION LEVELS. We first extend the second item in Lemma A.2.

LEMMA A.3.

$$(A.31) \quad \lambda(\theta) \leq 0 \quad \forall \theta \in \Theta \text{ (with equality at } \underline{\theta} \text{ and } \bar{\theta} \text{ only)}.$$

Proof of Lemma A.3. We know from Lemma A.2 that this property is true in a left-neighborhood of $\bar{\theta}$. Moreover, (A.27) and (A.29) together imply that $\dot{\lambda}(\bar{\theta}) > 0$. If $\lambda(\theta)$ changes sign over $(\underline{\theta}, \bar{\theta})$, it must be that $\dot{\lambda}(\theta)$ also changes sign more than once. We now prove that this cannot happen. To this end, consider a point $\theta_0 \in (\underline{\theta}, \bar{\theta})$ where $\lambda(\theta_0) = 0$. From (A.7), (A.13) and (A.14), we deduce that $-\frac{\dot{\lambda}(\theta_0)}{g(\bar{\omega}(\theta_0))} = c_0 k - \bar{\omega}(\theta_0) - \frac{G(\bar{\omega}(\theta_0))}{g(\bar{\omega}(\theta_0))} + \frac{\dot{\bar{\omega}}^2(\theta_0)}{2}$.

Define now $\mathcal{Z}(\theta) = c_0 k - \bar{\omega}(\theta) - \frac{G(\bar{\omega}(\theta))}{g(\bar{\omega}(\theta))} + \frac{\dot{\bar{\omega}}^2(\theta)}{2}$. Observe that $\dot{\mathcal{Z}}(\theta) = -\dot{\bar{\omega}}(\theta)(1 + \frac{d}{d\omega}(\frac{G(\bar{\omega}(\theta))}{g(\bar{\omega}(\theta))})) - \ddot{\bar{\omega}}(\theta) < 0$, where the inequality follows from MHRP and the fact that $\bar{\omega}$ is increasing and concave. Moreover, $\mathcal{Z}(\bar{\theta}) < 0$ from Lemma A.2 and the fact that $\dot{\bar{\omega}}(\bar{\theta}) = 0$. Hence, $\mathcal{Z}(\theta)$ changes sign at most once over Θ . It implies that $\dot{\lambda}(\theta_0)$ also changes sign at most once over all points θ_0 such that $\lambda(\theta_0) = 0$. Because of (A.4), we necessarily have $\dot{\lambda}(\underline{\theta}) < 0 < \dot{\lambda}(\bar{\theta})$ and there is no such interior point $\theta_0 \in (\underline{\theta}, \bar{\theta})$. Finally, we get $\lambda(\theta) < 0$ for $\theta \in (\underline{\theta}, \bar{\theta})$. $\square$

Inserting (A.31) into (A.5) and (A.6) immediately implies $Q_1(\theta) \leq Q_{fb}(\theta)$ for all $\theta \in \Theta$. Since $\dot{\bar{\omega}}(\theta) \geq 0$, it follows that $\mu(\theta) = \lambda(\theta) + G(\bar{\omega}(\theta))(\bar{\theta} - \theta) \geq 0$ for all $\theta \in \Theta$. Therefore, $Q_0(\theta) \leq Q_m(\theta)$ for all $\theta \in \Theta$ and thus (5.5) follows. Taking into account (A.4) yields (5.7). $\square$

PROOF OF COROLLARY 1. From Proposition 1, $\bar{Q}_i(\theta) \leq Q_{fb}(\theta)$ for $i = 0, 1$. The buyer's first-order condition gives $\bar{T}'_i(\bar{Q}_i(\theta)) = \theta - \bar{Q}_i(\theta)$, and $Q_{fb}(\theta) = \theta - c_0$, so $\bar{T}'_i(Q) \geq c_0$ on $\bar{Q}_i(\Theta)$, with equality where consumption is undistorted, in particular at $\bar{\theta}$, where $\bar{Q}_0(\bar{\theta}) = \bar{Q}_1(\bar{\theta}) = Q_{fb}(\bar{\theta})$.

Denote now by $\vartheta_i(Q)$ the assignment correspondence under market configuration i (i.e., the inverse correspondence of function $\bar{Q}_i$) when B consumes an overall quantity Q .³⁹ Since $\bar{Q}_1(\theta) \geq \bar{Q}_0(\theta)$, we have $\vartheta_1(Q) \leq \vartheta_0(Q)$. Writing the first-order condition as $\bar{T}'_i(Q) = c_0 + Q_{fb}(\vartheta_i(Q)) - Q$, for any $Q \in \bar{Q}_0(\Theta) \cap (\bar{Q}_1(\Theta) - k)$, we have $\bar{T}'_1(Q - k) = c_0 + Q_{fb}(\vartheta_1(Q)) - Q \leq c_0 + Q_{fb}(\vartheta_0(Q)) - Q = \bar{T}'_0(Q)$, where the inequality follows from $\vartheta_1(Q) \leq \vartheta_0(Q)$ and the monotonicity of Q_{fb} . $\square$

PROOF OF PROPOSITION 2. We characterize an optimal contract when bunching arises for $\bar{Q}_1(\theta)$ on $[\underline{\theta}, \theta_0]$. For simplicity, we assume that bunching does not arise for $\bar{Q}_0(\theta)$. We thus add to the maximization problem the monotonicity condition

$$(A.32) \quad \dot{Q}_1(\theta) = p_1(\theta) \geq 0.$$

$Q_1(\theta)$ is now a new state variable found within the class of arcs which are absolutely continuous and $p_1(\theta)$ is a non-negative control variable.⁴⁰ Denote also by ζ_1 the adjoint variable associated with (A.32). The Hamiltonian $\mathcal{H}^b$ for the optimization problem so constrained is

$$\mathcal{H}^b(\theta, Q_0, Q_1, \omega, \lambda, \zeta_1) = \mathcal{H}(\theta, Q_0, Q_1, \omega, \lambda) + \zeta_1 p_1.$$

Applying *Pontryagin's Maximum Principle* yields the following necessary conditions for a normal maximum $(\bar{\omega}(\theta), \bar{Q}_0(\theta), \bar{Q}_1(\theta), \bar{p}_1(\theta))$:

1. *Adjoint variables.* $\lambda(\theta)$ and $\zeta_1(\theta)$ are continuous. The first optimality condition is still given by (A.3). For $\zeta_1(\theta)$, we find

$$-\dot{\zeta}_1(\theta) = G(\bar{\omega}(\theta))(Q_{fb}(\theta) - \bar{Q}_1(\theta)) + \lambda(\theta).$$

2. *Transversality conditions.* The conditions in (A.4) are still valid. Since there are no transversality conditions on $Q_1(\theta)$ at both endpoints, we must also add

$$(A.33) \quad \zeta_1(\underline{\theta}) = \zeta_1(\bar{\theta}) = 0.$$

³⁹Where $\bar{Q}_i$ is constant, ϑ_i is set-valued and the inequalities are read through its largest element; this does not affect the argument.

⁴⁰Observe that this condition implicitly amounts to assuming that optimal consumption levels have no jump, an innocuous restriction in our sense.

3. *Optimality conditions for the control variables.* Optimizing w.r.t. Q_0 still yields (A.5). Optimizing with respect to p_1 , which enters the objective linearly, gives

$$(A.34) \quad \bar{p}_1(\theta) = 0 \text{ if } \zeta_1(\theta) < 0 \text{ and } \zeta_1(\theta) = 0 \text{ if } \bar{p}_1(\theta) > 0.$$

We look for bunching on an interval $[\underline{\theta}, \theta_0]$, i.e. $\bar{Q}_1(\theta) = \bar{Q}_1^b = Q_1^b + k$ on $[\underline{\theta}, \theta_0]$. On $[\theta_0, \bar{\theta}]$, there is no bunching and (A.34) implies that $\zeta_1(\theta) = 0$ there. By continuity of ζ_1 at θ_0 , we obtain $\zeta_1(\theta_0) = 0$. From this and (A.33), we compute

$$0 = - \int_{\underline{\theta}}^{\theta_0} \dot{\zeta}_1(\theta) d\theta = \int_{\underline{\theta}}^{\theta_0} \left(G(\bar{\omega}(\theta))(Q_{fb}(\theta) - \bar{Q}_1^b) + \lambda(\theta) \right) d\theta.$$

Finally, using (A.5), we obtain the following

$$0 = \int_{\underline{\theta}}^{\theta_0} \dot{\zeta}_1(\theta) d\theta = \int_{\underline{\theta}}^{\theta_0} \left(G(\bar{\omega}(\theta))(Q_{fb}(\theta) - \bar{Q}_1^b) - (\bar{\theta} - \theta) + (1 - G(\bar{\omega}(\theta)))(Q_{fb}(\theta) - \bar{Q}_0(\theta)) \right) d\theta,$$

which can be rewritten as (5.12). Finally (5.13) follows from continuity of $\bar{Q}_1(\theta)$ at θ_0 . $\square$

PROOF OF PROPOSITION 3. Suppose that $T'(Q) - c_0 \geq 0$ for all Q . Incentive compatibility implies that $Q_0(\theta - k) \leq Q_0(\theta)$ but also that $T(Q)$ is absolutely continuous and thus integrable. We then obtain $\int_{Q_0(\theta-k)}^{Q_0(\theta)} (T'(Q) - c_0) dQ \geq 0$, which leads to $T(Q_0(\theta)) - c_0 Q_0(\theta) \geq T(Q_0(\theta - k)) - c_0 Q_0(\theta - k)$. Expressing this condition in terms of B 's information rent, we obtain

$$\mathcal{U}_0(\theta - k) + \frac{Q_0^2(\theta - k)}{2} - Q_{fb}(\theta - k)Q_0(\theta - k) \geq \mathcal{U}_0(\theta) + \frac{Q_0^2(\theta)}{2} - Q_{fb}(\theta)Q_0(\theta).$$

Inserting $\omega(\theta)$ from (6.2) yields the result. $\square$

OPTIMALITY CONDITIONS FOR NON-CONDITIONAL TARIFFS. We first describe the constrained set. The profile $(\mathcal{U}_0, Q_0)$ must satisfy

$$(A.35) \quad \mathcal{U}_0(\theta) = Q_0(\theta), \text{ a.e.,}$$

and Q_0 must be non-decreasing. Because of its importance in the sequel, we make this monotonicity condition more explicit and express it in a differential form.⁴¹ Focusing on admissible arcs Q_0 which are absolutely continuous, the monotonicity requirement is thus captured by the following condition:

$$(A.36) \quad \dot{Q}_0(\theta) = p(\theta) \geq 0 \quad \forall \theta \in \Theta.$$

The optimization problem entails delayed variables and we adopt the techniques of the corresponding literature to solve for the optimal arc.⁴² We thus define

$$(A.37) \quad \tilde{\mathcal{U}}_0(\theta) = \mathcal{U}_0(\theta - k) \text{ and } \tilde{Q}_0(\theta) = Q_0(\theta - k) \quad \forall \theta \in \Theta.$$

From (A.37), it follows that the variables $\mathcal{U}_0$ and Q_0 should also be defined on Θ_{-k} . Equivalently, $\tilde{Q}_0$ is

⁴¹This formulation implies that Q_0 is now found within the class of absolutely continuous functions. This amounts to assuming that it has no upward jump, an innocuous restriction in this environment.

⁴²Kamien and Schwartz (1991, Chapter 19).

free on Θ and is chosen so as to optimize the maximand of I 's problem without any considerations other than those implied by incentive compatibility: $(\tilde{\mathcal{U}}_0, \tilde{Q}_0)$ satisfies

$$(A.38) \quad \dot{\tilde{\mathcal{U}}}_0(\theta) = \dot{\tilde{Q}}_0(\theta), \text{ a.e.,}$$

and the monotonicity requirement for $\dot{\tilde{Q}}_0$ is captured as

$$(A.39) \quad \dot{\tilde{Q}}_0(\theta) = \tilde{p}(\theta) \geq 0 \quad \forall \theta \in \Theta.$$

Monotonicity at $\underline{\theta}$ implies that the left-side and right-side limits of $Q_0(\theta)$ at $\underline{\theta}$ must satisfy (6.4). This condition can also be written as follows

$$(A.40) \quad \tilde{Q}_0(\bar{\theta}) \leq Q_0(\underline{\theta}).$$

We decompose (A.40) into a pair of end-point conditions

$$(A.41) \quad \tilde{Q}_0(\bar{\theta}) \leq r$$

$$(A.42) \quad Q_0(\underline{\theta}) \geq r$$

where r is an extra control variable. Together with the monotonicity conditions (A.36) and (A.39), (A.41) and (A.42) ensure monotonicity of Q_0 over $\Theta_{-k} \cup \Theta$.

Observe also that $\tilde{\mathcal{U}}_0(\underline{\theta} + k) = \tilde{\mathcal{U}}_0(\bar{\theta}) = \mathcal{U}_0(\underline{\theta})$ is anchored by the binding participation constraint at $\underline{\theta}$

$$\mathcal{U}_0(\underline{\theta}) = 0$$

and thus, we have also,

$$\tilde{\mathcal{U}}_0(\bar{\theta}) = 0.$$

B 's net payoff when purchasing also from E can be written as follows

$$(A.43) \quad \mathcal{U}_1(\theta) - c = \tilde{\mathcal{U}}_0(\theta) + \theta k - \frac{k^2}{2} - c = \mathcal{U}_0(\theta) + \omega(\theta) - c \quad \forall \theta \in \Theta.$$

(A.43) shows that $\mathcal{U}_1(\theta) - c \geq 0$ when $\mathcal{U}_0(\theta) \geq 0$ and entry occurs, i.e., $\omega(\theta) \geq c$.

Replacing $\omega(\theta)$ by its expression taken from (A.43) in terms of $\mathcal{U}_1(\theta) = \tilde{\mathcal{U}}_0(\theta) + \theta k - \frac{k^2}{2}$ and $\mathcal{U}_0(\theta)$, we rewrite $\mathcal{V}_I(T)$ as

$$\begin{aligned} \mathcal{V}_I(T) = \mathbb{E}_\theta \Big(G \Big(\tilde{\mathcal{U}}_0(\theta) + \theta k - \frac{k^2}{2} - \mathcal{U}_0(\theta) \Big) \Big(c_0 k + \left[S(\theta, q) \right]_{Q_0(\theta)}^{\tilde{Q}_0(\theta) + k} - \left(\tilde{\mathcal{U}}_0(\theta) + \theta k - \frac{k^2}{2} - \mathcal{U}_0(\theta) \right) \Big) \\ + S(\theta, Q_0(\theta)) - \mathcal{U}_0(\theta) \Big). \end{aligned}$$

Denote by $\lambda_0, \tilde{\lambda}_0, \zeta$ and $\tilde{\zeta}$ the adjoint variables associated with (A.35), (A.38), (A.36) and (A.39) respectively. Up to some multiplicative factor, the Hamiltonian $\mathcal{H}^u$ for the above optimization problem can be written as follows

$$\mathcal{H}^u(\theta, \mathcal{U}_0, \tilde{\mathcal{U}}_0, Q_0, \tilde{Q}_0, p, \tilde{p}, \lambda_0, \tilde{\lambda}_0, \zeta, \tilde{\zeta}) = G \left(\tilde{\mathcal{U}}_0 + \theta k - \frac{k^2}{2} - \mathcal{U}_0 \right) \left(c_0 k + \left[S(\theta, q) \right]_{Q_0}^{\tilde{Q}_0 + k} \right)$$

$$-\left(\tilde{\mathcal{U}}_0 + \theta k - \frac{k^2}{2} - \mathcal{U}_0\right) + S(\theta, Q_0) - \mathcal{U}_0 + \zeta p + \tilde{\zeta} \tilde{p} + \lambda_0 Q_0 + \tilde{\lambda}_0 \tilde{Q}_0.$$

Applying *Pontryagin's Maximum Principle* yields the following necessary optimality conditions that are satisfied by a normal maximum $(\bar{\mathcal{U}}_0, \bar{\mathcal{U}}_0, \bar{Q}_0, \bar{Q}_0, \bar{p}, \bar{p})$.

1. *Adjoint variable λ_0 .* λ_0 is piecewise continuously differentiable

$$(A.44) \quad -\dot{\lambda}_0(\theta) = G(\bar{\omega}(\theta)) - g(\bar{\omega}(\theta)) \left(c_0 k + [S(\theta, q)]_{\bar{Q}_0(\theta)}^{\bar{Q}_0(\theta)+k} - \bar{\omega}(\theta) \right) - 1 \quad \forall \theta \in \Theta.$$

2. *Adjoint variable $\tilde{\lambda}_0$.* $\tilde{\lambda}_0$ is piecewise continuously differentiable with

$$(A.45) \quad -\dot{\tilde{\lambda}}_0(\theta) = -G(\bar{\omega}(\theta)) + g(\bar{\omega}(\theta)) \left(c_0 k + [S(\theta, q)]_{\bar{Q}_0(\theta)}^{\bar{Q}_0(\theta)+k} - \bar{\omega}(\theta) \right) \quad \forall \theta \in \Theta.$$

3. *Transversality conditions.* There is no end-point constraint on $\bar{\mathcal{U}}_0$ at $\bar{\theta}$ and on $\tilde{\mathcal{U}}_0$ at $\underline{\theta}$, or

$$(A.46) \quad \lambda_0(\bar{\theta}) = 0,$$

$$(A.47) \quad \tilde{\lambda}_0(\underline{\theta}) = 0.$$

From (A.44) and (A.45), we deduce $\dot{\lambda}_0(\theta) + \dot{\tilde{\lambda}}_0(\theta) = 1$. Integrating yields $\lambda_0(\theta) + \tilde{\lambda}_0(\theta) = \theta - \kappa$ for some constant κ . Using (A.46) and (A.47) respectively yields $\tilde{\lambda}_0(\bar{\theta}) = \bar{\theta} - \kappa$ and $\lambda_0(\underline{\theta}) = \underline{\theta} - \kappa = \tilde{\lambda}_0(\bar{\theta}) - k$, where the last equality follows from Assumption 2. Finally, we also get $\lambda_0(\theta) = \theta - \bar{\theta} + \tilde{\lambda}_0(\bar{\theta}) - \tilde{\lambda}_0(\theta)$. The transversality condition $\tilde{\lambda}_0(\underline{\theta}) = 0$ encodes the optimality of $\bar{\mathcal{U}}_0(\underline{\theta})$. Since $\bar{\mathcal{U}}_0(\bar{\theta}) = \bar{\mathcal{U}}_0(\underline{\theta}) = 0$, using (A.38) leads to $\bar{\mathcal{U}}_0(\underline{\theta}) = -\int_{\underline{\theta}}^{\bar{\theta}} \bar{Q}_0(\tilde{\theta}) d\tilde{\theta}$. From (A.43), we also get $\bar{\mathcal{U}}_0(\underline{\theta}) + \underline{\theta} k - \frac{k^2}{2} = \bar{\omega}(\underline{\theta})$. These two conditions define $(\bar{\omega}(\underline{\theta}), \kappa)$. For future reference, the condition

$$(A.48) \quad \underline{\theta} k - \frac{k^2}{2} - \int_{\underline{\theta}}^{\bar{\theta}} \bar{Q}_0(\tilde{\theta}) d\tilde{\theta} = \bar{\omega}(\underline{\theta})$$

implicitly defines the constant κ .

4. *Adjoint variable ζ .* ζ is piecewise continuously differentiable with

$$(A.49) \quad -\dot{\zeta}(\theta) = (1 - G(\bar{\omega}(\theta)))(Q_{fb}(\theta) - \bar{Q}_0(\theta)) + \lambda_0(\theta) \quad \forall \theta \in \Theta.$$

Since there are no end-point constraints on $\bar{Q}_0$ at $\bar{\theta}$ and (A.40) holds, the transversality conditions can be written as follows

$$\zeta(\bar{\theta}) = 0 \geq \zeta(\underline{\theta}).$$

5. *Adjoint variable $\tilde{\zeta}$.* $\tilde{\zeta}$ is piecewise continuously differentiable with $-\dot{\tilde{\zeta}} = \frac{\partial \mathcal{H}^u}{\partial \bar{Q}_0}$, or

$$(A.50) \quad -\dot{\tilde{\zeta}}(\theta) = G(\bar{\omega}(\theta))(Q_{fb}(\theta) - \bar{Q}_0(\theta) - k) + \tilde{\lambda}_0(\theta) \quad \forall \theta \in \Theta.$$

Because there are no end-point constraints on $\bar{Q}_0$ at $\underline{\theta}$ and (A.40) holds, the transversality conditions read $\tilde{\zeta}(\underline{\theta}) = 0 \geq \tilde{\zeta}(\bar{\theta})$.

6. *Optimality conditions w.r.t. p .* For all $\theta \in \Theta$

$$p(\theta) = 0 \text{ if } \zeta(\theta) < 0 \text{ and } p(\theta) \geq 0 \text{ if } \zeta(\theta) = 0.$$

7. *Optimality conditions w.r.t. $\tilde{p}$.* For all $\theta \in \Theta$, $\tilde{p}(\theta) = 0$ if $\tilde{\zeta}(\theta) < 0$, and $\tilde{p}(\theta) \geq 0$ if $\tilde{\zeta}(\theta) = 0$.

8. *Optimality conditions w.r.t. r .* Since there are no constraints on r , we must have

$$(A.51) \quad \zeta(\underline{\theta}) = \tilde{\zeta}(\bar{\theta}).$$

This concludes the statement of the optimality conditions with non-conditional tariffs. $\square$

PROOF OF PROPOSITION 4. Assumption 3 amounts to $\frac{\partial \mathcal{H}^u}{\partial \mathcal{U}_0} < 0$ and $\frac{\partial \mathcal{H}^u}{\partial \mathcal{U}_0} < 0$ at the optimum. Hence, (A.44) and (A.45) imply that $\lambda_0(\theta)$ and $\tilde{\lambda}_0(\theta)$ are strictly increasing. This implies, in turn, that $\lambda_0(\bar{\theta}) = 0 > \lambda_0(\underline{\theta}) = \underline{\theta} - \kappa$ and $\tilde{\lambda}_0(\bar{\theta}) = \bar{\theta} - \kappa > \tilde{\lambda}_0(\underline{\theta}) = 0$. If $\bar{Q}_0$ is locally separating at $\underline{\theta}$ and $\tilde{Q}_0$ is locally separating at $\bar{\theta}$, the FOCs on separating zones ($\dot{\zeta} = \dot{\tilde{\zeta}} = 0$) give $\bar{Q}_0(\bar{\theta}) = Q_{fb}(\underline{\theta}) + \frac{\bar{\theta} - \kappa}{G(\bar{\omega}(\underline{\theta}))} > \bar{Q}_0(\underline{\theta}) = Q_{fb}(\underline{\theta}) + \frac{\bar{\theta} - \kappa - \Delta\theta}{1 - G(\bar{\omega}(\underline{\theta}))}$, which violates the monotonicity of $\bar{Q}_0$ on $\Theta_{-k} \cup \Theta$. $\square$

PROOF OF PROPOSITION 5. On any interval where $\bar{Q}_0(\theta)$ and $\tilde{Q}_0(\theta)$ are both increasing, we have

$$(A.52) \quad \bar{Q}_0(\theta) = Q_{fb}(\theta) - \frac{\kappa - \theta + \tilde{\lambda}_0(\theta)}{1 - G(\bar{\omega}(\theta))},$$

$$(A.53) \quad \bar{Q}_1(\theta) = \tilde{Q}_0(\theta) + k = Q_{fb}(\theta) + \frac{\tilde{\lambda}_0(\theta)}{G(\bar{\omega}(\theta))}.$$

Define now the new variable

$$(A.54) \quad \mu(\theta) = \tilde{\lambda}_0(\theta) + (\kappa - \theta)G(\bar{\omega}(\theta)).$$

Inserting into (A.52) and (A.53), we obtain

$$(A.55) \quad \bar{Q}_0(\theta) = Q_m(\theta) + \bar{\theta} - \kappa - \frac{\mu(\theta)}{1 - G(\bar{\omega}(\theta))},$$

$$(A.56) \quad \bar{Q}_1(\theta) = Q_m(\theta) + \bar{\theta} - \kappa + \frac{\mu(\theta)}{G(\bar{\omega}(\theta))}.$$

BUNCHING. Proposition 4 shows that bunching arises around $\bar{\theta} - k = \underline{\theta}$. We characterize the solution with one bunching area per profile: $\bar{Q}_0$ is constant and equal to Q^b on $[\underline{\theta}, \theta_0]$, and increasing on $[\theta_0, \bar{\theta}]$; $\tilde{Q}_0$ is constant and equal to Q^b on $[\theta_1, \bar{\theta}]$ and increasing on $[\underline{\theta}, \theta_1]$. The quantity levels (A.55) and (A.56) are thus valid everywhere except on the bunching area. On the boundaries of the bunching area, we thus have

$$\bar{Q}_0(\theta_1) = \bar{Q}_0(\theta_0) = Q^b.$$

Expressing this equality by means of (A.55) and (A.56) and using the continuity of $\bar{Q}_0$ at θ_0 and $\tilde{Q}_0$ at θ_1 , we obtain

$$(A.57) \quad Q^b = Q_m(\theta_1) - k + \bar{\theta} - \kappa + \frac{\mu(\theta_1)}{G(\bar{\omega}(\theta_1))} = Q_m(\theta_0) + \bar{\theta} - \kappa - \frac{\mu(\theta_0)}{1 - G(\bar{\omega}(\theta_0))}.$$

Since $\bar{Q}_0$ is strictly increasing on $(\theta_0, \bar{\theta}]$, we have $\zeta(\theta) = 0$ on this interval. Taking into account the

continuity of ζ at θ_0 , $\zeta(\theta_0) = 0$. Integrating (A.49) on $[\underline{\theta}, \theta_0]$ where $\bar{Q}_0(\theta) = Q^b$ yields

$$(A.58) \quad \zeta(\underline{\theta}) = \int_{\underline{\theta}}^{\theta_0} -\dot{\zeta}(\theta) d\theta = \int_{\underline{\theta}}^{\theta_0} \left((1 - G(\bar{\omega}(\theta)))(Q_m(\theta) + \bar{\theta} - \kappa - Q^b) - \mu(\theta) \right) d\theta.$$

Since $\bar{Q}_0(\theta)$ is also strictly increasing on $[\underline{\theta}, \theta_1)$, $\tilde{\zeta}(\theta) = 0$ for $\theta \in [\underline{\theta}, \theta_1)$. Taking into account the continuity of $\tilde{\zeta}$ at θ_1 , $\tilde{\zeta}(\theta_1) = 0$. Integrating (A.50) on $[\theta_1, \bar{\theta}]$ we obtain the following

$$(A.59) \quad -\tilde{\zeta}(\bar{\theta}) = \int_{\theta_1}^{\bar{\theta}} -\dot{\tilde{\zeta}}(\theta) d\theta = \int_{\theta_1}^{\bar{\theta}} \left(G(\bar{\omega}(\theta))(Q_m(\theta) - k + \bar{\theta} - \kappa - Q^b) + \mu(\theta) \right) d\theta.$$

Using (A.51), (A.58) and (A.59), we get

$$(A.60) \quad \int_{\underline{\theta}}^{\theta_0} \left((1 - G(\bar{\omega}(\theta)))(Q_m(\theta) + \bar{\theta} - \kappa - Q^b) - \mu(\theta) \right) d\theta \\ = \int_{\theta_1}^{\bar{\theta}} \left(G(\bar{\omega}(\theta))(Q^b + k - (\bar{\theta} - \kappa) - Q_m(\theta)) - \mu(\theta) \right) d\theta.$$

TRAJECTORIES OF THE SYSTEM. The transversality conditions (A.46) and (A.47) can be expressed in terms of μ as follows

$$(A.61) \quad \mu(\underline{\theta}) = (\kappa - \underline{\theta})G(\bar{\omega}(\underline{\theta})) \text{ and } \mu(\bar{\theta}) = (\bar{\theta} - \kappa)(1 - G(\bar{\omega}(\bar{\theta}))).$$

Assume now that $\theta_0 < \theta_1$.⁴³ Three subsets of the types interval must be considered.

MIDDLE-RANGE VALUES OF θ . Observe that (A.55) and (A.56) both apply on the interval $[\theta_0, \theta_1]$ which has non-empty interior. Inserting (A.55) and (A.56) into (6.3) yields the following

$$(A.62) \quad \dot{\bar{\omega}}(\theta) = \frac{\mu(\theta)}{G(\bar{\omega}(\theta))(1 - G(\bar{\omega}(\theta)))} \quad \forall \theta \in [\theta_0, \theta_1].$$

Inserting (A.62) into (A.55) and (A.56) yields (6.6). Differentiating (A.54) w.r.t. θ , we obtain

$$(A.63) \quad \dot{\mu}(\theta) = \dot{\lambda}_0(\theta) - G(\bar{\omega}(\theta)) + (\kappa - \theta)\dot{\bar{\omega}}(\theta)g(\bar{\omega}(\theta)) \quad \forall \theta \in [\theta_0, \theta_1].$$

Using (A.62), (A.63) and inserting into (A.45), we get

$$(A.64) \quad \dot{\mu}(\theta) = -g(\bar{\omega}(\theta)) \left(c_0 k - \bar{\omega}(\theta) + \frac{(2G(\bar{\omega}(\theta)) - 1)\mu^2(\theta)}{2G^2(\bar{\omega}(\theta))(1 - G(\bar{\omega}(\theta)))^2} \right) \quad \forall \theta \in [\theta_0, \theta_1].$$

The differential equations (A.62) and (A.64) are identical to (A.13) and (A.15) that we found in the case of market-share contracts although transversality conditions differ because these equations are now only valid on $[\theta_0, \theta_1]$. To determine the values of μ at θ_0 and θ_1 from (A.61), we need to analyze the trajectories on $[\theta_1, \bar{\theta}]$ and $[\underline{\theta}, \theta_0]$.

⁴³Two other configurations are possible. First, bunching may affect only one of the two profiles, with $\theta_0 = \underline{\theta}$ (no bunching of $\bar{Q}_0$) or $\theta_1 = \bar{\theta}$ (no bunching of $\bar{Q}_0$); this requires the FOC to remain active at one boundary. Second, when $\theta_0 > \theta_1$ the bunching intervals overlap and at least one of them spans an entire boundary endpoint: either $\theta_1 = \underline{\theta}$ (bunching of $\bar{Q}_0$ from $\underline{\theta}$ inwards) or $\theta_0 = \bar{\theta}$ (bunching of $\bar{Q}_0$ up to $\bar{\theta}$), again deactivating at least one boundary FOC. These boundary configurations follow as limits of the case $\theta_0 < \theta_1$; see the Online Appendix.

LEFT-NEIGHBORHOOD OF $\bar{\theta}$. On the interval $[\theta_1, \bar{\theta}]$, consumption levels are given by

$$(A.65) \quad \bar{Q}_0(\theta) = Q_m(\theta) + \bar{\theta} - \kappa - \frac{\mu(\theta)}{1 - G(\bar{\omega}(\theta))},$$

$$(A.66) \quad \bar{Q}_1(\theta) = Q^b + k.$$

Inserting (A.66) into (6.3) yields for all $\theta \in [\theta_1, \bar{\theta}]$

$$(A.67) \quad \dot{\bar{\omega}}(\theta) = Q^b + k - \bar{\theta} + \kappa - Q_m(\theta) + \frac{\mu(\theta)}{1 - G(\bar{\omega}(\theta))}.$$

Using (A.63), (A.66) and inserting into (A.45), we get for all $\theta \in [\theta_1, \bar{\theta}]$

$$(A.68) \quad \dot{\mu}(\theta) = -g(\bar{\omega}(\theta)) \left(c_0 k - \bar{\omega}(\theta) - \frac{1}{2} \left(Q^b + k - \bar{\theta} + \kappa - Q_m(\theta) \right)^2 + \frac{\mu^2(\theta)}{2(1 - G(\bar{\omega}(\theta)))^2} \right).$$

Together with the transversality conditions (A.61) at $\bar{\theta}$, the system of differential equations (A.67) and (A.68) is satisfied by the solution $(\bar{\omega}(\theta), \mu(\theta))$ on the interval $[\theta_1, \bar{\theta}]$. Notice that the difficulty here is that the first of the transversality conditions (A.61) pertains to a point outside this interval. It means that the solution has to be found by looking simultaneously at the three intervals, $[\underline{\theta}, \theta_0]$, $[\theta_0, \theta_1]$ and $[\theta_1, \bar{\theta}]$ and pasting those solutions together by using the continuity of $\bar{\omega}(\theta)$ and $\mu(\theta)$ at both θ_0 and θ_1 .

RIGHT-NEIGHBORHOOD OF $\underline{\theta}$. On the interval $[\underline{\theta}, \theta_0]$, we have

$$(A.69) \quad \bar{Q}_0(\theta) = Q^b,$$

$$(A.70) \quad \bar{Q}_1(\theta) = Q_m(\theta) + \bar{\theta} - \kappa + \frac{\mu(\theta)}{G(\bar{\omega}(\theta))}.$$

Inserting (A.69) and (A.70) into (6.3) now yields for all $\theta \in [\underline{\theta}, \theta_0]$

$$(A.71) \quad \dot{\bar{\omega}}(\theta) = Q_m(\theta) + \bar{\theta} - \kappa + \frac{\mu(\theta)}{G(\bar{\omega}(\theta))} - Q^b.$$

Using (A.63), (A.69) and (A.70) and inserting into (A.44), we get for all $\theta \in [\underline{\theta}, \theta_0]$

$$(A.72) \quad \dot{\mu}(\theta) = -g(\bar{\omega}(\theta)) \left(c_0 k - \bar{\omega}(\theta) + \frac{1}{2} \left(Q^b - Q_m(\theta) - \bar{\theta} + \kappa \right)^2 - \frac{\mu^2(\theta)}{2G^2(\bar{\omega}(\theta))} \right).$$

Together with the transversality conditions (A.61) at $\underline{\theta}$, the system of differential equations (A.71) and (A.72) is satisfied by the solution $(\bar{\omega}(\theta), \mu(\theta))$ on the interval $[\underline{\theta}, \theta_0]$.

When $(\bar{\omega}(\theta), \mu(\theta))$ is known on Θ , the system (A.57)-(A.60) together with the anchor condition (A.48) provides four equations with four unknowns, Q^b , θ_1 , θ_0 and κ . Because of a lack of tractability, this system can only be solved numerically.

We now turn to some comparative statics.

PROBABILITY OF ENTRY. The second transversality condition (A.61) together with (A.65) implies $\bar{Q}_0(\bar{\theta}) = Q_{fb}(\bar{\theta})$. Using the fact that $\bar{Q}_1(\theta) = \bar{Q}_0(\theta) + k = Q^b + k$ on $[\theta_1, \bar{\theta}]$, we get

$$\dot{\bar{\omega}}(\bar{\theta}) = \bar{Q}_1(\bar{\theta}) - \bar{Q}_0(\bar{\theta}) = Q^b + k - Q_{fb}(\bar{\theta}) = Q^b - Q_{fb}(\bar{\theta}).$$

The first transversality condition (A.61) together with (A.70) implies $\bar{Q}_1(\underline{\theta}) = Q_{fb}(\underline{\theta})$. Using the fact that $\bar{Q}_0(\theta) = Q^b$ on $[\underline{\theta}, \theta_0]$, we also get

$$(A.73) \quad \dot{\bar{\omega}}(\underline{\theta}) = \bar{Q}_1(\underline{\theta}) - \bar{Q}_0(\underline{\theta}) = Q_{fb}(\underline{\theta}) - Q^b = -\dot{\bar{\omega}}(\bar{\theta}).$$

Observe now that $Q_{fb}(\underline{\theta}) > Q^b$. This property follows from observing that, without bunching on the interval $[\theta_1, \bar{\theta}]$, (A.70) would apply, which together with the transversality condition (A.61), would imply $\bar{Q}_1(\bar{\theta}) = Q_{fb}(\bar{\theta}) = Q_{fb}(\underline{\theta}) + k$. That bunching arises for $\bar{Q}_1$ around $\bar{\theta}$ thus implies that $Q_{fb}(\underline{\theta}) + k > Q^b + k$ as required above. From this, it follows that $\dot{\bar{\omega}}(\theta)$ changes sign on Θ , with $\dot{\bar{\omega}}(\underline{\theta}) > 0 > \dot{\bar{\omega}}(\bar{\theta})$. Consumption levels and entry are no longer everywhere positively correlated.

SIGN OF $\mu(\theta)$. By (A.62), the sign of $\dot{\bar{\omega}}$ equals the sign of μ . From (A.73), it follows that $\mu(\theta) < 0$ in a left-neighborhood of $\bar{\theta}$ and $\mu(\theta) > 0$ in a right-neighborhood of $\underline{\theta}$.

SUB-MARTINGALE PROPERTY IN THE REGULAR CASE. Using the expressions of $\bar{Q}_0(\theta)$ and $\bar{Q}_1(\theta)$ on the intervals $[\underline{\theta}, \theta_0]$, $[\theta_0, \theta_1]$ and $[\theta_1, \bar{\theta}]$, we obtain

$$\int_{\underline{\theta}}^{\bar{\theta}} \left((1 - G(\bar{\omega}(\theta))) \bar{Q}_0(\theta) + G(\bar{\omega}(\theta)) \bar{Q}_1(\theta) \right) d\theta = \int_{\underline{\theta}}^{\bar{\theta}} (Q_m(\theta) + \bar{\theta} - \kappa) d\theta,$$

or, using $\gamma = \bar{\theta} - \kappa$, $\mathbb{E}_{\theta}[Q_m(\theta)] + \gamma = \mathbb{E}_{\theta}[(1 - G(\bar{\omega}(\theta))) \bar{Q}_0(\theta) + G(\bar{\omega}(\theta)) \bar{Q}_1(\theta)]$ with $\gamma \in (0, \Delta\theta)$. $\square$

PROOF OF COROLLARY 2. For $Q < Q^b$, there is a unique type $\theta \in \Theta$, say $\vartheta^-(Q + k)$, such that $\bar{Q}_0(\theta - k) = \bar{Q}_1(\theta) - k = Q$. That type also buys k units from E and his consumption of Q units from I satisfies the first-order condition $T'(Q) = \vartheta^-(Q + k) - Q - k$ where $\vartheta^-(Q + k)$ denotes the assignment function defined for $Q < Q^b$ as $\bar{Q}_1(\vartheta^-(Q + k)) = Q + k$. For $Q > Q^b$, there is a unique type $\theta \in \Theta$, say $\vartheta^+(Q)$, such that $\bar{Q}_0(\theta) = Q$. That type buys all those units from I and a similar first-order condition $\bar{T}'(Q) = \vartheta^+(Q) - Q$ determines his consumption. The marginal price thus satisfies⁴⁴

$$\bar{T}'(Q) = \begin{cases} \vartheta^+(Q) - Q = c_0 + \bar{\theta} - \gamma - \bar{\vartheta}^+(Q) + \frac{\mu(\bar{\vartheta}^+(Q))}{1 - G(\bar{\omega}(\bar{\vartheta}^+(Q)))} & \text{if } Q > Q^b, \\ \bar{\vartheta}^-(Q + k) - Q - k = c_0 + \bar{\theta} - \gamma - \bar{\vartheta}^-(Q + k) - \frac{\mu(\bar{\vartheta}^-(Q + k))}{G(\bar{\omega}(\bar{\vartheta}^-(Q + k)))} & \text{if } Q < Q^b. \end{cases}$$

$\bar{T}(Q)$ is not differentiable at Q^b . Instead, it admits right- and left-derivatives, denoted as $\bar{T}'(Q^{b+})$ and $\bar{T}'(Q^{b-})$, respectively. At this kink in Q^b , the marginal price for an additional unit is greater than the marginal price for one less unit $\bar{T}'(Q^{b+}) = \theta_0 - Q^b > \theta_1 - Q^b - k = \bar{T}'(Q^{b-})$. $\square$

PROOF OF PROPOSITION 6. From Proposition 1, the optimal market-share rebate satisfies $\bar{\omega}(\theta) < c_0 k$ for all θ , so the AEC test fails. By revealed preference, $\omega(\theta) \geq T_0(Q_0(\theta)) - T_1(Q_0(\theta) - k)$, so the EP margin is a lower bound on the AEC margin and fails a fortiori. For a non-conditional tariff $T(Q)$, the envelope theorem applied to $\mathcal{U}_0(\theta) = \max_q \theta q - \frac{1}{2} q^2 - T(q)$ gives $\mathcal{U}'_0(\theta) = Q_0(\theta)$, and, using the definition of ω and $\mathcal{U}_1$, the condition $\omega(\theta) \geq c_0 k$ can be rewritten as $\int_{\theta-k}^{\theta} Q_0(x) dx \leq \int_{\theta-k}^{\theta} (x - c_0) dx$, i.e. $\int_{\theta-k}^{\theta} [Q_0(x) - Q_{fb}(x)] dx \leq 0$. $\square$

⁴⁴We use Equations (A.55) and (A.56) to get an expression of the assignment functions $\vartheta^-(Q)$ and $\vartheta^+(Q)$ on the relevant domains for Q .

Online Appendix A

Numerical Simulations and Extensions

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OA.1. INTRODUCTION

This appendix details the numerical methods, additional theory, and the complete set of results behind the paper. Sections OA.2–OA.4 describe how the market-share (MS) and non-conditional (NC) contracts are solved. Sections OA.5 and OA.6 relate the NC contracts to all-unit discounts and fix the conventions of the antitrust tests. Section OA.7 contains the economic results, and Sections OA.8 to OA.10 the extensions and robustness checks.

Section OA.2 sets up the numerical framework: the calibration, the simulation grid $(k, \mu) \in [0.10, 2.00]^2$ at step 0.05 (1521 couples) for four values of the dispersion s of the entrant's fixed cost of entry. It presents the two independent solution pipelines. The first is a Pontryagin cascade built on a multi-regime approach, in which each pattern of bunching or saturation defines a regime-specific boundary-value problem; the second is a direct nonlinear program (NLP) that maximizes over discretized quantity profiles without imposing any regime structure a priori. Their concordance underlies the division of labor between them.

Sections OA.3 and OA.4 follow the same sequence: the catalog of admissible regimes, the theoretical derivation of the generic regime (MS-S for the market-share contract, SBA01 for the non-conditional one), the cascade algorithm, and the contract-specific concordance statistics of the two pipelines. Section OA.4 shows a structural argument (Proposition OA.1) for why boundary bunching is generic under the non-conditional contract.

Section OA.5 is devoted to all-unit discount (AUD) contracts. A descriptive analysis maps the schemes at issue in major antitrust cases onto three contractual classes (linear tiers, two-part tiers, minimum-purchase commitments). The optimal AUD contract is determined in two polar cases, no entry and certain entry. The section continues by approximating the optimal NC tariff numerically with three practical two-tier AUD families (linear, linear with a minimum-purchase requirement, two-part), to quantify how much of the optimum each contractual class recovers.

Section OA.6 fixes the conventions used in the antitrust analysis: the ex post reconstruction of tariffs from the quantity trajectories, the domain on which the efficient-price (EP) test is well defined, and the treatment of the kink of the NC tariff (one-sided derivatives and subdifferential), with its implications for the marginal below-cost pricing (mBPC) test.

Section OA.7 reports the antitrust and welfare results. It documents the position of the kink relative to marginal cost and the extent of below-cost pricing on each branch of the non-conditional tariff; compares the verdicts of the three nested tests (mBPC, EP, AEC) across the grid, including the role of the price cliff in the EP test applied to AUDs; quantifies the welfare comparison between the two contracts and its decomposition; and analyzes the AEC test in detail: what it tracks relative to a myopic benchmark, and how its verdict varies across buyer types and cost environments.

Section OA.8 relaxes the assumption $\Delta\theta = k$ maintained in the main text for the analysis of the optimal NC contract. The optimal NC contract is characterized theoretically in the two cases $\Delta\theta < k$ (Section OA.8.1) and $\Delta\theta > k$ (Section OA.8.2); numerical simulations on a dedicated grid in $(\Delta\theta, \Delta\theta/k, \mu)$ then verify that the welfare ranking and the antitrust verdicts extend away from $\Delta\theta = k$ (Section OA.8.3); Section OA.8.4 interprets the regime structure and the role of the assumption.

Section OA.9 relaxes the assumption that the incumbent is an essential trading partner. The optimality conditions of both contracts are derived theoretically when the buyer can purchase from the entrant alone, and the ETP and NETP solutions are then compared numerically.

Section OA.10 collects the remaining robustness checks, on the entry-cost distribution G and on the calibration $(\underline{\theta}, c_0)$.

OA.2. NUMERICAL SIMULATIONS: PRELIMINARIES

OA.2.1. Common framework

The incumbent I faces (i) a buyer B of type θ uniformly distributed on $[\underline{\theta}, \bar{\theta}]$, and (ii) a potential entrant E whose marginal cost c is random with cumulative distribution G (a normal distribution with parameters (μ, s) , left-truncated at zero). The entrant, if active, supplies an exogenous quantity k of the input. We compute the optimal contract numerically (i) for various values of s (0.1, 0.5, 1, 2) and (ii) along the grid $(k, \mu) \in [0.10, 2.00]^2$ at step 0.05, totaling 1521 couples per contract.

The two contracts.

MS (*market-share*): two distinct tariffs $T_0(Q)$ (no entry) and $T_1(Q)$ (entry) are offered ex ante; B pays the relevant one ex post.

NC (*non-conditional*): a single tariff $T(Q)$ is paid by B for the quantity q purchased from I , regardless of whether the entrant enters.

Calibration. Throughout the appendix, $\underline{\theta} = 4$, $c_0 = 2$, $\bar{\theta} = \underline{\theta} + \Delta\theta$.

Common notation. The buyer's first-best consumption is $Q_{fb}(\theta) = \theta - c_0$ and $Q_m(\theta) = 2\theta - \bar{\theta} - c_0$. Quantities purchased from I are denoted $Q_0(\theta)$ (no entry) and $\tilde{Q}_0(\theta)$ (entry); the total quantity consumed under entry is $Q_1(\theta) = \tilde{Q}_0(\theta) + k$. Buyer rents are $\mathcal{U}_0(\theta)$ (no entry) and $\mathcal{U}_1(\theta)$ (entry, gross of c); the entry rebate is $\omega(\theta) = \mathcal{U}_1(\theta) - \mathcal{U}_0(\theta)$. Entry occurs iff $c \leq \omega(\theta)$, so the probability of entry conditional on θ equals $G(\omega(\theta))$. The incumbent's expected profit is denoted $\mathcal{V}_I$.

Assumption on k . We maintain Assumption 1 ($Q_{fb}(\underline{\theta}) \geq k$ or $k \leq \underline{\theta} - c_0 = 2$). We also maintain Assumption 2 ($\Delta\theta = k$) throughout most of the analysis. Assumption 2 is relaxed in Section OA.8.

OA.2.2. Multi-regime approach

For both contracts, the incumbent's problem has the structure of an optimal-control problem with state ω . Depending on the parameters (k, μ) , the optimal trajectory may exhibit bunching (Q_0 and/or $\tilde{Q}_0$ constant on a sub-interval of Θ) and/or exclusion ($\tilde{Q}_0 \equiv 0$ or $Q_0 \equiv 0$ on a sub-interval). Each combination of these features defines a distinct regime with its own system of differential equations, boundary conditions, and scalar unknowns. The number of admissible regimes differs across contracts and which regime is active varies over the grid.

The numerical strategy is the same for both contracts:

- (i) For each candidate regime, we solve the regime-specific boundary-value problem (BVP).

- (ii) We verify ex post that the solution satisfies the regime’s qualifying conditions (ordering of thresholds, sign constraints on multipliers, monotonicity of quantities).
- (iii) The regimes are attempted in cascade and the first one whose solution passes all ex post checks is retained. Since the regimes correspond to nested programs, each adding a binding restriction to a less constrained one, the first admissible solution is also the one with the highest expected profit $\mathcal{V}_I$ among the admissible candidates.

Sections OA.3 and OA.4 describe each contract in turn: a catalog of admissible regimes, the technical detail of a single generic regime used as reference, the algorithm, and the selection criterion.

OA.2.3. Two independent pipelines and their concordance

Each contract is solved along two independent pipelines. The first is the Pontryagin cascade just described: a chain of regime-specific BVPs, each reduced to root-finding in a few scalar unknowns. The second is a direct nonlinear program (NLP), in which $\mathcal{V}_I$ is maximized over discretized quantity profiles subject to monotonicity, positivity, and the relevant constraints, solved with `scipy.optimize.minimize` (SLSQP), initialized with a smooth heuristic quantity profile scaled from the first-best (NC) or Maskin–Riley (MS) schedules.

The two pipelines play complementary roles, and their agreement justifies dividing the work between them. The Pontryagin solver delivers the structural decomposition (which regime is active, where bunching or saturation occurs, and the pointwise quantities and marginal tariffs in θ), but it inherits the numerical fragility of multi-zone shooting problems. The NLP solver imposes no regime structure, covers the entire grid, and is robust, but returns only the discretized profiles. The number of regimes makes the NLP attractive: it handles every configuration with a single formulation, whereas the Pontryagin approach requires a separate optimality system for each. We therefore treat the NLP as the canonical pipeline for aggregate metrics, and reserve the Pontryagin solver for quantities that require pointwise structural precision (such as the marginal tariff).

This division is justified by the close agreement of the two methods wherever both converge. Denoting by $\mathcal{V}_{\text{Pont}}$ and $\mathcal{V}_{\text{NLP}}$ the incumbent’s expected profit computed by the Pontryagin cascade and by the direct NLP, the relative discrepancy $|\mathcal{V}_{\text{Pont}} - \mathcal{V}_{\text{NLP}}|/|\mathcal{V}_{\text{Pont}}|$ has a median of order 10^{-6} under both contracts, and the same holds for every other aggregate metric: expected buyer surplus, total welfare, deadweight loss, entry probability, and expected quantities. These discrepancies are several orders of magnitude below the precision required for any economic conclusion.

OA.3. CONTRACT MS

Regime catalog. The MS contract offers two independent tariffs $T_0(Q)$ and $T_1(Q)$. The state is the rebate $\omega(\theta)$, with a single co-state $\lambda(\theta)$ satisfying transversalities $\lambda(\underline{\theta}) = \lambda(\bar{\theta}) = 0$. Five regimes are a priori possible, as shown in Table OA.1, where $q_1 = Q_1 - k$.

The cascade proceeds in the order MS-S $\rightarrow$ MS-B1 $\rightarrow$ MS-B1-aug $\rightarrow$ MS-S-aug $\rightarrow$ MS-double-sat: each regime is attempted only if the preceding one fails its ex post validation (typically the monotonicity of Q_1 , the non-negativity of Q_0 , or the non-negativity of q_1). The first valid regime is retained.

Regime	Structure	# unknowns	Where active
MS-S	Q_0, Q_1 separating throughout	1	generic
MS-B1	Q_1 bunched on left $[\underline{\theta}, \theta_0]$	3	moderate k
MS-B1-aug	MS-B1 + saturation $Q_0 \equiv 0$ on $[\underline{\theta}, \theta_a]$	4	large k
MS-S-aug	$Q_0 \equiv 0$ on $[\underline{\theta}, \theta_a]$, Q_1 separating	2	large k , small μ
MS-double-sat	double saturation $Q_0 \equiv 0$ and $q_1 \equiv 0$	5	k very large

Table OA.1: Regimes of the MS contract.

OA.3.1. Generic regime: MS-S

MS-S is the fully separating contract: Q_0 and Q_1 are strictly increasing on Θ , with $Q_0(\theta) > 0$ everywhere.

Change of variable. Define $\mu(\theta) = \lambda(\theta) + G(\omega(\theta))(\bar{\theta} - \theta)$.⁴⁵ The transversalities become $\mu(\underline{\theta}) = \Delta\theta G(\omega(\underline{\theta}))$ and $\mu(\bar{\theta}) = 0$. On separating zones,

$$Q_0(\theta) = Q_m(\theta) - \frac{\mu(\theta)}{1 - G(\omega(\theta))}, \quad Q_1(\theta) = Q_m(\theta) + \frac{\mu(\theta)}{G(\omega(\theta))}.$$

ODEs.

$$\dot{\omega}(\theta) = \frac{\mu(\theta)}{G(\omega)(1 - G(\omega))}, \quad \dot{\mu}(\theta) = -g(\omega) \left[c_0 k - \omega + \frac{(2G(\omega) - 1)\mu^2}{2G^2(\omega)(1 - G(\omega))^2} \right].$$

Ex post validation. The MS-S solution is accepted iff:

- Q_0, Q_1 are strictly increasing on Θ ;
- $Q_0(\theta) \geq 0$ on Θ ;
- $\mu \geq 0$, with $\mu(\bar{\theta}) = 0$;
- $\omega(\bar{\theta}) \in (\omega_m, c_0 k)$, where ω_m is the monopolistic rebate defined in Section 4.2.

Other regimes.

- *MS-B1*: Q_1 is bunched on a left sub-interval $[\underline{\theta}, \theta_0]$ at a common value Q_1^b . Three scalar unknowns $(\omega(\bar{\theta}), Q_1^b, \theta_0)$. Triggered when MS-S delivers a non-monotone Q_1 .
- *MS-B1-aug*: in addition to MS-B1, $Q_0 \equiv 0$ on $[\underline{\theta}, \theta_a]$ (exclusion of the lowest types). Four scalar unknowns. Triggered when MS-B1 violates $Q_0 \geq 0$. Two different sub-regimes have to be considered (since they do not involve the same ODEs on the middle interval) depending on whether Q_1 or Q_0 is relaxed first.
- *MS-S-aug*: $Q_0 \equiv 0$ on $[\underline{\theta}, \theta_a]$ (exclusion of the lowest types) while Q_1 remains separating throughout, with no bunching. Two scalar unknowns $(\omega(\bar{\theta}), \theta_a)$. Triggered when MS-B1-aug would force $Q_1^b < 0$.
- *MS-double-sat*: $q_1 \equiv 0$ on $[\underline{\theta}, \theta_b]$ in addition to $Q_0 \equiv 0$ on $[\underline{\theta}, \theta_a]$. Five scalar unknowns. Two different sub-regimes have to be considered (since they do not involve the same ODEs on the middle interval) depending on whether $\theta_a \leq \theta_b$ or $\theta_a \geq \theta_b$.

⁴⁵The function $\mu(\theta)$, distinguished by its argument θ , is a reduced co-state and is unrelated to the scalar μ indexing the grid (k, μ) , which denotes the mean of the entrant's cost distribution G ; the two are kept apart by context throughout.

For each regime, the augmented Pontryagin formalism introduces one additional state-constraint multiplier per saturated zone;⁴⁶ the system remains exactly determined.

OA.3.2. MS: algorithm and solver concordance

The coupled ODE system is solved as a two-point BVP on $(\omega, \dot{\omega})$, with the two transversalities enforced as boundary residuals, through the regime cascade above (a chain of root-finding sub-problems). The BVP is numerically delicate, for reasons classical to multi-zone shooting: stiffness on saturated zones, narrow basins of attraction near regime boundaries (a bunching threshold may collapse to within 10^{-3} of a zone endpoint), and sensitivity of state-constraint multipliers as a saturation length goes to zero. A regime is retained only if two conditions are jointly met: the residual norm of the root-finding system (transversalities, junction continuities, integral identities for bunched zones) falls below 10^{-5} , with typical values in the range 10^{-10} to 10^{-13} ; and the ex post profile passes a strict point-by-point admissibility check at tolerance 10^{-8} , requiring $Q_0 \geq 0$, $Q_1 \geq k$, monotonicity of both profiles, and non-negativity of the state-constraint multiplier on saturated zones. With NLP-based warm-starts feeding the cascade, the solver converges on 1,519 of the 1,521 grid points (99.9%).^{47,48}

Figure OA.1 shows the regimes recovered by the Pontryagin solver, with the two non-converged couples in grey; the regimes form contiguous regions. Four of the five regimes of Table OA.1 are selected on the grid: as k rises, bunching of Q_1 sets in before the exclusion of Q_0 , so MS-S-aug is never selected at $s = 1$. The two pipelines agree closely where both converge: the relative profit gap (Table OA.2) has a median of 1.2×10^{-6} and a maximum of 8.3×10^{-5} . This validates the cascade against the separate NLP pipeline (Figure OA.2).

Statistic	Relative gap
Median	1.2×10^{-6}
Mean	1.5×10^{-6}
P90	2.0×10^{-6}
P99	7.1×10^{-6}
Max	8.3×10^{-5}
Number of couples	1519

Table OA.2: Relative profit gap under MS between the Pontryagin and NLP solvers, $|\mathcal{V}_{\text{Pont}} - \mathcal{V}_{\text{NLP}}|/|\mathcal{V}_{\text{Pont}}|$, across the (k, μ) grid at $s = 1$.

⁴⁶Hartl, R. F., Sethi, S. P., & Vickson, R. G. (1995). A survey of the maximum principles for optimal control problems with state constraints. *SIAM Review*, 37(2), 181–218.

⁴⁷The two residual failures are structurally benign isolated points at sharp regime transitions ($k = 1.45$, $\mu = 1.95$, on the MS-B1/MS-B1-aug boundary; and $k = 1.60$, $\mu = 0.45$, on the MS-B1-aug/MS-double-sat boundary, a quasi-singular point where the intermediate zone collapses to width $\approx 5 \times 10^{-3}$).

⁴⁸To assess whether our regime classification depends on the initialization of the Pontryagin solver, we recomputed the entire (k, μ) grid with an independent reference run (“mode B”) that solves each couple from scratch, without using the NLP solution as a starting point. On the couples where both the main run and the reference run converge (1,355 of 1,521), they agree on the regime for 1,348 of them, with profit differences at machine precision (median 0, maximum 1.3×10^{-15}). The seven remaining discrepancies all lie on regime boundaries where two regimes attain the same profit (profit gap $\leq 10^{-4}$) and the classification is intrinsically ambiguous. The from-scratch run fails to converge within the time budget on 166 couples, of which 94% are doubly-saturated regimes, the most constrained ones; every one of these couples is nonetheless solved in the main run, which uses the NLP solution to initialize the Pontryagin solver. Initialization from the NLP optimum is therefore not necessary for the result (where the from-scratch solver converges, it reproduces the same solution), but it is what makes the most constrained regimes numerically reachable. The reference run is available upon request.

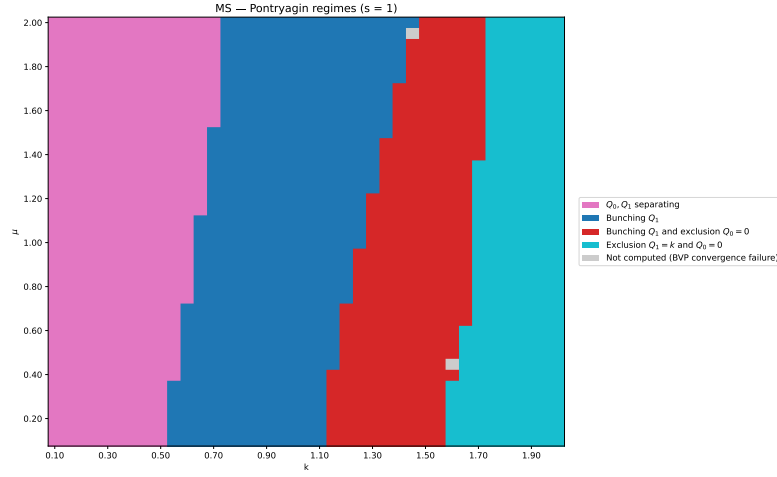


Figure OA.1: Map of the regimes under the optimal MS contract, across the (k, μ) grid at $s = 1$.

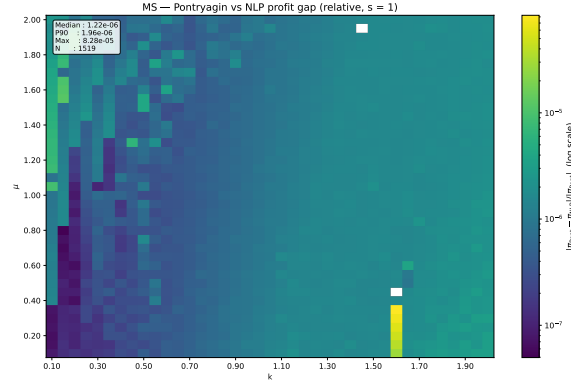


Figure OA.2: Map of the relative profit gap under MS between the Pontryagin and NLP solvers, across the (k, μ) grid at $s = 1$.

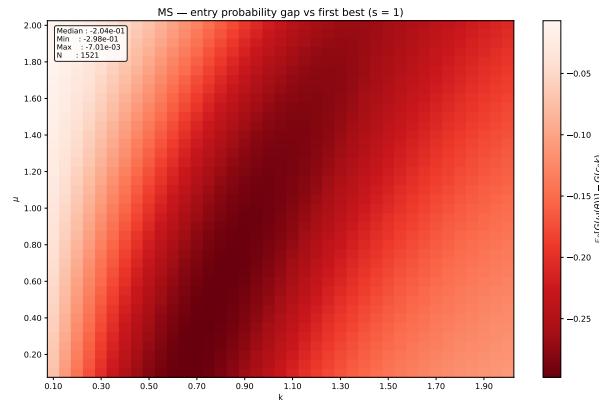


Figure OA.3: Entry probability gap relative to the first best under the optimal MS contract, $\mathbb{E}_\theta[G(\omega(\theta))] - G(c_0 k)$, in the (k, μ) plane ($s = 1$). The gap is negative throughout: the MS contract induces insufficient entry relative to the social optimum.

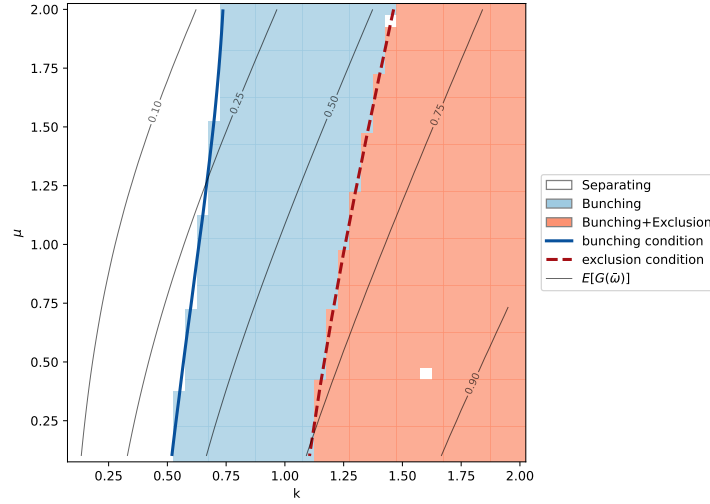


Figure OA.4: Validation of the analytical conditions for MS-S admissibility (see Section 5). The solid blue (resp. dashed red) line marks the analytical frontier beyond which Q_1 ceases to be separating (resp. Q_0 ceases to be non-negative). Level curves: expected probability of entry. $s = 1$.

Table OA.3 relates the entry distortion to the regime taxonomy of Figure OA.4. Entry falls short of the socially optimal level in every regime and at every couple of the grid, consistent with the result $\omega(\theta) < c_0 k$ throughout the type space (Figure OA.3). The size of the shortfall is not monotone in how exclusionary the contract appears. It is largest in MS-B1, the regime of a pure minimum-purchase requirement (median -0.273 , and the grid-wide maximum of -0.298 at $k = 0.65$), and it decreases as the contract takes on more visibly exclusionary features: adding the exclusion of the lowest types ($Q_0 \equiv 0$, MS-B1-aug) brings the median to -0.212 , and full withdrawal of supply under entry ($Q_1 \equiv k$, MS-double-sat) to -0.157 .

The features that most resemble foreclosure (a refusal to serve absent entry, and a withdrawal of supply even under entry) are thus associated with the smallest, not the largest, distortion of entry. This regime ordering reflects a single comparative static in the entrant's capacity. Pooling across μ , the median distortion deepens from -0.030 at $k = 0.10$ to -0.279 near $k = 0.80$ – 0.85 , then recovers to -0.152 at $k = 2.00$: its magnitude is hump-shaped in k , peaking at intermediate capacity. Because the regimes are themselves ordered by k (the capacity ranges in Table OA.3 are nearly disjoint and increasing), they inherit this ordering, so the regime classification and the capacity axis convey the same fact.

The dependence on μ (the mean of the entrant's cost distribution) is by contrast weak. The median distortion varies only between -0.178 and -0.213 as μ ranges over $[0.10, 2.00]$, narrowing slightly as μ rises: the entry shortfall is nearly flat in μ . The distortion of entry under the MS contract is therefore governed by the entrant's capacity, not by how likely entry is.⁴⁹

OA.4. CONTRACT NC

A genericity argument for boundary bunching. Assumption 3 gives a sufficient condition for bunching. Assumption 3 is always satisfied on $(k, \mu) \in [0.1, 2]^2$ for all values of s tested ($s = 0.1, 0.5, 1, 2$).

⁴⁹One caveat applies to the measurement. The distortion is computed against the first-best entry probability $G(c_0 k)$, which itself shifts with k . Isolating the contract-induced component would require a counterfactual that holds entry behavior fixed, which we do not pursue here.

Regime	Structure	n	k range	Median (IQR)	Range
MS-S	separating	438	[0.10, 0.70]	-0.117 (-0.182, -0.062)	[-0.280, -0.007]
MS-B1	bunching Q_1	494	[0.55, 1.45]	-0.273 (-0.286, -0.256)	[-0.298, -0.146]
MS-B1-aug	bunching + $Q_0 \equiv 0$	310	[1.15, 1.70]	-0.212 (-0.236, -0.183)	[-0.265, -0.137]
MS-double-sat	$Q_1 \equiv k, Q_0 \equiv 0$	277	[1.60, 2.00]	-0.157 (-0.187, -0.136)	[-0.233, -0.110]

Table OA.3: Entry distortion (realized minus socially optimal entry probability) by MS regime, $s = 1$. Regimes are listed in increasing order of how strongly their form suggests foreclosure. Columns report the median, interquartile range, and full range of the distortion within each regime, pooled across μ over the 1,519 converged couples of the (k, μ) grid.

We add here an independent, structural reason to expect boundary bunching to be generic. The argument is heuristic: it establishes an algebraic obstruction to fully separating regimes, but does not rule out the non-generic coincidence under which that obstruction is met.

The NC contract requires a single tariff $T(Q)$ to apply identically under both market structures, which imposes the continuity condition $\tilde{Q}_0(\bar{\theta}) = Q_0(\underline{\theta})$ between the two trajectories.

PROPOSITION OA.1. *If Q_0 is locally separating at $\underline{\theta}$ and $\tilde{Q}_0$ is locally separating at $\bar{\theta}$, then the identity*

$$(OA.4.1) \quad \frac{\bar{\theta} - \kappa}{G(\omega(\bar{\theta}))} = \frac{\underline{\theta} - \kappa}{1 - G(\omega(\underline{\theta}))}$$

must hold.

Proof. The FOCs on separating zones give, using $\tilde{\lambda}_0(\underline{\theta}) = 0$ and $\tilde{\lambda}_0(\bar{\theta}) = \bar{\theta} - \kappa$:

$$Q_0(\underline{\theta}) = Q_{fb}(\underline{\theta}) - \frac{\kappa - \underline{\theta}}{1 - G(\omega(\underline{\theta}))}, \quad Q_1(\bar{\theta}) = Q_{fb}(\bar{\theta}) + \frac{\bar{\theta} - \kappa}{G(\omega(\bar{\theta}))}.$$

Substituting into $Q_1(\bar{\theta}) = Q_0(\underline{\theta}) + k$ and using $Q_{fb}(\bar{\theta}) - Q_{fb}(\underline{\theta}) = \Delta\theta = k$ yields (OA.4.1). $\square$

The Pontryagin BVP has two free scalar parameters $(w_0, \kappa) = (\omega(\underline{\theta}), \kappa)$, closed by two conditions: the right transversality on $\tilde{\lambda}_0$ and the anchor condition on w_0 . The system is thus exactly determined without (OA.4.1). A fully separating regime would therefore require the exactly-determined solution (w_0^*, κ^*) to satisfy the additional relation (OA.4.1), which carves out a codimension-1 subvariety of $\mathbb{R}_{(w_0, \kappa)}^2$. We do not have a proof that (w_0^*, κ^*) avoids this subvariety for all (k, μ) ; the coincidence is, however, non-generic, and on our entire simulation grid it never occurs. We read this as a second, structural reason to regard boundary bunching as generic.

Implications. At least one of the FOCs at $\underline{\theta}$ or at $\bar{\theta}$ must be deactivated, which requires bunching of Q_0 near $\underline{\theta}$, of $\tilde{Q}_0$ near $\bar{\theta}$, or both. This rules out two families: pure separating regimes on Θ , and strictly interior bunching regimes (where Q_0 and $\tilde{Q}_0$ are both separating near both boundaries, with bunching strictly inside $(\underline{\theta}, \bar{\theta})$).

Regime catalog. The NC contract admits five base regimes, distinguished by the pattern of bunching of Q_0 and $\tilde{Q}_0$ on Θ . Naming convention: SBA stands for small bunching area (disjoint or single-quantity bunching, $\theta_0 < \theta_1$), LBA for large bunching area (overlapping or full-quantity bunching, $\theta_0 > \theta_1$). The suffix 0, 1, 01 indicates which quantity is bunched (Q_0 , $\tilde{Q}_0$, or both).

Regime	Structure of bunching	# unknowns	Where active
SBA01	$Q_0 = Q^b$ on $[\underline{\theta}, \theta_0]$ and $\tilde{Q}_0 = Q^b$ on $[\theta_1, \bar{\theta}]$, with $\theta_0 \leq \theta_1$	5	generic
SBA1	$\theta_0 = \underline{\theta}$ (only the right bunching zone)	4	boundary case
SBA0	$\theta_1 = \bar{\theta}$ (only the left bunching zone)	4	boundary case
LBA1	$\theta_1 = \underline{\theta}$, with $Q_0 = Q^b$ and $Q_1 = Q^b + k$ on a middle zone reaching the left boundary	4	boundary case
LBA0	$\theta_0 = \bar{\theta}$, symmetric of LBA1 reaching the right boundary	4	boundary case

Table OA.4: Regimes of the NC contract. “# unknowns” counts the scalar parameters of the BVP.

The cascade implements these five base regimes. As under MS, each could in principle be refined into augmented sub-cases (a left saturation $Q_0 \equiv 0$ excluding the lowest types, or an additional interior bunching zone) whenever a candidate solution returns a non-monotone or negative quantity.⁵⁰

OA.4.1. Generic regime: SBA01

We describe SBA01 in detail; the other regimes share the same overall structure, differing by which scalar unknowns are present and which boundary conditions replace the integral identity below.

Change of variable. The incumbent’s problem is reformulated through $\mu(\theta) = \tilde{\lambda}_0(\theta) + (\kappa - \theta) G(\omega(\theta))$ and $\hat{Q}(\theta) = 2\theta - c_0 - \kappa$, where $\tilde{\lambda}_0$ is the Pontryagin co-state and κ is a scalar dual variable (the FOC on r couples it to the entire trajectory). The variable μ plays the same role as in MS, with a definition adapted to the presence of κ . The state pair (ω, μ) evolves on Θ according to a piecewise-defined ODE system.

Three zones. SBA01 partitions Θ into:

- $[\underline{\theta}, \theta_0]$: $Q_0 = Q^b$ (bunched), $\tilde{Q}_0$ separating.
- $[\theta_0, \theta_1]$: both Q_0 and $\tilde{Q}_0$ separating.
- $[\theta_1, \bar{\theta}]$: $\tilde{Q}_0 = Q^b$ (bunched), Q_0 separating.

ODE on the middle (fully separating) zone.

$$\dot{\omega}(\theta) = \frac{\mu(\theta)}{G(\omega(\theta))(1 - G(\omega(\theta)))}, \quad \dot{\mu}(\theta) = -g(\omega(\theta)) \left[c_0 k - \omega(\theta) + \frac{(2G(\omega(\theta)) - 1)\mu^2}{2G^2(\omega(\theta))(1 - G(\omega(\theta)))^2} \right],$$

with quantities

$$Q_0(\theta) = \hat{Q}(\theta) - \frac{\mu(\theta)}{1 - G(\omega(\theta))}, \quad Q_1(\theta) = \hat{Q}(\theta) + \frac{\mu(\theta)}{G(\omega(\theta))}.$$

The ODE expressions on the two bunched zones follow the same template with Q_0 or $\tilde{Q}_0$ replaced by Q^b ; we omit the full forms.

⁵⁰We do not code these sub-cases: at $s = 1$ the generic regime SBA01, the least constrained of the catalog, is optimal at every grid point, so no more constrained regime can dominate it, and the validation against the NLP pipeline (which imposes no regime structure) confirms that nothing is lost by stopping at the base catalog.

Scalar unknowns and conditions. The unknowns are κ , $w_0 = \omega(\underline{\theta})$, Q^b , θ_0 , θ_1 , closed by five scalar conditions:

(C1) *Right transversality*: $\mu(\bar{\theta}) = (\bar{\theta} - \kappa) (1 - G(\omega(\bar{\theta})))$.

(C2) *Junction continuity at θ_0* : $Q^b = \hat{Q}(\theta_0) - \mu(\theta_0)/(1 - G(\omega(\theta_0)))$.

(C3) *Junction continuity at θ_1* : $Q^b + k = \hat{Q}(\theta_1) + \mu(\theta_1)/G(\omega(\theta_1))$.

(C4) *Anchor condition*:

$$w_0 = \underline{\theta} k + \frac{1}{2} k^2 - \int_{\underline{\theta}}^{\bar{\theta}} Q_1(\theta) d\theta.$$

(C5) *FOC on the scalar control r* :

$$\int_{\underline{\theta}}^{\theta_0} [(1 - G(\omega(\theta))) (\hat{Q}(\theta) - Q^b) - \mu(\theta)] d\theta = \int_{\theta_1}^{\bar{\theta}} [G(\omega(\theta)) (Q^b + k - \hat{Q}(\theta)) - \mu(\theta)] d\theta.$$

The left transversality $\mu(\underline{\theta}) = (\kappa - \underline{\theta}) G(w_0)$ defines the ODE initial value given (κ, w_0) , not a residual; the system is exactly determined.

Ex post validation. A solution of SBA01 is accepted iff: the ordering $\underline{\theta} < \theta_0 \leq \theta_1 < \bar{\theta}$ holds; $Q_0, \tilde{Q}_0$ are nonnegative and weakly increasing in θ ; $Q^b > 0$, $\omega(\theta) \geq 0$, $\mu(\theta) \geq 0$ on Θ ; and the auxiliary co-state signs are consistent with the regime.

Other regimes. The remaining regimes differ from SBA01 by which threshold collapses or how the bunching zone reaches a boundary. In SBA1, SBA0 (4 unknowns each), one bunching zone collapses to the boundary ($\theta_0 = \underline{\theta}$ for SBA1, $\theta_1 = \bar{\theta}$ for SBA0); the FOC on r disappears and is replaced by a junction continuity equation determining Q^b algebraically. In LBA1, LBA0 (4 unknowns each), a middle zone with both quantities bunched ($Q_0 = Q^b$ and $Q_1 = Q^b + k$) reaches one of the two boundaries. Each regime has the same exact-determination property; full ODE specifications are available upon request.

OA.4.2. NC: algorithm and solver concordance

For SBA01, three distinct ODE systems govern the state (ω, μ) on the three sub-intervals, with right-hand sides corresponding to the bunching pattern of each zone. The endpoints θ_0, θ_1 are themselves unknowns determined by junction continuity, so the partition of Θ is part of the solution; the bunching value Q^b enters the ODEs as a parameter shared across zones. Given a trial vector $(\kappa, w_0, Q^b, \theta_0, \theta_1)$, the solver sets $\mu(\underline{\theta})$ from the left transversality, integrates the three zones in sequence (adaptive Runge–Kutta, each zone initialized from the previous one's endpoint), and evaluates the five residuals C1–C5. A Newton-type root finder drives the residuals to zero. Because the residual map is highly non-linear (it composes a stiff ODE flow with junction conditions at endogenously located thresholds), convergence is sensitive to the initial guess; we mitigate this with a multistart strategy, drawing initializations from the no-entry limit and from neighboring grid points, running the root finder from each in parallel, and retaining the highest-profit solution that passes ex post validation. Figure OA.5 maps the regimes recovered over the grid.

Saturation. Comparing the two regime maps brings out an asymmetry that the focus on the generic regimes (MS-S, SBA01) would hide. Under MS, the no-entry quantity saturates, $Q_0 \equiv 0$ on a low-type

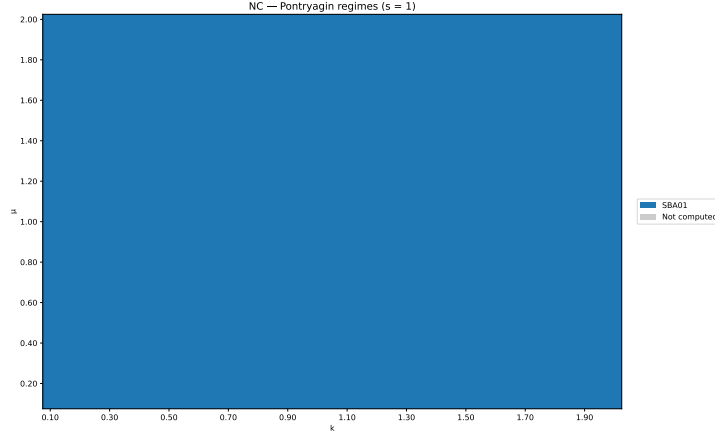


Figure OA.5: Map of the regimes under the optimal NC contract ($s = 1$).

interval, over a substantial part of the grid (the augmented regimes of Table OA.1). Under NC, by contrast, Q_0 never saturates. This is not a feature of $s = 1$: re-solving the NLP across $s \in \{0.1, 0.5, 1, 2\}$, the contrast is unchanged at every dispersion level: MS saturates Q_0 on 44 to 63% of couples (on intervals reaching 44% of the type space), while NC saturates on none.⁵¹ The mechanism is the coupling/decoupling distinction. MS uses two independent tariffs T_0 and T_1 , so it can drive the no-entry quantity to zero (excluding the low types' outside option to widen the wedge $Q_1 - Q_0$ and extract more through ω) while keeping the entry schedule separating. NC has a single tariff, which couples the two branches through the matching floor $Q^b = Q_0(\underline{\theta}) = \tilde{Q}_0(\bar{\theta})$: saturating the bottom would also collapse the top of the entry branch, so the optimum keeps Q^b strictly positive.

The independent NLP pipeline maximizes $\mathcal{V}_I$ over discretized profiles $(Q_0, \tilde{Q}_0)$ subject to monotonicity, positivity, the junction equality $\tilde{Q}_0(\bar{\theta}) = Q_0(\underline{\theta})$, and $\omega \geq 0$, warm-started from the Pontryagin solution when available. As stated in Section OA.2.3, the two pipelines agree to machine precision: the relative profit gap (Table OA.5) has a median of 3.0×10^{-7} and never exceeds 1.8×10^{-6} , and the same holds for all other aggregate metrics.⁵²

Statistic	Relative gap
Median	3.0×10^{-7}
Mean	3.8×10^{-7}
P90	8.2×10^{-7}
P99	1.4×10^{-6}
Max	1.8×10^{-6}
Number of couples	1521

Table OA.5: Relative profit gap under NC between the Pontryagin and NLP solvers, $|\mathcal{V}_{\text{Pont}} - \mathcal{V}_{\text{NLP}}|/|\mathcal{V}_{\text{Pont}}|$, across the (k, μ) grid at $s = 1$.

⁵¹The pattern is robust to s because it is driven by the coupling structure, not by cost dispersion; on the analysis range $k \leq 2$, NC saturates neither Q_0 nor the entry quantity $\tilde{Q}_0$.

⁵²This agreement also has a further implication. The Pontryagin formulation restricts attention to absolutely continuous profiles, since the maximum principle requires differentiability of the state almost everywhere. The NLP imposes no such restriction (only monotonicity and positivity); the fact that its solution is nonetheless continuous in θ across the entire grid, and coincides with the Pontryagin one at machine precision, is evidence that the absolute-continuity restriction is without loss on the parameter range considered.

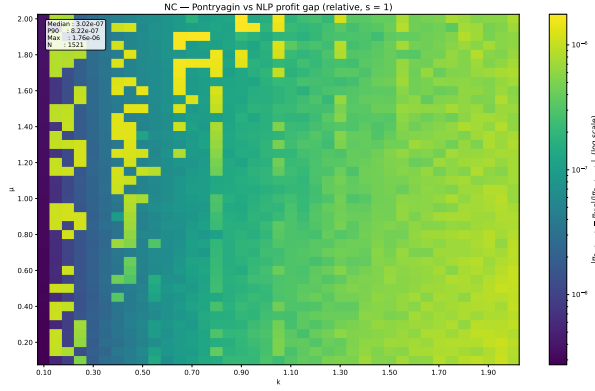


Figure OA.6: Map of the relative profit gap between the Pontryagin and NLP solvers, across the (k, μ) grid at $s = 1$.

Entry distortions. Figure OA.7 maps the entry distortion (realized minus socially optimal entry probability) under the NC contract. Unlike the MS contract, which forecloses entry everywhere on the grid, the non-conditional tariff distorts entry in both directions: entry falls short of its first-best level on 71% of the grid and exceeds it on the remaining 29% (Table OA.6). The sign is governed by the entrant's capacity. Pooling across μ , the median distortion is positive for small k and turns negative around $k \approx 0.6$, then declines to a trough near $k \approx 1.45$; the sign frontier is a diagonal, with the capacity at which entry switches from excessive to deficient rising in μ (from $k \approx 0.4$ at $\mu = 0.1$ to $k \approx 1.05$ at $\mu = 2.0$). Over-entry is thus the rule at low capacity and under-entry at high capacity, with μ shifting the boundary between the two.

Both distortions are small in magnitude (over-entry reaches at most +0.025 and under-entry -0.095), an order of magnitude below the uniform under-entry of the MS contract (Table OA.3). The contrast is qualitative as well as quantitative: where the MS contract systematically restricts entry, the non-conditional tariff does not control the level of entry. Constrained to a single schedule, the incumbent can no longer set the rebate $\omega(\theta)$ freely; it is tied to the consumption profile through (6.3) and must serve several screening objectives at once, so its net effect on entry has no definite sign.

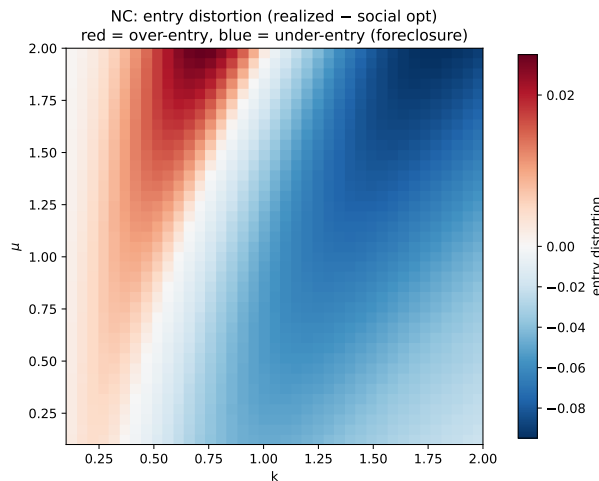


Figure OA.7: Entry distortion under NC (realized minus socially optimal entry probability), over the (k, μ) grid at $s = 1$. Red: over-entry; blue: under-entry.

Panel A. Median entry distortion by capacity (pooled over μ)								
k	0.2	0.4	0.6	0.8	1.0	1.45	1.7	2.0
median	+0.005	+0.010	+0.002	−0.022	−0.048	−0.070	−0.061	−0.046

Panel B. Sign frontier: smallest k at which entry turns deficient, by μ					
μ	0.1	0.5	1.0	1.5	2.0
k^*	0.40	0.45	0.60	0.80	1.05

Table OA.6: Entry distortion under NC, across the (k, μ) grid at $s = 1$. Entry is excessive on 29% of the grid and deficient on 71%; the distortion ranges from +0.025 (at $k = 0.7, \mu = 2.0$) to −0.095 (at $k = 1.8, \mu = 2.0$), with grid median −0.037. Panel A reports the median over μ at selected capacities; Panel B reports the capacity k^* at which the median distortion changes sign, for selected μ .

Wedge γ . The wedge $\gamma = \bar{\theta} - \kappa$ measures the cost of the single-tariff constraint imposed by the NC contract. Empirically (see Table OA.7), γ remains strictly in $(0, \Delta\theta)$ throughout the grid: positive on all 1521 solutions (median 0.75) and below $\Delta\theta$ everywhere, with a maximum normalized value of $\gamma/\Delta\theta = 0.895$.

The normalized wedge $\gamma/\Delta\theta$ increases monotonically with k (recall $\Delta\theta = k$ under Assumption 2), from a median of 0.06 at $k = 0.10$ to 0.83 at $k = 2.00$. The growth is concave: most of the increase occurs over $k \in [0.10, 1.00]$, after which $\gamma/\Delta\theta$ saturates around 0.83. The empirical maximum 0.895 is reached for $k \geq 1.50$ and remains constant beyond, suggesting that even for large entry segments, the wedge does not exhaust the theoretical range $(0, \Delta\theta)$. The cost of the single-tariff constraint grows with the size of the entry segment but converges to a fraction of $\Delta\theta$ strictly less than one. The wedge also decreases mildly with μ (from approximately 0.86 at $\mu = 0.10$ to 0.72 at $\mu = 1.05$): when entry becomes less likely (higher mean entry cost), the strategic stakes of the single-tariff constraint diminish.

	γ	$\gamma/\Delta\theta$
Mean	0.76	0.61
Median	0.75	0.71
Min	0.001	0.014
Max	1.79	0.895

Table OA.7: Distribution of the wedge $\gamma = \bar{\theta} - \kappa$ across 1521 Pontryagin solutions on the (k, μ) grid at $s = 1$ (with $\Delta\theta = k$). The wedge remains strictly in $(0, \Delta\theta)$ on all 1521 couples, with a normalized maximum of 0.895, approached for $k \geq 1.50$.

OA.5. AUD CONTRACTS

OA.5.1. Descriptive analysis

The all-unit discount (AUD) contracts scrutinized in major antitrust cases are structurally heterogeneous, and this heterogeneity maps onto the three contractual classes we simulate: their within-tier pricing (linear or two-part), their use of minimum-purchase commitments, and their number of tiers.

Within-tier pricing: linear or two-part. Some contracts match the canonical three-parameter form (p_1, p_0, Q^b) of the theoretical literature,⁵³ with purely linear within-tier prices: the Mars retail rebate of Conlon and Mortimer (2021) is a single retroactive volume threshold with no fixed component,⁵⁴ and the single-cliff market-share rebates in *Eaton* (2012, 70–97.5% thresholds) and *Eisai–Sanofi* (2016, a 75% threshold on Lovenox) fit the same stylization.⁵⁵ Others add quasi-fixed components: the Intel–Dell MOAP/MCP program combined percentage rebates with quarterly lump-sum payments (20–40% of total transfers).⁵⁶

Minimum-purchase commitments. The major loyalty-rebate cases pair the variable rebate with *minimum-purchase commitments*: *Intel* (2009) required Dell to source “all or almost all” of its CPUs from Intel, with 80–100% targets on other OEMs; *Tomra* (2006) combined retroactive rebates with individualized quantity commitments; *Eaton* conditioned its rebates on 70–97.5% market-share targets.⁵⁷ These commitments prevent small residual purchases from rivals while the buyer still claims the rebate—the loyalty-inducing function of conditional rebates in EU doctrine (*Hoffmann-La Roche*, 1979).⁵⁸

Number of tiers. Strictly two-tier (single-cliff) schemes—*Eaton*, *Eisai*, the Mars rebate, Brooke Group—contrast with finely graduated ones: *Michelin II* (2001) stacked an annual volume rebate of about sixty thresholds with a service bonus and a progression bonus, and *Tomra*’s scheme used increments of 0.2–0.5% per tier.⁵⁹

Accordingly, we simulate three contractual classes: (a) AUDs with a linear price per tier ($T(Q) = p_i Q$, the Mars/Eaton stylization); (b) AUDs with a two-part structure per tier ($T(Q) = p_i Q + F_i$, the Intel–Dell structure); and (c) any of these with an added minimum-purchase requirement $q_{\min}$ (the loyalty-threshold dimension of *Intel*, *Tomra*, *Eaton*). Comparing outcomes across classes identifies which conclusions are robust to the contractual options available to the incumbent.

OA.5.2. Theory: two polar cases

Setup. An incumbent I with constant marginal cost c_0 faces a buyer B whose taste θ is uniform on $[\underline{\theta}, \bar{\theta}]$, $\Delta\theta = \bar{\theta} - \underline{\theta}$, with quadratic preferences $u(q, \theta) = \theta q - q^2/2$. A potential entrant E of capacity k supplies, if entry occurs, k units at zero unit price in exchange for B covering its fixed cost c . We impose $\Delta\theta = k$. The incumbent designs an AUD

$$(OA.5.1) \quad T_{AUD}(q) = \begin{cases} p_1 q & \text{if } q < Q^b, \\ p_0 q & \text{if } q \geq Q^b, \end{cases} \quad p_1 \geq p_0.$$

⁵³See Kolay et al. (2004) and Chao et al. (2018).

⁵⁴Conlon and Mortimer (2021).

⁵⁵*ZF Meritor LLC v. Eaton Corp.*, 696 F.3d 254 (3d Cir. 2012); *Eisai, Inc. v. Sanofi-Aventis U.S., LLC*, 821 F.3d 394 (3d Cir. 2016).

⁵⁶*SEC v. Dell Inc.*, Litigation Release No. 21599 (July 22, 2010); the MCP program transferred about \$6 billion to Dell over 2002–2007.

⁵⁷*Commission Decision of 13 May 2009*, Case COMP/C-3/37.990 — *Intel*; *Commission Decision of 29 March 2006*, Case COMP/E-1/38.113 — *Prokent-Tomra*.

⁵⁸*Hoffmann-La Roche & Co. AG v. Commission*, Case 85/76, ECLI:EU:C:1979:36.

⁵⁹*Commission Decision of 20 June 2001*, Case COMP/E-2/36.041/PO — *Michelin*, aff’d Case T-203/01 (CFI 2003).

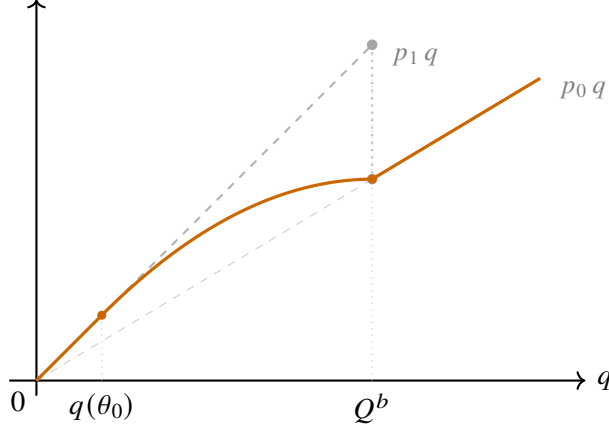


Figure OA.8: The two-price AUD tariff T_{AUD} ($p_1 q$ for $q < Q^b$, $p_0 q$ for $q \geq Q^b$; dashed grey, with its retroactive drop at Q^b) and its convex envelope (orange).

The no-entry case. The optimal AUD absent any entrant shows that the contract is a screening device: the kink induces bunching at the plateau, the cliff $p_1 - p_0$ sustains it, and exclusion considerations, when present, are added on top.

Given (p_0, p_1, Q^b) , a buyer of type θ maximizes $\theta q - q^2/2 - T_{\text{AUD}}(q)$. With

$$S_1(\theta) = \frac{1}{2}(\theta - p_1)^2, \quad S_b(\theta) = \theta Q^b - \frac{1}{2}Q^{b2} - p_0 Q^b, \quad S_0(\theta) = \frac{1}{2}(\theta - p_0)^2,$$

the indifference thresholds are $\theta_1 = p_0 + Q^b$ (from $S_b = S_0$) and $\theta_0 = p_1 + Q^b - \sqrt{2(p_1 - p_0)Q^b}$ (from $S_1 = S_b$). Assuming full market coverage (which requires $\Delta \leq \underline{\theta} - c_0$), when $\underline{\theta} < \theta_0 < \theta_1 < \bar{\theta}$, three regions are active: $[\underline{\theta}, \theta_0]$ on the p_1 -segment, $[\theta_0, \theta_1]$ at the plateau, $[\theta_1, \bar{\theta}]$ on the p_0 -segment. With θ uniform,

$$(OA.5.2) \quad \mathcal{V}_I = \frac{1}{\Delta \theta} \left[\int_{\underline{\theta}}^{\theta_0} (p_1 - c_0)(\theta - p_1) d\theta + \int_{\theta_0}^{\theta_1} (p_0 - c_0)Q^b d\theta + \int_{\theta_1}^{\bar{\theta}} (p_0 - c_0)(\theta - p_0) d\theta \right].$$

PROPOSITION OA.2 (Optimal AUD without entry). *The optimal AUD without entry is:*

- A: if $\Delta \theta \leq (\underline{\theta} - c_0)/2$, then $Q^b = \underline{\theta} - c_0$, $p_0 = (\underline{\theta} + c_0)/2$, and p_1 free subject to $\theta_0 \leq \underline{\theta}$.
- B: if $\Delta \theta \geq (\underline{\theta} - c_0)/2$, then $p_0 = (\bar{\theta} + 2c_0)/3$, $Q^b = 2(3\underline{\theta} - \bar{\theta} - 2c_0)/3$, and $p_1 \geq \underline{\theta}$.

The profit is $\mathcal{V}_I = (\underline{\theta} - c_0)^2/2$ in A, and $(p_0 - c_0)(\bar{\theta} - p_0)^2/(2\Delta\theta)$ in B. In both, the p_1 -segment is inactive in equilibrium: no type purchases on it. The cliff $p_1 - p_0$ is an off-equilibrium threat to sustain pooling at Q^b .

Proof. Step 1: the p_1 -segment is inactive at the optimum. Reparametrize with $\Delta := \sqrt{2(p_1 - p_0)Q^b} \geq 0$, so $\theta_0 = p_1 + Q^b - \Delta$ and $p_1 - p_0 = \Delta^2/(2Q^b)$. Eliminating p_0 via $\text{FOC}_{p_0} = 0$,

$$\partial_{\Delta} \mathcal{V}_I + \partial_{Q^b} \mathcal{V}_I \propto \Delta (\Delta - 2Q^b) P(\Delta, Q^b, c_0, \underline{\theta}, \bar{\theta}),$$

where P 's zeros produce no admissible interior critical point.⁶⁰ The two factors give two candidate branches. On $\Delta = Q^b$ (so $\theta_0 = p_1$), the system reduces to $p_0 = (\bar{\theta} + 2c_0)/3$, $Q^b = 2(3\underline{\theta} - \bar{\theta} - 2c_0)/3$, $p_1 = \underline{\theta}$, giving $\theta_0 = \underline{\theta}$ (boundary, not interior). On $\Delta = 2Q^b$ the bunching zone collapses ($\theta_0 = \theta_1 = \theta_{\text{eq}}$),

⁶⁰Verified by elimination of p_0 via $\text{FOC}_{p_0} = 0$ and numerical exploration of the parameter space.

and

$$\theta_{\text{eq}} - \underline{\theta} = \frac{4Q^b (c_0 - \underline{\theta} + \Delta\theta) - \Delta\theta [2(\underline{\theta} - c_0) - \Delta\theta]}{4(3Q^b + \Delta\theta)} < 0,$$

so $\theta_{\text{eq}} < \underline{\theta}$ (inadmissible). Hence the optimum lies on $\theta_0 \leq \underline{\theta}$.

Step 2: the restricted program ($\theta_0 \leq \underline{\theta}$). Treat p_1 as free and solve in (p_0, Q^b) with $\theta_1 = p_0 + Q^b$. In *Configuration A* ($\theta_1 \geq \bar{\theta}$), all types pool at Q^b ; the IR constraint at $\underline{\theta}$ binds ($p_0 = \underline{\theta} - Q^b/2$), and maximizing $(\underline{\theta} - Q^b/2 - c_0)Q^b$ gives $Q^b = \underline{\theta} - c_0$, $p_0 = (\underline{\theta} + c_0)/2$, feasible iff $\Delta\theta \leq (\underline{\theta} - c_0)/2$. In *Configuration B* ($\underline{\theta} < \theta_1 < \bar{\theta}$), $\mathcal{V}_I$ is convex in Q^b at fixed p_0 , so the optimum is on the binding IR boundary $Q^b = 2(\underline{\theta} - p_0)$;⁶¹ the bracket then simplifies to $(\bar{\theta} - p_0)^2/2$, and the FOC in p_0 yields $p_0 = (\bar{\theta} + 2c_0)/3$, feasible iff $\Delta\theta \geq (\underline{\theta} - c_0)/2$. $\square$

The kink partitions types (low types pool at the plateau with IR binding at $\underline{\theta}$, high types sort onto the p_0 -segment when $\Delta\theta$ is large enough), and the cliff $p_1 - p_0$ supports the kink.

The full-entry case. Let entry occur with probability one ($c = 0$). Receiving k units, the buyer's choice q at the incumbent maximizes $(\theta - k)q - \frac{1}{2}q^2 - T_{\text{AUD}}(q) + (\theta k - \frac{1}{2}k^2)$, i.e. he optimizes as if facing the AUD alone with effective type $\theta_{\text{eff}} = \theta - k$. The incumbent's per-buyer profit is unchanged, so Proposition OA.2 applies verbatim under $\theta \mapsto \theta - k$, on a uniform effective type interval $[\underline{\theta} - k, \underline{\theta}]$ (assuming again full market coverage, which now requires $k = \Delta\theta \leq \underline{\theta} - k - c_0$ or $k \leq (\underline{\theta} - c_0)/2$).

COROLLARY OA.1. Under $k \leq (\underline{\theta} - c_0)/2$, Proposition OA.2 carries over with $\underline{\theta}$ replaced by $\underline{\theta} - k$: for $k \in [0, (\underline{\theta} - c_0)/3]$, $Q^b = \underline{\theta} - k - c_0$, $p_0 = (\underline{\theta} - k + c_0)/2$; for $k \in [(\underline{\theta} - c_0)/3, (\underline{\theta} - c_0)/2]$, $p_0 = (\underline{\theta} + 2c_0)/3$, $Q^b = 2(2\underline{\theta} - 3k - 2c_0)/3$; in both, $p_1 \geq \underline{\theta} - k$ and the p_1 -segment is inactive.

A certain entrant thus does not turn the AUD into an entry-deterrence device; it only re-calibrates the contract to a lower effective type. In particular, the p_1 -segment is again never active, so the three-segment shape (p_0 + plateau + p_1), is absent in both polar limits $c \rightarrow 0$ and $c \rightarrow \infty$; our simulations show it emerges only under uncertain entry, where the incumbent uses the p_1 -branch to influence the entry decision.

COROLLARY OA.2. In both polar cases, adding a minimum-purchase requirement is of no use to the incumbent: since the p_1 -segment is inactive, a quantity floor only restricts a region no type traverses, leaving quantities, transfers, and profit unchanged.

Whether this instrument acquires a strategic role under uncertain entry is addressed in our simulations.

OA.5.3. Numerical approximation of the optimal non-conditional tariff by AUD

We now ask whether the optimal NC tariff, T , can be approximated by practical AUDs of the type seen in *Intel–Dell* or *Michelin*: few tiers, with an endogenous minimum-purchase requirement (MPR).

We consider three AUD families on a domain $q \geq q_{\text{inf}}$, with $q_{\text{inf}} \geq 0$ chosen optimally and acting as an MPR:

- a *purely linear* family (AUD-L-pur), $T_{\text{AUD}}(q) = p_i q$ on 2 tiers and $q_{\text{inf}} = 0$;
- a *linear* family with MPR (AUD-L-MPR), $T_{\text{AUD}}(q) = p_i q$ on 2 tiers and $q_{\text{inf}} \geq 0$ chosen endogenously;

⁶¹The other boundary $Q^b \rightarrow 0$ degenerates to linear pricing $p_0 q$, strictly dominated by any active plateau.

- a *two-part* family (AUD-NF), $T_{\text{AUD}}(q) = F_i + p_i q$ on 2 tiers and $q_{\text{inf}} = 0$.

The MPR is the only infra-marginal instrument in AUD-L-MPR; in AUD-NF the fixed fees provide a partial substitute. We impose $\mathcal{U}_0(\theta) \geq 0$ and no monotonicity on the p_i , optimizing over the prices, the threshold between tiers Q^b , the fixed fees (AUD-NF only), and q_{inf} .

Methodology. The problem is non-concave and non-differentiable; we first run differential evolution for global exploration, then refine locally with Nelder–Mead, with the buyer’s within-tier problem in closed form and expected profit averaged over 3,000 types, on the grid $(k, \mu) \in [0.1, 2.0]^2$ at step 0.05 and $s = 1$ (1521 points).

Remark. In simulations not reported here, we study the number of tiers N . There are rapidly diminishing returns in N : in AUD-NF, $N=2 \rightarrow 3$ moves the median ratio by less than 0.001; in AUD-L-MPR, the mean ratio rises by +0.008 ($N=2 \rightarrow 3$) and +0.002 ($N=3 \rightarrow 4$). We report $N = 2$, the most parsimonious approximation. Likewise, adding a quantity ceiling q_{sup} has a negligible impact on profit.

Profit comparison. Table OA.8 reports the distribution of the profit ratio $\mathcal{V}_I^{\text{AUD}}/\mathcal{V}_I^{\text{NC}}$ across the grid $(k, \mu) \in [0.10, 2.00]^2$ at step 0.05 and $s = 1$, for the three AUD families. The non-conditional profit $\mathcal{V}_I^{\text{NC}}$ is the Pontryagin solution at each (k, μ) , and the ratio is computed pointwise on the 1521 grid points where both contracts are solved.

Family	Mean	Median	Min	Max	% > 0.99	% > 0.95
AUD-L-pur (linear, no MPR)	0.393	0.312	0.124	0.977	0.0%	85.5%
AUD-L-MPR (linear, with MPR)	0.986	0.988	0.954	1.000	43.7%	100.0%
AUD-NF (two-part)	0.996	0.998	0.984	1.000	90.5%	100.0%

Table OA.8: Distribution of $\mathcal{V}_I^{\text{AUD}}/\mathcal{V}_I^{\text{NC}}$ across the 1521-point grid, $s = 1$.

The three families fall into two groups. The purely linear AUD without an infra-marginal instrument captures less than half of the NC profit on average, never exceeds 0.99, and falls as low as 0.12 at some grid points. Adding an infra-marginal instrument closes almost the entire gap: the linear family with an endogenous minimum-purchase requirement reaches a mean ratio of 0.986 and exceeds 0.95 at every grid point, and the two-part family reaches 0.996 and exceeds 0.99 on 90.5% of the grid. The improvement from AUD-L-pur to AUD-L-MPR is the relevant comparison, since the two families share the same linear within-tier prices and the same number of tiers and differ only by the presence of the minimum-purchase requirement. The minimum-purchase requirement is what brings the AUD close to the NC benchmark. The two-part family, in which the fixed fees provide a partial substitute for the requirement, yields a further but smaller gain, consistent with the fixed fees and the minimum-purchase requirement playing the same infra-marginal role.

A common convex kink. One feature is common to all three contracts and does not depend on the form of the rebate. The NC tariff, the linear AUD, and the linear AUD with a minimum-purchase requirement all generate a convex kink at Q^b , and the bunching of an interval of types at the plateau is a consequence of that kink rather than of the rebate being retroactive (AUD) or incremental (NC). For the NC contract this follows from Corollary 2 and is confirmed by the numerical analysis of T_{NC} ; for the two linear AUD families the simulations reproduce the same pattern on the entire grid (the low-type interval

bunches at Q^b without entry, the high-type interval bunches at Q^b under entry). What matters for behavior is thus the convex envelope and the kink at Q^b , not the form of the rebate that supports it.

The contribution of the minimum-purchase requirement. The minimum-purchase requirement does not act on the kink itself. Table OA.9 reports the average share of types bunched at the floor q_{inf} and at the threshold Q^b , separately without and with entry, for the two linear families. Two points follow. First, bunching at Q^b under entry is almost identical across the two families (38.0% for the pure AUD, 41.8% with the requirement), so the requirement leaves the role of the kink unchanged. Second, the requirement adds a second locus of bunching at the floor. Under entry, 58.2% of types bunch at q_{inf} under the requirement, against 11.7% for the pure AUD; but these two figures do not measure the same object. For the pure AUD the floor is $q_{\text{inf}} = 0$, so a type “at the floor” buys nothing from the incumbent and sources entirely from the entrant, which is exclusivity of the entrant; for the requirement the floor is $q_{\text{inf}} > 0$, so a type at the floor is forced to a minimum purchase from the incumbent. The requirement thus converts the entrant exclusivity left open by the pure AUD into a contracted minimum purchase: the kink at Q^b governs the upper types in both families, while the floor, present only with the requirement, captures the lower types that the pure AUD would let switch to the entrant.

Family	Without entry		With entry	
	at q_{inf}	at Q^b	at q_{inf}	at Q^b
AUD-L-pur (no MPR)	0.0%	42.7%	11.7%	38.0%
AUD-L-MPR (with MPR)	14.5%	75.6%	58.2%	41.8%

Table OA.9: Average share of types bunched at the floor q_{inf} and at the threshold Q^b , by entry branch, over the 1521-point grid, $s = 1$. For the pure AUD $q_{\text{inf}} = 0$, so the floor column under entry measures types buying nothing from the incumbent (entrant exclusivity), not a contracted minimum purchase.

The two-part family: three regimes. The two-part AUD reaches the NC benchmark as closely as the linear family with a minimum-purchase requirement (mean ratio 0.996), but it does so in a way that is not uniform across the grid. Depending on (k, μ) , the optimal two-part contract takes one of three qualitative forms, according to the use made of the small-quantity tier (p_1, F_1) . In the first form (73.1% of the grid) the small-quantity tier is used only under entry, and the threshold lies beyond the crossing point of the two tariffs ($Q^b < q_{\times}$, with $q_{\times} = (F_1 - F_0)/(p_0 - p_1)$), so that crossing Q^b lowers the bill: the contract takes the form of a threshold discount. In the second form (21.0%) the small-quantity tier is also used without entry and the threshold lies before the crossing point ($Q^b \geq q_{\times}$), so there is no discount at the threshold. In the third form (5.9%) the small-quantity tier is never chosen by any type; it is an inactive tier. Figure OA.9 shows that the three forms occupy distinct regions of the (k, μ) plane: the discount form covers large k and small μ , the no-discount form a band at moderate k and high μ , and the inactive-tier form a vertical strip at small k .

The two forms share the same tariff structure. In the discount regime the marginal price is higher on large quantities and the fixed fee is lower ($p_0 > p_1$ and $F_0 < F_1$ on 100% of the grid points); in the no-discount regime the same ordering holds (98.1% of points). The two forms therefore both display a convex kink at Q^b , and they differ only by the position of the threshold relative to the crossing point of

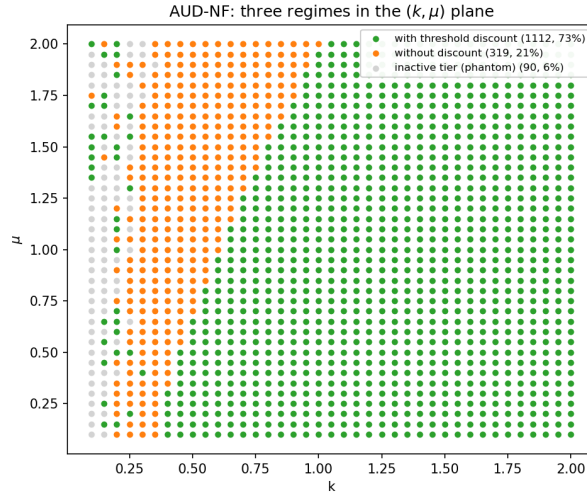


Figure OA.9: The three regimes of the optimal two-part AUD in the (k, μ) plane, $s = 1$: with threshold discount ($Q^b < q_\times$), without discount ($Q^b \geq q_\times$), and inactive small-quantity tier.

the two tariffs, not by the structure of the prices. What distinguishes a contract with a threshold discount from one without is thus where the threshold falls, not whether the schedule is convex: the convex kink is common to both, and the discount is a property of the location of Q^b .

Table OA.10 reports, for each regime, the average share of types bunched at the floor ($q_{\text{inf}} = 0$ for the two-part family) and at the threshold Q^b , without and with entry. No type buys nothing from the incumbent in any regime or branch, so the two-part family, like the linear family with a minimum-purchase requirement, leaves no room for exclusivity of the entrant. In the two active-tier regimes bunching at Q^b is comparable across branches and accounts for a similar share of types, consistent with the common convex kink. The inactive-tier regime is different: without entry only a small share of types is at Q^b , but under entry all types bunch at Q^b .

Regime	n	Without entry		With entry	
		at 0	at Q^b	at 0	at Q^b
With threshold discount	1112	0.0%	45.1%	0.0%	43.3%
Without discount	319	0.0%	50.2%	0.0%	43.8%
Inactive tier	90	0.0%	14.2%	0.0%	100.0%

Table OA.10: Average share of types bunched at the floor and at Q^b , by regime and entry branch, for the AUD-NF, over the 1521-point grid, $s = 1$. The floor is $q_{\text{inf}} = 0$ for the two-part family.

The inactive-tier contracts arise at small k (Figure OA.9), where the capacity captured by the entrant is small and entry is correspondingly infrequent. In this regime the small-quantity tier is chosen by no type, and its parameters are not identified by the equilibrium. The role of this tier is not to allocate any type but to support bunching at Q^b in the infrequent event of entry: under entry all types bunch at Q^b , and the unvisited small-quantity tier is the off-equilibrium threat that makes this bunching incentive-compatible. The inactive tier is thus an instrument that disciplines the entry branch.

Entry relative to the first best. Table OA.11 compares the entry probability induced by each family, $\mathbb{E}[G(\omega)]$, with the first-best entry probability $G(c_0 k)$. The table also reports the NC contract, which is

the benchmark the AUD families approximate. The contrast is in the direction of the distortion. The purely linear AUD induces over-entry at every grid point: entry is more frequent than is efficient on 100% of the grid. The NC contract instead induces under-entry on 71.1% of the grid, with a median gap of -0.037 . Adding an infra-marginal instrument to the linear AUD reverses the sign of its distortion and aligns it with the NC contract: the linear family with a minimum-purchase requirement under-enters on 71.7% of the grid and the two-part family on 70.8%, with median gaps of -0.031 and -0.034 , both close to the -0.037 of the NC contract. The magnitudes are moderate because the grid includes a large region of high μ where entry is infrequent under any contract and the gap is mechanically small; what is systematic is the sign, not the size. The families with an infra-marginal instrument thus reproduce not only the profit of the NC contract but also its entry distortion, while the pure linear AUD departs from it in both respects.

Contract	n	Over-entry	Under-entry	Median gap	Mean gap
Non-conditional (T_{NC})	1521	28.9%	71.1%	-0.037	-0.033
AUD-L-pur (no MPR)	1521	100.0%	0.0%	$+0.057$	$+0.066$
AUD-L-MPR (with MPR)	1521	26.9%	71.7%	-0.031	-0.027
AUD-NF (two-part)	1521	29.2%	70.8%	-0.034	-0.031

Table OA.11: Realized entry probability $\mathbb{E}[G(\omega)]$ minus first-best entry probability $G(c_0 k)$, over the 1521-point grid, $s = 1$. Over-entry and under-entry give the share of grid points with a positive and a negative gap.

In what follows we work directly with T_{NC} as the reference contract, economic and antitrust conclusions extending qualitatively to the practical two-tier AUD-with-MPR, whose profit stays within about 5% of the NC benchmark at every grid point and within 1.5% on average (Table OA.8).

OA.6. TARIFF RECONSTRUCTION AND ANTITRUST TEST CONVENTIONS

Tariff reconstruction from quantity trajectories For NC and MS, the optimal contract is characterized as quantity profiles $(Q_0, \tilde{Q}_0, \mathcal{U}_0, \mathcal{U}_1)$ rather than as a tariff T . The tariff is reconstructed ex post: a buyer of type θ choosing $Q_0(\theta)$ under no entry satisfies

$$T(Q_0(\theta)) = \theta Q_0(\theta) - \frac{1}{2} Q_0(\theta)^2 - \mathcal{U}_0(\theta),$$

and similarly for $\tilde{Q}_0$ under entry: $T(\tilde{Q}_0(\theta)) = \theta Q_1(\theta) - \frac{1}{2} Q_1(\theta)^2 - \mathcal{U}_1(\theta)$ with $Q_1(\theta) = \tilde{Q}_0(\theta) + k$.

NC. A *single* tariff T is recovered, defined on the union of images $\text{Im}(Q_0) \cup \text{Im}(\tilde{Q}_0)$. Junction continuity at $\tilde{Q}_0(\bar{\theta}) = Q_0(\bar{\theta})$ ensures that the two branches connect smoothly.

MS. The tariffs T_0 and T_1 are independent and reconstructed separately on $\text{Im}(Q_0)$ and $\text{Im}(\tilde{Q}_0)$ respectively. No continuity requirement links them.

The marginal tariff $T'(Q)$ is then obtained by finite differences along the reconstructed T (NC, MS).

Convention for the EP-form test The form-based efficient price (EP) test compares the tariff against the cost of replicating the k units at the incumbent's cost:

$$T(Q_0(\theta)) - T(Q_0(\theta) - k) \geq c_0 k.$$

This test is only defined for types θ such that $Q_0(\theta) - k$ falls within the domain of the menu offered by I . Outside this subset (denoted Θ_{valid}) the strategy “buy $Q_0 - k$ from I and k units at cost c_0 from an equally-efficient competitor” is not implementable. To compare AEC and EP on an equal footing, we restrict both tests to Θ_{valid} .

Bunching and non-differentiability of the non-conditional tariff. The NC tariff $T(Q)$ must apply identically whether the buyer sources exclusively from I or splits purchases with the entrant. This requirement imposes the matching condition

$$\tilde{Q}_0(\bar{\theta}) = Q_0(\underline{\theta}) = Q^b,$$

which ties the upper end of the “with-entry” branch (parametrized by $\tilde{Q}_0$) to the lower end of the “no-entry” branch (parametrized by Q_0). This matching condition rules out a fully separating solution: at least one of the two allocation functions must exhibit bunching near a boundary. Concretely, $Q_0(\theta) = Q^b$ on an interval $[\underline{\theta}, \theta_0]$ and $\tilde{Q}_0(\theta) = Q^b$ on an interval $[\theta_1, \bar{\theta}]$, with $\theta_0 \leq \theta_1$. A non-degenerate mass of types therefore chooses the same quantity Q^b : the types in $[\underline{\theta}, \theta_0]$ who buy exclusively from I , and the types in $[\theta_1, \bar{\theta}]$ who split their purchases.

Kink and one-sided derivatives. At the matching point Q^b , the tariff T is continuous but not differentiable: it admits distinct left- and right-derivatives. These one-sided derivatives are obtained from the buyer’s first-order conditions on each of the two branches that meet at Q^b . To the right ($Q > Q^b$), the no-entry branch becomes separating from type θ_0 onward, whose FOC gives $T'(Q) = \theta - Q_0(\theta)$; evaluated at the matching point,

$$T'(Q^{b+}) = \theta_0 - Q^b.$$

To the left ($Q < Q^b$), the with-entry branch becomes separating below type θ_1 . Since a buyer who splits purchases consumes $Q + k$ units in total (with k supplied by the entrant), his FOC is taken with respect to total consumption and reads $T'(Q) = \theta - (Q + k)$; evaluated at the matching point,

$$T'(Q^{b-}) = \theta_1 - Q^b - k.$$

The term $-k$ reflects precisely the fact that these buyers already hold k units: their marginal willingness to pay for additional units bought from I is lowered accordingly. The kink is convex, that is $T'(Q^{b+}) > T'(Q^{b-})$, which amounts to $\theta_0 - Q^b > \theta_1 - Q^b - k$, i.e. $\theta_0 - \theta_1 > -k$; this inequality holds whenever bunching is present (it fails only in the degenerate case $\theta_0 = \underline{\theta}$, $\theta_1 = \bar{\theta}$ where the separating region spans the entire interval).

Subdifferential. At a point where a convex function fails to be differentiable, the classical derivative is replaced by the subdifferential: the set of slopes of all supporting lines that touch the graph from below at that point. For the tariff T at the kink Q^b , the subdifferential is the closed interval

$$\partial T(Q^b) = [T'(Q^{b-}), T'(Q^{b+})] = [\theta_1 - Q^b - k, \theta_0 - Q^b].$$

This subdifferential has a direct economic interpretation. Each type bunched at Q^b has a marginal utility of consumption at Q^b (the slope of his indifference curve): $\theta - Q^b$ for a no-entry type, and $\theta - Q^b - k$ for a with-entry type. For a type to actually choose Q^b , it is necessary and sufficient that his indifference slope belong to the subdifferential, i.e. that his indifference curve can “nest” in the corner formed by the kink. The subdifferential is thus exactly the range of marginal utilities of the types pooled at Q^b ; its bounds are attained by the types located at the boundaries θ_0 and θ_1 of the bunching regions.⁶²

Implications for the marginal below-cost pricing (mBPC) test. The marginal below-cost pricing (mBPC) test examines whether $T'(Q) \geq c_0$. On separating regions, where the tariff is differentiable, the slope is unique and the comparison with c_0 is unambiguous; the test then bears on a well-defined property of the tariff function. At the kink Q^b , by contrast, the “slope” is no longer a scalar but the interval $\partial T(Q^b)$. The relevant criterion for detecting below-cost pricing is then to look at the lower bound of the subdifferential, i.e. the left-derivative $T'(Q^{b-}) = \theta_1 - Q^b - k$: if it is strictly below c_0 , then there exist units (just below Q^b) charged at a marginal rate below cost. Under MS this difficulty does not arise.

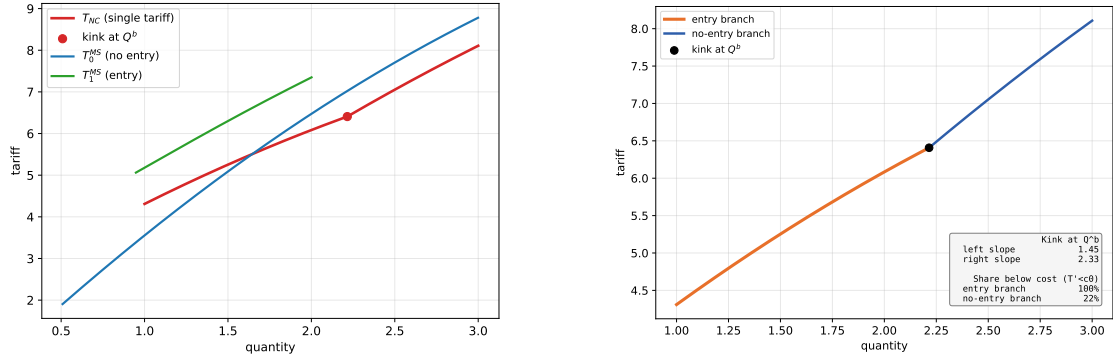
From a test on the tariff function to the share of affected buyers. The mBPC test, as stated above, bears on the tariff function: it establishes the existence of quantities charged below marginal cost, irrespective of who purchases them. One may wish to complement this diagnosis with a measure of the mass of buyers exposed to a marginal price below cost, which is a distinct information. On separating regions this measure is immediate: each type maps to a unique quantity and a unique slope, and one counts the types for which $T'(\cdot) < c_0$. At the bunching point, however, a mass of types shares the same quantity Q^b without any single marginal price being attached to them; a convention is then required. The convention we adopt relies on the subdifferential: a bunched type is counted as “affected” if his marginal utility at Q^b (the element of the subdifferential corresponding to him, namely $\theta - Q^b$ without entry or $\theta - Q^b - k$ with entry) is below c_0 . This convention identifies the buyers whose last unit purchased from I is valued below marginal cost, hence those for whom the tariff structure induces a downward distortion of consumption. It thereby links the test on the tariff function (existence of below-cost pricing) to its reach over the buyer population (extent of the phenomenon).

OA.7. ANTITRUST AND WELFARE RESULTS

OA.7.1. Antitrust tests

Figure OA.10 shows the structural difference between the two contracts. The NC contract offers a single tariff T made of two branches (an entry branch below the matching point Q^b and a no-entry branch above it) that join continuously at Q^b but with a discontinuous change in slope, producing a kink. The MS contract instead offers two smooth conditional tariffs T_0 and T_1 .

⁶²It is different from the bunching that arises under market-share (MS) contracts. Under MS, the incumbent uses two distinct conditional tariffs, T_0 and T_1 . The bunching that may occur under MS is bunching along a single branch: a range of types chooses the same quantity, but the conditional tariff in question remains smooth (C^1) at that point. The allocation function $\theta \mapsto \bar{Q}_0(\theta)$ is flat there, yet this does not prevent the tariff slope $T'_1(q)$ from being uniquely defined at that quantity. There is therefore no kink, and the subdifferential reduces to a singleton. The non-differentiability of the NC tariff stems specifically from the matching of two branches of a single tariff.



(a) Optimal tariffs under the two regimes.

(b) Below-cost pricing on the non-conditional tariff.

Figure OA.10: Optimal tariffs for a representative configuration ($k = 1$, $\mu = 1$, $s = 1$), as tariff levels against the quantity purchased from the incumbent. Panel (a) contrasts the two regimes; panel (b) isolates the non-conditional tariff and reports the marginal slopes at the kink and the share of buyers below cost on each branch.

The kink crosses marginal cost. Panel (b) shows the role of the kink: the left derivative of T at Q^b lies strictly below the incumbent's marginal cost c_0 while the right derivative lies above it, so the entry branch is priced below cost at the margin. This is systematic. Table OA.12 reports that $T'(Q^{b-}) < c_0 < T'(Q^{b+})$ on 100% of the grid, with a median jump of 0.91. The kink is the tariff instrument by which the incumbent translates the entry-versus-no-entry decision into a discontinuity in marginal price that crosses c_0 . A pointwise marginal price test applied on either side of Q^b therefore yields opposite verdicts (suspect on the entry side, innocuous on the no-entry side) even though the two branches belong to the same optimal tariff. The average tariff $T(q)/q$, by contrast, stays above c_0 on both branches everywhere on the grid, so an average-price test clears the tariff that a marginal-price test flags. The antitrust verdict depends on the chosen test.

	$T'(Q^{b-}) - c_0$	$T'(Q^{b+}) - c_0$	Jump
Mean	-0.59	+0.31	0.90
Median	-0.57	+0.32	0.91
Min	-1.42	+0.07	0.10
Max	-0.01	+0.51	1.75

Table OA.12: Position of the kink at Q^b relative to marginal cost c_0 . Left slope $T'(Q^{b-}) = \theta_1 - Q^b - k$ corresponds to the entry branch terminating at the kink; right slope $T'(Q^{b+}) = \theta_0 - Q^b$ to the no-entry branch starting at the kink. The jump equals $k - (\theta_1 - \theta_0) > 0$. Across the (k, μ) grid at $s = 1$.

Below-cost pricing by branch. The below-cost region is not confined to the kink. Table OA.13 reports, for each branch, the fraction of types $\theta \in \Theta$ with $T'(\theta) < c_0$ and the fraction of the branch's quantity support $[q_{lo}, q_{hi}]$ where $T'(Q) < c_0$; the two measures coincide on the smooth portion of T but diverge at the kink, where a positive-measure interval of types maps to a single quantity Q^b . On the entry branch ($Q < Q^b$), $T' < c_0$ holds essentially everywhere: 99.9% of the quantity support is below cost at the median, with a worst violation of -0.57 on average. On the no-entry branch ($Q > Q^b$), below-cost pricing is virtually absent in terms of quantity support (0.02% at the median) but affects about 20% of

the types. The dissymmetry reflects bunching: types in $[\underline{\theta}, \theta_0]$ are all assigned to Q^b and inherit the left slope $T'(Q^{b-})$, which is itself below cost. Away from the kink the no-entry branch is above cost almost everywhere, consistent with the standard screening distortion for exclusive purchases. The gap between fraction of types and fraction of support on the no-entry branch is thus a feature of the test, not of the schedule: any marginal-price test registers the kink as a below-cost zone for the bunched types even though the no-entry branch itself prices above cost. Weighting each type by its probability of effectively choosing a branch, the entry branch accounts for about 76% (median) of the total below-cost mass. Weighting each type by its probability of effectively choosing the entry branch, a median of 83% of buyers are served below cost (Figure OA.11), a coverage measure distinct from the 76% above, which is the entry branch's share of the total below-cost mass rather than the fraction of buyers that below-cost pricing reaches.

	Entry branch			No-entry branch		
	Frac. types	Frac. support	Worst	Frac. types	Frac. support	Worst
Mean	0.92	0.998	-0.59	0.20	0.001	-0.29
Median	0.999	0.999	-0.57	0.20	0.0002	-0.22
Min	0.21	0.991	-1.42	0.00	0.000	-1.04
Max	0.999	0.999	-0.01	0.52	0.005	0.00

Table OA.13: Below-cost pricing on the entry and no-entry branches of T , across 1521 Pontryagin solutions on the grid $(k, \mu) \in [0.10, 2.00]^2$ at $s = 1$. Frac. types: proportion of $\theta \in \Theta$ with $T'(\theta) < c_0$. Frac. support: proportion of the branch's Q -extent where $T'(Q) < c_0$. Worst: $\min_{\theta}(T'(\theta) - c_0)$ on the branch.

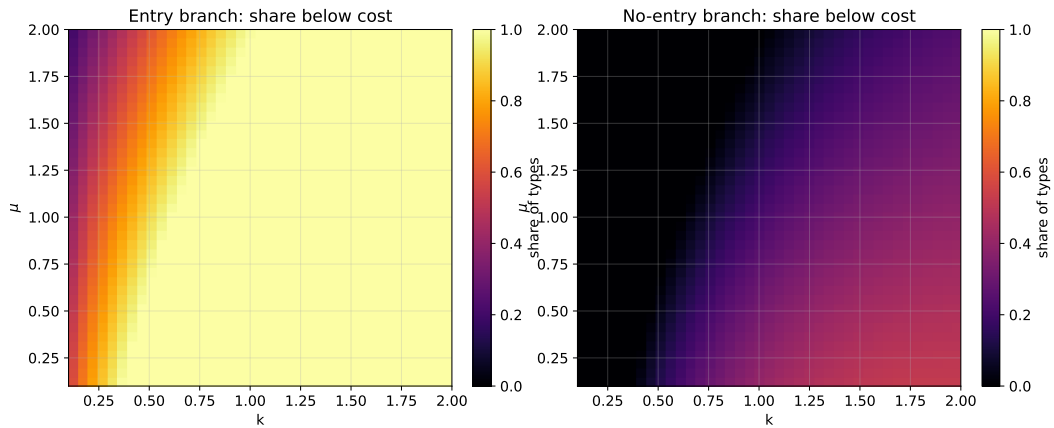


Figure OA.11: Share of buyers facing below-cost marginal pricing ($T'(Q) < c_0$) on each branch of the non-conditional tariff, against k and μ (Pontryagin, $s = 1$). Left: entry branch; right: no-entry branch; common color scale.

Figure OA.11 maps this over the grid. On the entry branch below-cost pricing is essentially universal, falling off only in the corner of small k and large μ where the screening problem is nearly degenerate. On the no-entry branch the share is zero over a large region and rises only as k increases and μ decreases, reaching at most about one half. The contrast measures the asymmetric propagation of the kink: below-cost pricing originates on the entry branch and reaches the no-entry branch only when the kink is pronounced and entry is likely.

Below-cost pricing accommodates entry. The marginal below-cost pricing documented above is not, in our setting, a predatory or exclusionary device; it is the instrument through which the incumbent accommodates entry under a non-conditional tariff. Recall the net gain to I when entry occurs,

$$(OA.7.1) \quad \Gamma(\theta) = c_0 k + \left[S(\theta, q) \right]_{Q_0(\theta)}^{Q_1(\theta)} - \omega(\theta),$$

the bracketed term weighting $G(\omega(\theta))$ in the incumbent's payoff (4.1). A rational incumbent raises the rebate ω (and hence the likelihood of entry) when $\Gamma > 0$, and lowers it where $\Gamma < 0$. The sign of Γ separates an exploitative posture, in which entry is encouraged and its gains captured, from an exclusionary one, in which entry is costly and the incumbent works to prevent it. Proposition 3 ties this sign to the shape of the tariff: whenever $T'(Q) \geq c_0$ everywhere, $\Gamma(\theta) \leq 0$ for all θ ; conversely, the incumbent can obtain positive entry gains only by pricing some units below cost. The below-cost region is thus the marginal counterpart of the rebate-based extraction that an MS contract performs through the level of ω alone, leaving both conditional schedules above cost. Where the non-conditional tariff remains above cost (on the no-entry branch, serving high-demand buyers), it cannot extract entry gains and is exclusionary instead, consistent with the rise of $\omega(\theta)$ at low consumption and its decline at high consumption. The antitrust reading is the reverse of the usual one: a marginal price below cost, which an effects-based screen would read as exclusionary, is here the sign of a contract that accommodates the rival in order to extract part of its efficiency gains.

A consequence of this allocation-based reading concerns the exclusion of low-demand types that the MS contract imposes at large k , where $Q_0(\theta) = 0$ on a left sub-interval. Read on its form (a refusal to serve a fringe of buyers), this resembles vertical foreclosure; read through Γ , the diagnosis reverses. The force driving Q_0 to zero is the incumbent's incentive to widen the wedge $Q_1 - Q_0$ and capture more incremental surplus through ω when entry occurs; the no-entry option becomes so unattractive that the buyer–entrant coalition is pushed toward triggering entry, and $\Gamma(\theta) > 0$ for these types as well. Exclusion here makes entry more, not less, likely: the contract works with the entrant, not against it.

Comparison of antitrust tests. Figure OA.12 summarises the three tests. The tests are nested (mBPC implies EP, which implies AEC, a chain established formally in the main text), so each configuration falls into exactly one ordered category, coloured by the strictest test it passes. All tests are evaluated on the same types Θ_{valid} (those for which $T(Q_0) - T(Q_0 - k)$ is defined); this share is close to one over most of the grid and falls to about one half only in the small- k , large- μ corner. The categories form nested bands ordered in k : the tariff passes even the marginal test in the upper-left corner, then passes only EP, only AEC, and finally fails AEC once k exceeds roughly 0.7. Across the grid the tariff fails AEC for about 73% of configurations, passes AEC only for 12%, EP but not mBPC for 10%, and the marginal test for 5%, with EP intermediate between the marginal and as-efficient-competitor criteria.⁶³

Antitrust tests: the special case of AUDs. The all-unit discount tariff illustrates the gap between a form-based test and an effects-based reading. We focus on the purely linear AUD, where the price cliff at Q^b is the only feature that separates the tariff from its lower convex envelope, so its effect on the

⁶³These shares are computed on Θ_{valid} , the common domain required to compare the three nested tests; they are not directly comparable to as-efficient-competitor shares evaluated on the full type space.

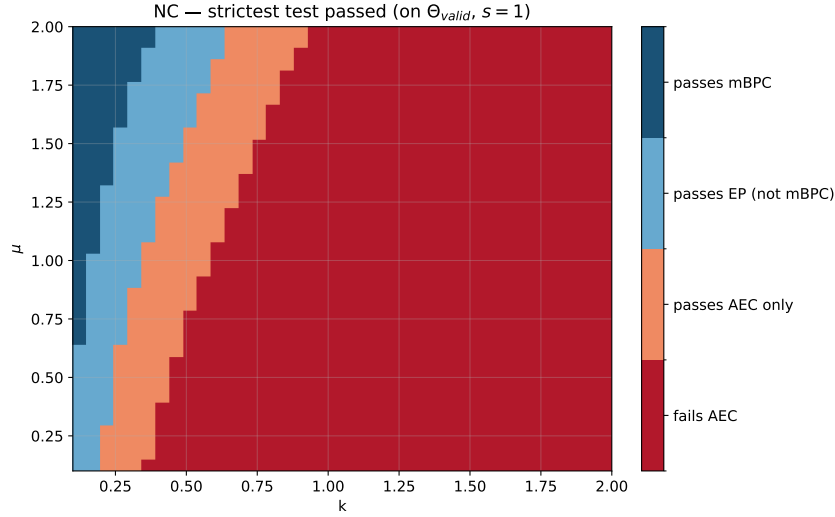


Figure OA.12: Strictest test passed by the non-conditional tariff, against k and μ (Pontryagin, $s = 1$). The three tests are evaluated on the common set of types Θ_{valid} for which $Q_0(\theta) - k$ lies in the support of the tariff, so that the EP test is well defined.

tests can be isolated. We compare the EP test computed on the raw schedule T_{AUD} with the same test computed on the envelope $\bar{T}_{\text{AUD}}$, which induces the same allocation, the same payments and hence the same as-efficient-competitor margin.⁶⁴ Any difference between the two verdicts is therefore attributable to the cliff alone, since it is the only object on which the two tariffs differ, and only at the out-of-equilibrium quantity $Q_0 - k$ that the test evaluates. Table OA.14 reports the two tests on the grid, at $s = 1$, on the domain where $Q_0 - k$ is an admissible quantity.

	Raw schedule	Convex envelope
Fails AEC	0.0%	—
Fails EP	35.6%	0.0%
AEC passes & raw EP fails	542 configurations (35.6%)	
of which the envelope passes	100%	

Table OA.14: AEC and EP tests on the raw AUD schedule versus on its lower convex envelope, AUD-L-pur, $s = 1$, on the grid where $Q_0 - k$ is admissible. The dash indicates that the AEC verdict is identical on both schedules by construction: the envelope leaves allocation and payments unchanged.

The purely linear AUD never fails the AEC test: it over-supplies entry, so the rebate always exceeds the entrant's entry cost. Yet the EP test applied to the raw schedule fails on 35.6% of the grid. These two verdicts are contradictory under the nesting of the tests, and the contradiction is resolved by the envelope: on the same configurations the EP test computed on $\bar{T}_{\text{AUD}}$ never fails, and it coincides with the AEC verdict everywhere. The cliff alone generates the failing verdict. Because the buyer makes the same choices under T_{AUD} and under $\bar{T}_{\text{AUD}}$, that reading rests on a quantity the buyer never purchases: it is an artefact of the form of the rebate, not of its effects. A test that scores the raw cliff returns an exclusionary

⁶⁴On a tariff with an unvisited tier, the price of that tier is not uniquely pinned down: any value above a threshold supports the same equilibrium, and the solver returns one such value. This non-identification is immaterial here, as it concerns out-of-equilibrium quantities only: the allocation, the payments, the profit and the as-efficient-competitor margin are unchanged, and the test on the raw schedule evaluates T_{AUD} at $Q_0 - k$, a quantity that lies on a visited tier in the pure case.

verdict that the effects-equivalent envelope does not.⁶⁵

OA.7.2. Welfare analysis

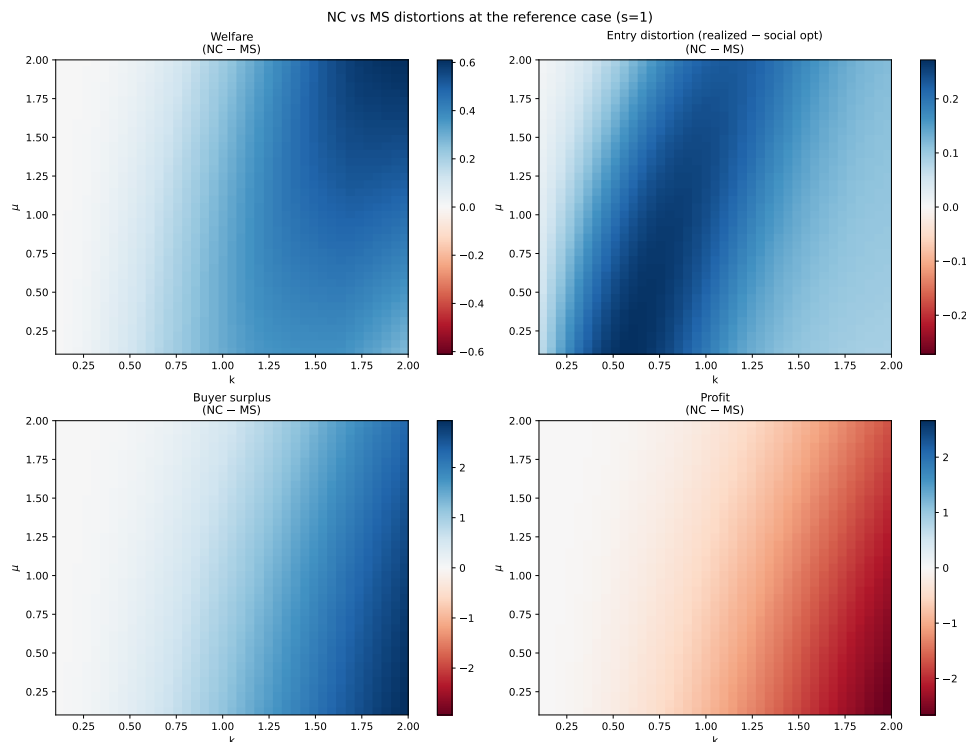


Figure OA.13: Distortions of the two contracts at $s = 1$, as the difference $NC - MS$ over (k, μ) . Diverging scale centered at zero (blue: NC higher; red: MS higher). Panels: total welfare, entry distortion (realized minus socially optimal entry probability), buyer surplus, incumbent profit.

Figure OA.13 compares the contracts at $s = 1$. The four panels point in the same direction: the MS contract extracts more profit than the non-conditional one (red, increasingly so for large k) at the buyer's expense (the buyer-surplus panel is the mirror image), and it forecloses entry more heavily (the entry-distortion panel is positive throughout). The welfare panel is also positive: the NC contract delivers strictly higher welfare at $s = 1$, so the extra foreclosure is a social loss, not a pure transfer. The welfare panel follows the entry-distortion panel in space, which indicates that the welfare cost is driven by foreclosure rather than by within-contract quantity distortions. The MS contract thus combines rent extraction and entry foreclosure into a welfare loss that grows with the entrant's capacity k .

Table OA.15 puts magnitudes on the comparison at the reference case $s = 1$. The welfare advantage of the NC contract is the small net of two large transfers of opposite sign. Across the grid the buyer gains a median of 22.6% of first-best welfare under the NC contract, the incumbent loses a median of 14.9%, and the welfare difference $\Delta \mathcal{W} = \Delta \mathcal{U} + \Delta \mathcal{V}$ is the residual, a median of 6.5%. Both transfers grow with the entrant's capacity: at $k \in (1.5, 2]$ the buyer gains 35% of first-best and the incumbent loses 28%, while the net $\Delta \mathcal{W}$ rises and then settles near 8%. The difference between the two contracts is thus largely distributive, a matter of who captures the surplus rather than how much of it survives.

The ranking is strict at the reference dispersion but not unconditionally. Table OA.16 follows it across the cost dispersion s : $\mathcal{W}_{NC} > \mathcal{W}_{MS}$ on every couple at $s = 1$ and on 98% at $s = 0.5$, but on 69% at

⁶⁵Aggregating the test margin in expectation over types rather than by the strict criterion gives 32.3% of configurations failing the raw EP test, with the same qualitative structure: the envelope clears all of them.

$s = 0.1$, where the order reverses on high- k , low- μ couples. The reversal does not reflect a new mechanism: the welfare gap remains the small residual of two large opposing transfers, and at low dispersion it is small enough for its sign to flip on part of the grid. As s falls the entry cost becomes nearly deterministic and both contracts obtain almost the efficient entry, so the entry channel that gives the NC contract its advantage is muted; what remains is the quantity distortion of its minimum-purchase plateau, largest at high k , and this tips the balance to the MS contract. The advantage of the NC contract is therefore a foreclosure-margin advantage, present wherever entry is uncertain enough for that margin to matter.

Panel A. Share of first-best welfare (%)

k	n	$\mathcal{W}_{\text{NC}} > \mathcal{W}_{\text{MS}}$	$\Delta\mathcal{W}$	$\Delta\mathcal{U}$	$\Delta\mathcal{V}$
[0.10, 0.50]	351	100%	+0.9 [+0.3, +1.9]	+2.2 [+0.8, +4.7]	−1.3 [−2.8, −0.5]
(0.50, 1.00]	390	100%	+5.6 [+4.3, +6.6]	+14.8 [+10.8, +19.0]	−9.1 [−12.4, −6.4]
(1.00, 1.50]	390	100%	+8.1 [+7.6, +8.7]	+28.1 [+24.8, +31.0]	−19.6 [−22.9, −16.2]
(1.50, 2.00]	390	100%	+7.6 [+6.2, +8.9]	+35.3 [+33.8, +36.5]	−27.8 [−30.3, −25.0]
All	1521	100%	+6.5 [+3.4, +8.0]	+22.6 [+8.4, +32.8]	−14.9 [−24.2, −5.0]

Panel B. Model units

k	$\Delta\mathcal{W}$	$\Delta\mathcal{U}$	$\Delta\mathcal{V}$
[0.10, 0.50]	+0.022 [+0.007, +0.047]	+0.053 [+0.018, +0.118]	−0.032 [−0.070, −0.011]
(0.50, 1.00]	+0.185 [+0.131, +0.241]	+0.489 [+0.330, +0.690]	−0.302 [−0.451, −0.195]
(1.00, 1.50]	+0.388 [+0.348, +0.432]	+1.346 [+1.087, +1.613]	−0.932 [−1.186, −0.714]
(1.50, 2.00]	+0.489 [+0.416, +0.536]	+2.298 [+2.054, +2.535]	−1.802 [−2.095, −1.529]
All	+0.306 [+0.096, +0.430]	+0.921 [+0.239, +1.854]	−0.606 [−1.374, −0.141]

Table OA.15: Welfare gap $\mathcal{W}_{\text{NC}} - \mathcal{W}_{\text{MS}}$ and its decomposition $\Delta\mathcal{W} = \Delta\mathcal{U} + \Delta\mathcal{V}$ at the reference dispersion $s = 1$, by entrant capacity k . Median [interquartile range] over admissible (k, μ) couples; $\Delta\mathcal{U}$ and $\Delta\mathcal{V}$ are the buyer-surplus and incumbent-profit differences (non-conditional minus market-share). Panel A normalises by first-best welfare; Panel B is in model units. $\mathcal{W}_{\text{NC}} > \mathcal{W}_{\text{MS}}$: share of couples on which the NC contract dominates.

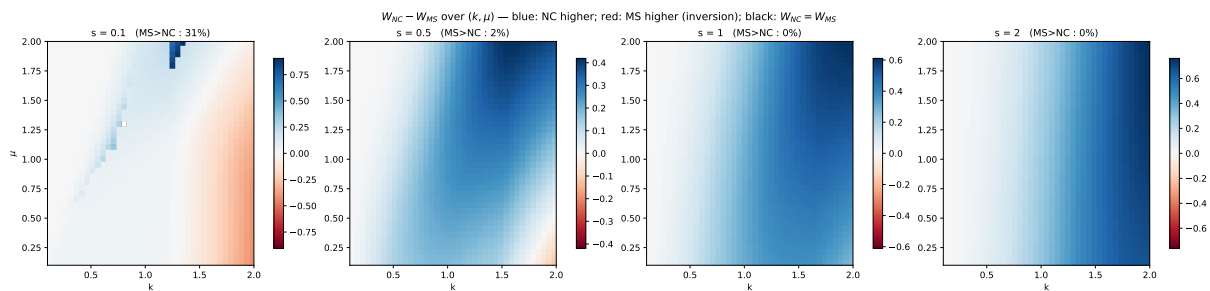


Figure OA.14: Welfare gap $\mathcal{W}_{\text{NC}} - \mathcal{W}_{\text{MS}}$ over the (k, μ) grid, by cost dispersion s (diverging scale centred at zero; blue: the NC contract delivers higher welfare, red: the MS contract does). The inversion region (red) is confined to high capacity and low μ at $s = 0.1$ and vanishes as s rises, disappearing entirely by $s = 1$.

OA.7.3. What does the AEC test track?

The myopic benchmark. Throughout, the strategic reaction is measured against a myopic incumbent that ignores entry and posts its no-entry monopoly tariff T_m (the Maskin–Riley schedule implementing

Panel A. Share of first-best welfare (%)

s	n	$\mathcal{W}_{\text{NC}} > \mathcal{W}_{\text{MS}}$	$\Delta\mathcal{W}$	$\Delta\mathcal{U}$	$\Delta\mathcal{V}$
0.1	1520	69%	+0.4 [−0.7, +1.5]	+24.0 [+6.5, +30.3]	−22.5 [−31.7, −2.1]
0.5	1521	98%	+4.0 [+1.7, +5.5]	+24.0 [+8.0, +32.4]	−17.7 [−27.8, −4.7]
1	1521	100%	+6.5 [+3.4, +8.0]	+22.6 [+8.4, +32.8]	−14.9 [−24.2, −5.0]
2	1521	100%	+7.6 [+2.9, +10.4]	+19.4 [+7.0, +30.5]	−11.8 [−19.7, −4.1]
All	6083	92%	+3.9 [+0.9, +7.1]	+22.6 [+7.5, +31.7]	−15.7 [−25.5, −4.3]

Panel B. Model units

s	$\Delta\mathcal{W}$	$\Delta\mathcal{U}$	$\Delta\mathcal{V}$
0.1	+0.017 [−0.043, +0.047]	+1.054 [+0.176, +1.791]	−0.969 [−1.865, −0.064]
0.5	+0.181 [+0.063, +0.260]	+1.019 [+0.225, +1.929]	−0.745 [−1.646, −0.132]
1	+0.306 [+0.096, +0.430]	+0.921 [+0.239, +1.854]	−0.606 [−1.374, −0.141]
2	+0.291 [+0.079, +0.561]	+0.742 [+0.194, +1.584]	−0.452 [−1.021, −0.115]
All	+0.146 [+0.029, +0.339]	+0.924 [+0.210, +1.793]	−0.648 [−1.453, −0.119]

Table OA.16: Welfare gap $\mathcal{W}_{\text{NC}} - \mathcal{W}_{\text{MS}}$ and its decomposition across the cost-dispersion parameter s , pooled over capacity. Median [interquartile range] over admissible couples; conventions as in Table OA.15. The non-conditional ranking is strict from $s = 0.5$ upward and reverses on a minority of high-capacity couples only at the lowest dispersion.

$Q_m(\theta) = \max\{0, 2\theta - \bar{\theta} - c_0\}$, taken in its closed quadratic form on all $q \geq 0$, with a strictly positive access fee $T_m(0) = T_0 > 0$. The constant is pinned by the participation constraint of the marginal served type $\theta^* = \max\{\underline{\theta}, (\bar{\theta} + c_0)/2\}$, the lowest type the schedule serves; on the canonical grid ($k \leq 2$) this is $\underline{\theta}$, the two coinciding exactly at the essentiality boundary $k = 2$. The buyer remains bound by T_m when entry occurs: declining I altogether yields zero and forecloses the entrant, whereas staying with I and buying $q = 0$ still pays T_0 . With k units secured from the entrant, the buyer purchases from I what a type $\theta - k$ would buy absent entry, so total consumption is $Q_1(\theta) = Q_m(\theta - k) + k = \max\{k, Q_m(\theta) - k\}$ and the induced rebate is $\omega_m(\theta) = c_0 k + (\bar{\theta} - \theta)k + \frac{1}{2}k^2$ at interior types; the lowest types reach the corner $Q_1 = k$ (nothing bought from I) when $Q_m(\underline{\theta}) < 2k$. Profit, buyer surplus and welfare under the myopic tariff are computed from this allocation on the same basis as the optimal contracts. Two checks validate the construction: as $k \rightarrow 0$ the equilibrium and myopic outcomes coincide (under the NC contract, welfare, profit and entry agree to three digits at $k = 0.1$), and the strategic reaction $\mathcal{V}_{\text{real}} - \mathcal{V}_{\text{myopic}}$ is non-negative, as revealed preference requires, since the myopic tariff remains available to the incumbent.

The strategic reaction by contract. Table OA.17 reports, for each contract and each cost dispersion s , the median strategic reaction in model units and as a share of first-best welfare (common to all contracts at each couple), its partial correlation with the welfare gap $\Delta\mathcal{W}$, and the incidence and size of welfare losses relative to the myopic benchmark. Under the NC contract the reaction is welfare-productive: $\Delta\mathcal{W} > 0$ on 99 to 100% of couples at every s , and the partial correlation between the reaction and the welfare gap ranges from +0.47 to +0.77, so the configurations where the incumbent gains most from adapting are also those where welfare rises most. The reaction is localized: its grid-wide median at $s = 1$ is 1.52 (35% of first-best welfare), but it concentrates on the under-entry configurations, where the median ranges from 1.91 to 1.94 across s (36 to 39% of first-best), against 0.04 to 0.15 (1.5 to 5.6%) under over-entry; Table OA.19 reports the corresponding level decomposition by regime.

The MS contract reacts more strongly and less productively. Its median reaction exceeds the non-conditional one at every s (2.14 against 1.52 at $s = 1$, or 53 against 35% of first-best), but the partial correlation with the welfare gap drops to +0.10 to +0.30: the additional extraction is largely absorbed by buyer-surplus losses rather than translated into welfare gains. The adaptation is welfare-destructive on 19.7% of couples at $s = 1$ and 20.0% at $s = 2$, concentrated at extreme values of the entrant's capacity, with a median shortfall of 0.030 to 0.043 in model units (0.5 to 0.7% of first-best); at low dispersion the share is smaller and the shortfall negligible. The MS contract under-enters on the whole grid, so the under-/over-entry split is degenerate for it; the by-contract comparison of Table OA.17 is therefore the counterpart, across contracts, of the within-contract entry-regime decomposition of Table OA.19.

Contract	s	n	Median reaction		Partial corr.	$\Delta\mathcal{W} < 0$ (%)	Median shortfall	
			model	% FB			model	% FB
NC	0.1	1521	+1.66	+34.3	+0.597	0.8	-0.000	-0.00
NC	0.5	1521	+1.61	+34.7	+0.470	0.5	-0.000	-0.01
NC	1	1521	+1.52	+34.8	+0.615	0.0	—	—
NC	2	1521	+1.20	+31.4	+0.771	0.0	—	—
MS	0.1	1520	+2.60	+60.8	+0.255	11.9	-0.000	-0.00
MS	0.5	1521	+2.35	+55.9	+0.095	6.2	-0.001	-0.03
MS	1	1521	+2.14	+52.5	+0.110	19.7	-0.030	-0.46
MS	2	1521	+1.65	+43.2	+0.297	20.0	-0.043	-0.70

Table OA.17: Strategic reaction $\mathcal{V}_{\text{real}} - \mathcal{V}_{\text{myopic}}$ against the myopic benchmark, by contract and cost dispersion s (canonical grid; the market-share solver fails to converge on one couple at $s = 0.1$). Median reaction in model units and in percent of first-best welfare, which is common to all contracts at each couple; partial correlation of the reaction with the welfare gap $\Delta\mathcal{W}$, controlling for k and k^2 ; share of couples with $\Delta\mathcal{W} < 0$, and the median shortfall among them (absent under the NC contract at $s \geq 1$, negligible at low dispersion).

We now ask whether the AEC test is associated with the configurations in which the incumbent reacts most strategically, and with the resulting welfare loss. We measure association by the partial correlation of the continuous AEC failure fraction with each real-minus-myopic gap, controlling for k (regression on k and k^2), since both the test and the strategic reaction rise mechanically with k . The analysis is confined to the NC contract: under the MS contract the AEC test fails for every type, so its failure fraction is degenerate and cannot discriminate.

The as-efficient-competitor test passes when $\omega(\theta) \geq c_0 k$ for every θ , while entry by type θ occurs whenever the entrant's cost falls below $\omega(\theta)$, and entry is socially efficient whenever that cost falls below $c_0 k$. When the test passes, the realized entry threshold $\omega(\theta)$ lies above the efficient threshold $c_0 k$ for every type, so realized entry is more frequent than is efficient: the contract over-enters. The test can therefore pass only under over-entry; under under-entry it fails necessarily. The grid bears this out: across all values of s , no under-entry configuration passes the test, and every passing configuration over-enters.

The cost of tolerated over-entry. Under the bargaining assumption of the main text (the buyer holds all bargaining power vis-à-vis the entrant and captures all gains from entry), for every type with $\omega(\theta) > c_0 k$ the discipline rent $(\omega(\theta) - c_0 k) [G(\omega(\theta)) - G(c_0 k)]$ splits exactly into a transfer to the buyer, $\int_{c_0 k}^{\omega(\theta)} (\omega(\theta) - c) dG(c)$, and a duplication loss, $\int_{c_0 k}^{\omega(\theta)} (c - c_0 k) dG(c)$. Both are negligible on the

canonical grid at $s = 1$: the median duplication loss is 0.001% of first-best welfare on the AEC-passing couples, its maximum over the entire grid is 0.08%, and its grid aggregate is 0.007%. Over-entry, where it occurs, is marginal in value: $\omega(\theta)$ exceeds $c_0 k$ only slightly, and the test passes at almost no social cost. Figure OA.15 maps the duplication loss against its under-entry counterpart, the foreclosure loss from missed efficient entry, $\int_{\omega(\theta)}^{c_0 k} (c_0 k - c) dG(c)$, on a common color scale: the duplication panel is nearly blank, including inside the AEC-pass region (maxima 0.08 against 0.84% of first-best; grid medians 0.0001 against 0.0083 in model units). The discipline the entrant exerts on the incumbent operates through the threat of entry (the strategic reaction of Table OA.17, of the order of 35% of first-best), not through realized inefficient entry.

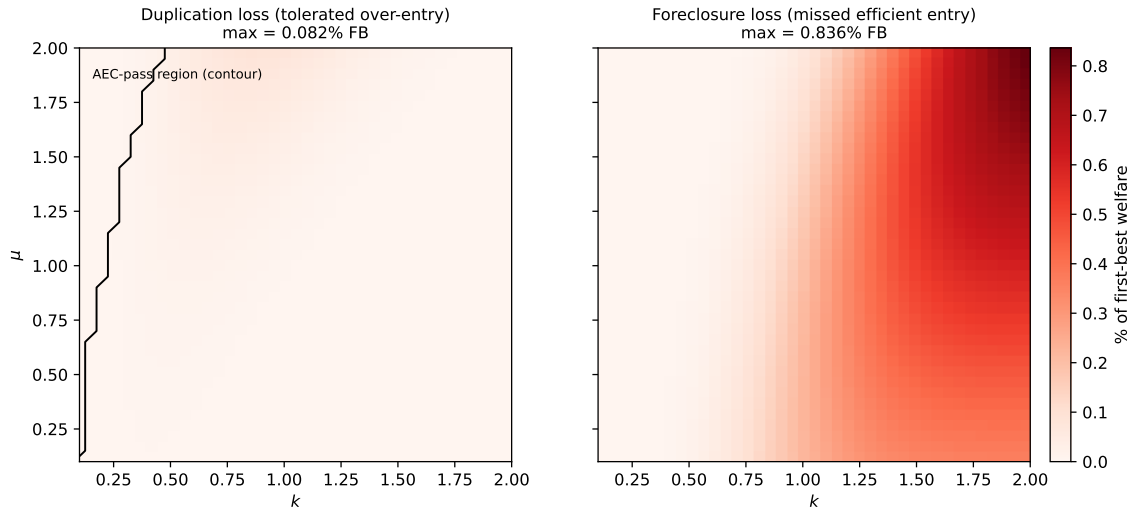


Figure OA.15: Duplication loss from tolerated over-entry (left) and foreclosure loss from missed efficient entry (right), in percent of first-best welfare, on a common color scale; NC contract, $s = 1$. The black contour on the left panel delimits the AEC-pass region. The duplication loss is negligible everywhere, including where the test passes; the foreclosure loss is an order of magnitude larger and grows with the entrant's capacity.

At the aggregate level the AEC failure fraction is strongly associated with the loss of buyer surplus (-0.58) and with the strategic reaction ($+0.42$), but almost uncorrelated with the welfare loss (-0.05 , all partial correlations controlling for k and k^2 , $s = 1$). The near-zero welfare correlation is a cancellation, not indifference: since $\Delta \mathcal{W} = \Delta \mathcal{U} + (\mathcal{V}_{\text{real}} - \mathcal{V}_{\text{myopic}})$, the negative association with buyer surplus and the positive association with the strategic reaction offset each other. The test responds to the redistribution from buyer to incumbent, not to the efficiency loss. Its association with the entry distortion is likewise weak in the aggregate (-0.18).

The aggregate cancellation hides a sign reversal that the partition between under- and over-entry brings out. Classifying each couple by the sign of realized minus socially optimal entry, the NC contract under-enters in 71% of cases (foreclosure) and over-enters in the remaining 29%, the latter concentrated in the small- k , high- μ region; the MS contract under-enters everywhere, as theory predicts. Controlling for k , the AEC failure fraction correlates with the welfare gap with opposite signs across the two regimes: -0.60 under under-entry but $+0.74$ under over-entry. Its association with the buyer-surplus gap, by contrast, stays negative throughout (-0.87 and -0.66). The reversal is robust to cost dispersion: the welfare correlation is negative under under-entry and positive under over-entry at every value of s (Table OA.18). The mechanism

is the asymmetry between the two distortions: under-entry forecloses an efficient entrant, hurting buyer and welfare alike, so the buyer-oriented test is incidentally aligned with welfare; over-entry lets an inefficient entrant in, which lowers welfare but benefits the buyer, because it disciplines an incumbent pricing above marginal cost. The AEC test, based on buyer surplus, therefore moves opposite to welfare where the two diverge.

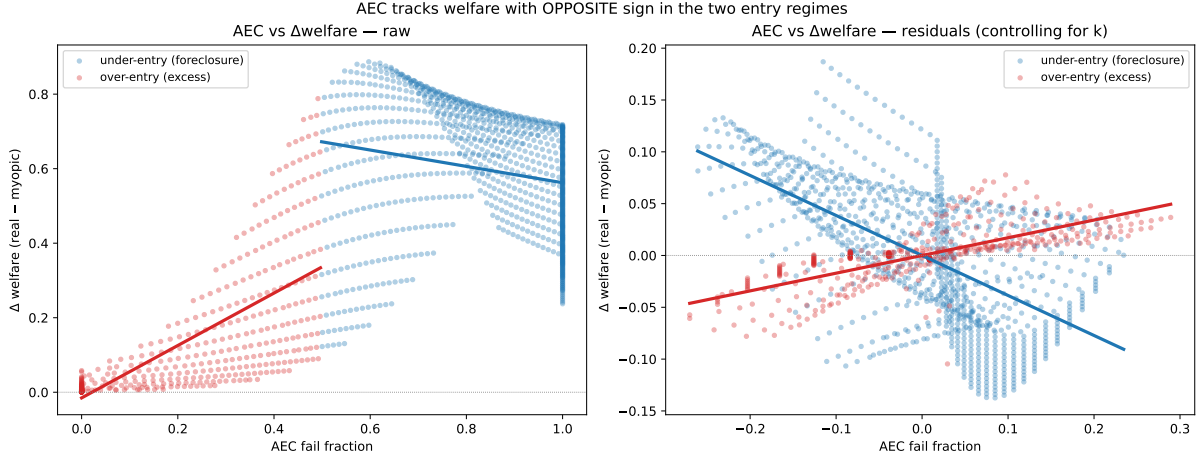


Figure OA.16: AEC failure fraction vs. welfare gap, residuals after controlling for k , by entry regime.

s	AEC $\times \Delta \mathcal{W}$		AEC $\times \Delta \mathcal{U}$	
	under-entry	over-entry	under-entry	over-entry
0.1	-0.489	+0.253	-0.596	-0.748
0.5	-0.513	+0.712	-0.632	-0.612
1.0	-0.600	+0.735	-0.870	-0.664
2.0	-0.535	+0.631	-0.979	-0.599

Table OA.18: Partial correlation of the AEC failure fraction with the welfare gap and the buyer-surplus gap, by entry regime, controlling for k and k^2 (NC contract). The welfare correlation reverses sign across regimes at every s ; the buyer-surplus correlation stays negative throughout.

The strategic reaction shows the same pattern. Partitioning the NC contract by entry regime, the reaction $\mathcal{V}_{\text{real}} - \mathcal{V}_{\text{myopic}}$ is concentrated almost entirely on the under-entry configurations: its median is an order of magnitude larger there than under over-entry (Table OA.19). The incumbent departs from the myopic tariff to foreclose; where it over-enters, it barely departs from it. The welfare and buyer-surplus gaps decompose the same way: under under-entry the buyer loses heavily while welfare rises, while under over-entry both gaps are small. This is the mechanism behind the sign reversal seen above, read on the levels rather than the correlations: the test tracks the buyer, the buyer and welfare diverge under under-entry, and the strategic reaction is concentrated there.

OA.7.4. The AEC-test across types and cost environments

Under private information the as-efficient-competitor margin is a profile across types, $\omega(\theta) - c_0 k$, rather than a single number, so its verdict depends on the type to which the test is applied. Table OA.20 reports, for the NC contract at $s = 1$, the fraction of admissible (k, μ) couples that fail the test within each decile of θ ; failure declines monotonically from 90.6% on the lowest decile to 26.1% on the highest.

s	Regime	Strategic reaction	$\Delta\mathcal{W}$	$\Delta\mathcal{U}$
0.1	under-entry	+1.931	+0.490	-1.274
	over-entry	+0.085	+0.071	-0.006
0.5	under-entry	+1.943	+0.561	-1.276
	over-entry	+0.035	+0.030	-0.004
1.0	under-entry	+1.939	+0.633	-1.211
	over-entry	+0.068	+0.055	-0.012
2.0	under-entry	+1.906	+0.751	-1.090
	over-entry	+0.149	+0.119	-0.030

Table OA.19: Strategic reaction $\mathcal{V}_{\text{real}} - \mathcal{V}_{\text{myopic}}$, welfare gap $\Delta\mathcal{W}$ and buyer-surplus gap $\Delta\mathcal{U}$ (real minus myopic), median over each entry regime, NC contract. Across all s the strategic reaction under under-entry is an order of magnitude above its over-entry counterpart; the buyer-surplus gap is strongly negative under under-entry while the welfare gap is positive.

The test is hardest on the lower deciles and slack on the upper ones, and the type selected is therefore not innocuous: a screen run on the lowest decile condemns the conduct more than three times as often as one run on the highest. The failure region, when non-empty, almost always includes the bottom of the support, even though the binding type itself is typically interior, a point we return to below.

Because no type is canonical, an aggregate reading is natural, $\mathbb{E}_\theta[\omega] \geq c_0 k$. Table OA.21 reports pass and fail rates under the strict ($\forall\theta$) and expectation criteria as the dispersion parameter s of the entry-cost distribution varies. The expectation criterion loosens as s rises, its failure rate falling from 76.9% at $s = 0.1$ to 56.2% at $s = 2$, since dispersing the cost distribution shifts mass towards higher realizations and makes the average admitted entry appear efficient; it nonetheless condemns a majority of couples at every s . Figure OA.17 draws the frontier $\mathbb{E}_\theta[\omega] - c_0 k = 0$ in the (k, μ) plane: the pass region, at low capacity k , widens with s but never covers the grid. The strict criterion remains high, between 82 and 92%, and does not loosen as s rises.

This contrast rests on the monotonicity of ω . Under the MS contract $\omega(\theta)$ is increasing in θ , so the margin is monotone and the binding type is always the lowest, $\underline{\theta}$: the worst-customer test and the lowest-decile test coincide, and the screen is hardest at $\underline{\theta}$ throughout. Under the NC contract ω is not monotone, and the binding type (the minimizer of $\omega(\theta) - c_0 k$) is interior on 51 to 74% of admissible couples rather than pinned to a boundary. Its location moves with the dispersion of the entry cost: the median position of $\arg \min_\theta \omega(\theta)$ within the support slides from the middle (0.46 at $s = 0.1$) towards the lower types (0.03 at $s = 2$) as s rises. The identity of the worst-served type is therefore endogenous to the cost environment, which is why the strict criterion is non-monotone in s and why no single type can be designated a priori as the one on which to run the test.

θ -decile	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10
AEC pass (%)	9.4	12.4	15.9	20.3	26.0	32.5	40.6	49.9	61.5	73.9
AEC fail (%)	90.6	87.6	84.1	79.7	74.0	67.5	59.4	50.1	38.5	26.1

Table OA.20: NC contract, $s = 1$: share of admissible (k, μ) couples passing/failing the AEC test ($\omega(\theta) \geq c_0 k$) within each decile of θ (D1 lowest, D10 highest); 1521 couples.

s	Strict ($\forall\theta$)		Expectation	
	pass	fail	pass	fail
0.1	17.9%	82.1%	23.1%	76.9%
0.5	11.2%	88.8%	24.9%	75.1%
1.0	8.5%	91.5%	29.1%	70.9%
2.0	9.8%	90.2%	43.8%	56.2%

Table OA.21: NC contract: AEC pass/fail rates under the strict ($\forall\theta$) and expectation criteria, by the dispersion parameter s of the entry-cost distribution; 1521 admissible couples per s .

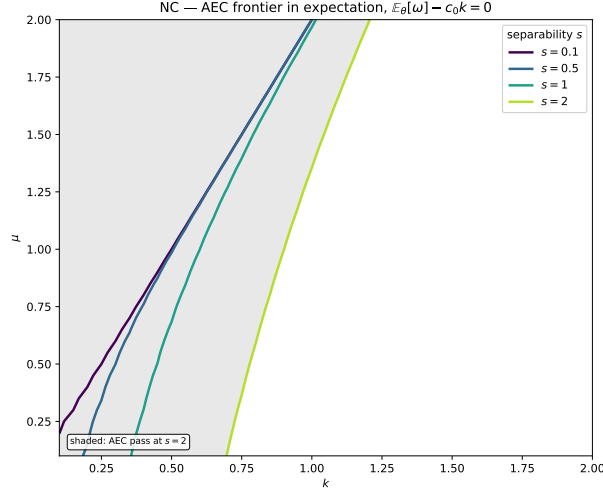


Figure OA.17: NC contract: the AEC frontier in expectation, $\mathbb{E}_\theta[\omega] - c_0k = 0$, in the (k, μ) plane for $s \in \{0.1, 0.5, 1, 2\}$. The pass region (low k) widens as s increases.

Robustness of the diagnostic inversion. The central diagnostic, that the as-efficient-competitor test condemns the NC contract precisely where it raises total welfare relative to the myopic benchmark, was established under the strict criterion at $s = 1$. Table OA.22 shows it survives both the aggregation format and the cost environment. Taking the expectation criterion $\mathbb{E}_\theta[\omega] \geq c_0k$, and across $s \in \{0.1, 0.5, 1, 2\}$, total welfare exceeds the myopic counterfactual on 99 to 100% of admissible couples, while the test fails on 56 to 77% of these welfare-improving couples. The strength of the inversion lies not in the prevalence of $\Delta\mathcal{W} > 0$ but in how often the test condemns conduct that raises welfare. That prevalence is not mechanical: although the strategic reaction $\mathcal{V}_{\text{real}} - \mathcal{V}_{\text{myopic}} \geq 0$ by revealed preference, welfare also nets the buyer-surplus change, $\Delta\mathcal{W} = \Delta\mathcal{U} + (\mathcal{V}_{\text{real}} - \mathcal{V}_{\text{myopic}})$, and $\Delta\mathcal{U}$ is large and negative in the entry-barrier regime; that welfare still rises is the substantive finding, and it can fail: at the essentiality boundary $\Delta\mathcal{U}$ nearly offsets the profit gain and $\Delta\mathcal{W}$ tips negative. What the test does, by contrast, is condemn this welfare-improving conduct on a majority of couples at every s , even under the laxer expectation reading: a structural feature of the test, not an artifact of how the margin is aggregated or of the assumed dispersion of entry costs.

OA.8. $\Delta\theta \neq k$

This appendix relaxes Assumption 2 ($\Delta\theta = k$) and characterizes the optimal NC contract for arbitrary $\Delta\theta$ and k . Section OA.8.1 treats $\Delta\theta < k$, Section OA.8.2 treats $\Delta\theta > k$, and Section OA.8.4 interprets the regime structure and the role of $\Delta\theta = k$.

s	n	$\Delta\mathcal{W} > 0$	AEC fail	AEC fail $\Delta\mathcal{W} > 0$
0.1	1521	99.2%	76.9%	77.5%
0.5	1521	99.5%	75.1%	75.5%
1.0	1521	100.0%	70.9%	70.9%
2.0	1521	100.0%	56.2%	56.2%

Table OA.22: NC contract, robustness of the diagnostic inversion. For each dispersion s of the entry-cost distribution: share of admissible (k, μ) couples with $\Delta\mathcal{W} = \mathcal{W}_{\text{real}} - \mathcal{W}_{\text{myopic}} > 0$; share failing the AEC test in expectation ($\mathbb{E}_\theta[\omega] < c_0 k$); and the latter restricted to welfare-improving couples. 1521 admissible couples per s .

OA.8.1. The case $\Delta\theta < k$

Geometry. The function Q_0 is defined on $\Theta_{-k} \cup \Theta$. When $\Delta\theta < k$, the inequality $\bar{\theta} - k < \underline{\theta}$ makes the two intervals disjoint,

$$\Theta_{-k} \cup \Theta = [\underline{\theta} - k, \bar{\theta} - k] \cup [\underline{\theta}, \bar{\theta}],$$

separated by a gap of length $k - \Delta\theta > 0$. Each real type θ thus appears twice, in two roles that are never adjacent: an exclusive role in Θ and a fictitious split role at $\theta - k \in \Theta_{-k}$.

Differences with the analysis under $\Delta\theta = k$. Three modifications follow from the disconnected support:

- (i) The monotonicity condition (6.4) becomes an inequality between *two distinct* types, $Q_0(\bar{\theta} - k) \leq Q_0(\underline{\theta})$, rather than an equality at a single joining point.
- (ii) The rent $\mathcal{U}_0(\bar{\theta} - k)$, which under $\Delta\theta = k$ coincides with $\mathcal{U}_0(\underline{\theta}) = 0$ through the joining $\bar{\theta} - k = \underline{\theta}$, now sits at an endpoint of the lower branch interior to Θ_{-k} , free of any participation requirement: types in Θ_{-k} earn strictly more than their reservation payoff, since the surplus $\theta k - k^2/2$ from the k units bought from E relaxes their participation. Since a single tariff $T(Q)$ must serve both the no-entry profile Q_0 and the entry profile $\tilde{Q}_0$, the two branches carry distinct co-states λ_0 and $\tilde{\lambda}_0$, with transversality conditions $\lambda_0(\bar{\theta}) = 0$ and $\tilde{\lambda}_0(\underline{\theta}) = 0$ at the free endpoints; when (CG) (defined below) is slack this leaves $\tilde{\lambda}_0(\bar{\theta}) = 0$ as well.
- (iii) No interior point carries a binding state constraint: the participation constraint $\mathcal{U}_0(\underline{\theta}) \geq 0$ acts at the left endpoint of Θ , so λ_0 is continuous on each connected component of $\Theta_{-k} \cup \Theta$.

Optimal control formulation. The optimality conditions below derive from the same Hamiltonian as the $\Delta\theta = k$ analysis, now read on the disconnected support. Writing $\mathcal{U}_0$ and $\tilde{\mathcal{U}}_0$ for the rents on the two branches, with $\dot{\mathcal{U}}_0 = Q_0$ and $\dot{\tilde{\mathcal{U}}}_0 = \tilde{Q}_0$, the incumbent's problem admits the Hamiltonian

$$\mathcal{H}^u = G(\omega) \left(c_0 k + [S(\theta, q)]_{Q_0}^{\tilde{Q}_0 + k} - \omega \right) + S(\theta, Q_0) - \mathcal{U}_0 + \lambda_0 Q_0 + \tilde{\lambda}_0 \tilde{Q}_0,$$

where $\omega = \tilde{\mathcal{U}}_0 + \theta k - \frac{k^2}{2} - \mathcal{U}_0$ and $\lambda_0, \tilde{\lambda}_0$ are the co-states of the two rent equations (we suppress the monotonicity multipliers, inactive on a separating arc).⁶⁶ The single non-conditional tariff couples the two branches through one control: Q_0 enters both the no-entry payoff and, via $\tilde{Q}_0(\theta) = Q_0(\theta - k)$, the

⁶⁶This is the Hamiltonian of the $\Delta\theta = k$ problem derived in the Appendix (cf. (A.44) and following); it is here carried to the extended support $\Theta_{-k} \cup \Theta$.

entry payoff. The dynamics of the two co-states satisfy $\dot{\lambda}_0 + \dot{\tilde{\lambda}}_0 = 1$, so that

$$\lambda_0(\theta) + \tilde{\lambda}_0(\theta) = \theta - \kappa$$

for a constant κ fixed by the transversality conditions. When (CG) is slack the relevant condition is the free endpoint $\tilde{\lambda}_0(\bar{\theta}) = 0$; when it binds, the level identity below makes this co-state active, $\tilde{\lambda}_0(\bar{\theta}) = \gamma$, and pins $\kappa = \bar{\theta} - \gamma$. The wedge γ in the sub-martingale property is thus exactly the multiplier of the bunching constraint.

Optimality under separation: equivalence with MS contracts. Provided the inequality in (i) is slack, the first-order conditions yield

$$Q_0(\theta) = Q_m(\theta) - G(\omega(\theta)) \dot{\omega}(\theta), \quad Q_1(\theta) = Q_m(\theta) + (1 - G(\omega(\theta))) \dot{\omega}(\theta),$$

hence the exact martingale property

$$G(\omega(\theta)) Q_1(\theta) + (1 - G(\omega(\theta))) Q_0(\theta) = Q_m(\theta), \quad \forall \theta \in \Theta.$$

Following similar steps as in the proof of Proposition 1, the rebate ω coincides with equation (5.11) of the market-share analysis, as do the boundary conditions $\dot{\omega}(\bar{\theta}) = 0$ and $\dot{\omega}(\underline{\theta}) = \Delta\theta/(1 - G(\omega(\underline{\theta})))$. Consequently $Q_i^{\text{NC}} = Q_i^{\text{MS}}$ for $i \in \{0, 1\}$, and the optimal non-conditional profit equals the market-share profit. When $Q_0(\Theta)$ and $Q_0(\Theta_{-k})$ are disjoint quantity ranges, the tariff $T(Q)$ can be specified independently on each, with an arbitrary value on the intervening gap that no buyer selects; the incumbent thereby recovers the two degrees of freedom of the market-share scenario.

Cross-gap admissibility condition. Substituting the boundary values $Q_1(\bar{\theta}) = Q_{\text{fb}}(\bar{\theta})$ and $Q_0(\underline{\theta}) = (\underline{\theta} - c_0) - \Delta\theta/(1 - G(\omega(\underline{\theta})))$ into the constraint of (i) gives the condition for the separating allocation to be admissible:

$$(CG) \quad (k - \Delta\theta) (1 - G(\omega(\underline{\theta}))) \geq \Delta\theta.$$

Condition (CG) is endogenous through the value $\omega(\underline{\theta})$. The $\Delta\theta = k$ analysis of Section 6 faces an analogous endogeneity in the binding of its joining condition. Because $1 - G(\omega(\underline{\theta})) \leq 1$ and $k > \Delta\theta$ in the under-entry regime, (CG) can hold only if $k \geq 2\Delta\theta$, that is $\Delta\theta/k \leq 1/2$: for $\Delta\theta/k > 1/2$ the cross-gap constraint binds at every μ and only Case 2 arises. Below this threshold it holds once entry at the lowest type is unlikely enough that $1 - G(\omega(\underline{\theta})) \geq \Delta\theta/(k - \Delta\theta)$, an inequality reached at large μ and meeting the boundary $\Delta\theta/k = 1/2$ only as $G(\omega(\underline{\theta})) \rightarrow 0$. Two cases follow.

Case 1: (CG) holds. The market-share allocation is admissible as a non-conditional one, $\mathcal{V}_I^{\text{NC}} = \mathcal{V}_I^{\text{MS}}$. The cross-gap range of quantities can carry any tariff value that attracts no buyer, leaving the incumbent with two independent degrees of freedom.

Case 2: (CG) fails. When the cross-gap constraint binds, both branches of Q_0 share a common plateau Q^b ,

$$Q_0(\theta) = Q^b \text{ for } \theta \in [\underline{\theta}, \theta_0], \quad Q_0(\theta - k) = Q^b \text{ for } \theta \in [\theta_1, \bar{\theta}],$$

with smooth pasting $Q_0(\theta_0^+) = Q^b$ and $Q_1(\theta_1^-) = Q^b + k$. Since both branches share the same tariff point Q^b , the value $T(Q^b)$ is pinned by the participation of the lowest exclusive type, and tariff consistency forces the rent of the highest fictitious type to satisfy the level identity

$$\mathcal{U}_0(\bar{\theta} - k) = -Q^b(k - \Delta\theta).$$

This identity is the new element when bunching is active: it turns the equality $Q_0(\bar{\theta} - k) = Q_0(\underline{\theta}) = Q^b$ into a constraint on the lower-branch terminal rent. As anticipated in the control formulation above, this activates the co-state, $\tilde{\lambda}_0(\bar{\theta}) = \gamma$ with $\gamma \geq 0$ the multiplier of the level identity (so that $\kappa = \bar{\theta} - \gamma$), and the martingale relation acquires a uniform wedge,

$$G(\omega(\theta)) Q_1(\theta) + (1 - G(\omega(\theta))) Q_0(\theta) = Q_m(\theta) + \gamma,$$

the sub-martingale property of Proposition 5, with $\gamma \in (0, \Delta\theta)$. The unknowns $(Q^b, \theta_0, \theta_1, \gamma)$ are jointly determined by smooth pasting at θ_0 and θ_1 , co-state continuity at the bunching boundaries, and the level identity.

Continuous limit at $\Delta\theta = k^-$ and numerical illustration. As $\Delta\theta \rightarrow k^-$, the level identity gives $\mathcal{U}_0(\bar{\theta} - k) = -Q^b(k - \Delta\theta) \rightarrow 0 = \mathcal{U}_0(\underline{\theta})$, the participation-driven matching condition that holds at $\Delta\theta = k$. The wedge γ rises to its maximal value of Proposition 5, and the features of Section 6 are recovered. The case $\Delta\theta = k$ is thus the right boundary of the bunching regime, attained continuously.

Figure OA.18 illustrates this. Fixing $\mu = 6$, so that entry is rare enough for (CG) to be slack at small $\Delta\theta$, and raising $\Delta\theta$ from 0.5 to $k = 2$, the lower branch on Θ_{-k} and the upper branch on Θ are separated by a gap that closes into a single joining point as $\Delta\theta \rightarrow k$. The annotations track the two quantities that drive the continuity: the averaged wedge γ rises from ≈ 0 in the slack, market-share-equivalent case to its maximum at $\Delta\theta = k$, while the terminal rent $\mathcal{U}_0(\bar{\theta} - k)$ rises from a negative value to exactly zero.

The same transition is visible on the reconstructed tariff $T(Q)$ in Figure OA.19. When (CG) is slack ($\Delta\theta = 0.5$), the tariff splits into two disjoint segments (a with-entry branch and a no-entry branch) separated by a range of quantities that no buyer selects and on which T can take any value; this is the market-share allocation in non-conditional form, with the gap encoding the two independent fixed fees. As $\Delta\theta$ increases and (CG) binds, the gap closes and the two segments join into a single continuous tariff with an inward kink at Q^b , the minimum-purchase quantity of Section 6.

OA.8.2. The case $\Delta\theta > k$

Geometry. When $\Delta\theta > k$, the inequality $\bar{\theta} - k > \underline{\theta}$ makes Θ_{-k} and Θ overlap on $[\underline{\theta}, \bar{\theta} - k]$, of length $\Delta\theta - k > 0$. The support $\Theta_{-k} \cup \Theta = [\underline{\theta} - k, \bar{\theta}]$ is connected, and Q_0 has a dual interpretation on the overlap: at any $\theta \in [\underline{\theta}, \bar{\theta} - k]$, the value $Q_0(\theta)$ is selected both by the exclusive buyer of type θ and by the split buyer of type $\theta + k$, via $\tilde{Q}_0(\theta + k) = Q_0(\theta)$.

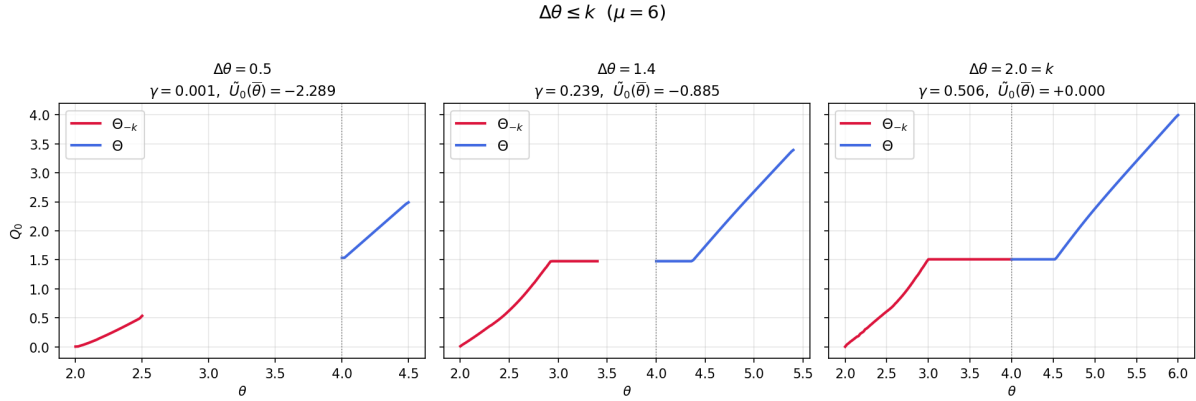


Figure OA.18: Quantity profile Q_0 on the extended support $\Theta_{-k} \cup \Theta$ for $\Delta\theta \leq k$ ($\mu = 6$, $s = 1$). Red: lower branch on Θ_{-k} ; blue: upper branch on Θ ; dotted line at $\bar{\theta}$. As $\Delta\theta \rightarrow k$ the gap closes and the branches join at a single point. Annotations: averaged wedge γ and terminal rent $\mathcal{U}_0(\bar{\theta} - k)$, both reaching their $\Delta\theta = k$ values continuously.

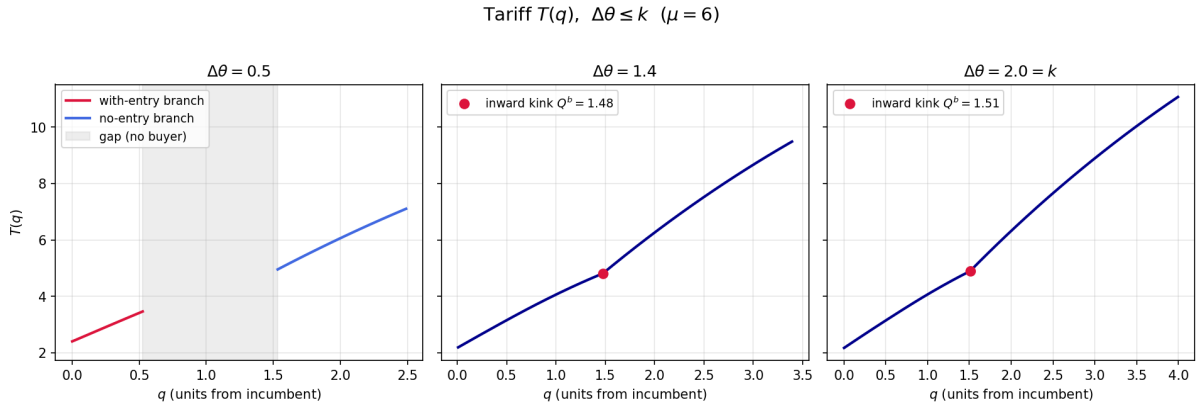


Figure OA.19: Reconstructed tariff $T(Q)$ for $\Delta\theta \leq k$ ($\mu = 6$, $s = 1$). At $\Delta\theta = 0.5$ the with-entry branch (red) and no-entry branch (blue) are disjoint, separated by a gap of quantities selected by no buyer (shaded); the tariff replicates the market-share allocation. As $\Delta\theta \rightarrow k$, the gap closes and the tariff becomes a single continuous schedule with an inward kink at Q^b .

Optimality conditions and interior participation. The problem can be cast as designing a single rent function $\mathcal{U}_0$ on $[\underline{\theta} - k, \bar{\theta}]$ with $\mathcal{U}_0 = Q_0$. Its boundary structure is: the participation constraint $\mathcal{U}_0(\underline{\theta}) \geq 0$, now interior to the support, which induces a discontinuity in λ_0 at $\underline{\theta}$ when binding; and the free-endpoint transversality conditions $\lambda_0(\underline{\theta} - k) = \lambda_0(\bar{\theta}) = 0$. This parallels $\Delta\theta = k$, where participation also acts at the interior point $\underline{\theta} = \bar{\theta} - k$; here that point lies strictly inside the support. On the overlap, the first-order condition for $Q_0(\theta)$ involves the integrand at both θ and $\theta + k$:

$$Q_0(\theta) = (\theta - c_0) + \frac{\lambda_0(\theta)}{1 - G(\omega(\theta)) + G(\omega(\theta + k))}, \quad \theta \in [\underline{\theta}, \bar{\theta} - k].$$

The denominator is the effective probability that $Q_0(\theta)$ is selected: a type- θ buyer faces no entry, or a type- $(\theta + k)$ buyer faces entry. The pure shadow region $[\underline{\theta} - k, \underline{\theta}]$ and the pure real region $(\bar{\theta} - k, \bar{\theta}]$ retain the single-event denominators $G(\omega(\theta + k))$ and $1 - G(\omega(\theta))$.

Obstruction to closed-form characterization. The merged condition couples $\omega(\theta)$ to $\omega(\theta + k)$ over the whole overlap; equivalently, the consistency constraint $\tilde{\mathcal{U}}_0(\theta) = \mathcal{U}_0(\theta - k)$ holds on an interval of positive length and its multiplier enters the co-state dynamics non-locally. The identity $\dot{\lambda}_0 + \dot{\tilde{\lambda}}_0 = 1$ underlying the martingale property fails on the overlap, and the contract is governed by a coupled delay differential equation in $\omega(\theta)$, $\omega(\theta + k)$, $\lambda_0(\theta)$, $\lambda_0(\theta - k)$. No first integral analogous to (5.11) reduces it to an ODE, and standard special cases (uniform or exponential G , specific ratios $\Delta\theta/k$) do not decouple it; numerical resolution is required. As $\Delta\theta \rightarrow k^+$, the overlap collapses to the point $\underline{\theta} = \bar{\theta} - k$, the merged condition reduces to a boundary identity, and Section 6 is recovered.

Figure OA.20 illustrates the overlap regime, again at $\mu = 6$ and starting from $\Delta\theta = k$. As $\Delta\theta$ grows beyond k , the overlap region (shaded) widens from a point. Two features are visible. First, the terminal rent $\mathcal{U}_0(\bar{\theta} - k)$ leaves its value of zero at $\Delta\theta = k$ continuously, confirming the $\Delta\theta \rightarrow k^+$ limit. Second, the upper branch changes slope at the overlap edge, reflecting the merged condition: its denominator switches from $1 - G(\omega(\theta)) + G(\omega(\theta + k))$ inside the overlap to $1 - G(\omega(\theta))$ in the pure real region.

$$\Delta\theta \geq k \quad (\mu = 6)$$

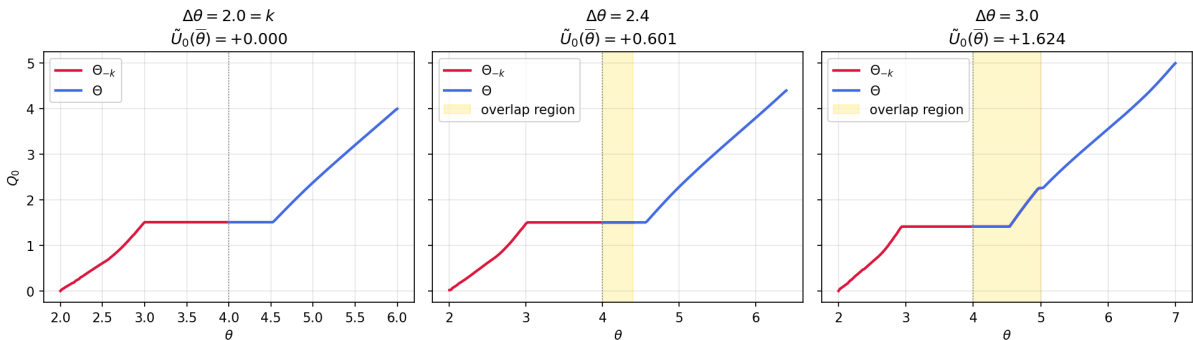


Figure OA.20: Quantity profile Q_0 on the extended support for $\Delta\theta \geq k$ ($\mu = 6$, $s = 1$). Red: lower branch on Θ_{-k} ; blue: upper branch on Θ ; shaded band: overlap region $[\underline{\theta}, \bar{\theta} - k]$, widening with $\Delta\theta$. The terminal rent $\mathcal{U}_0(\bar{\theta} - k)$ leaves zero continuously at $\Delta\theta = k$; the slope change of the upper branch at the overlap edge reflects the regime switch in the first-order condition.

The reconstructed tariff, shown in Figure OA.21, shows that the inward kink at Q^b persists throughout the overlap regime. Increasing $\Delta\theta$ beyond k does not smooth the tariff into a regular schedule; the

minimum-purchase feature of Section 6 survives the loss of the clean branch separation. The dual use of quantities on the overlap thus coexists with the kink rather than replacing it: the main qualitative feature of the non-conditional tariff is not an artifact of the special case $\Delta\theta = k$ but a property of the whole bunching-and-overlap range.

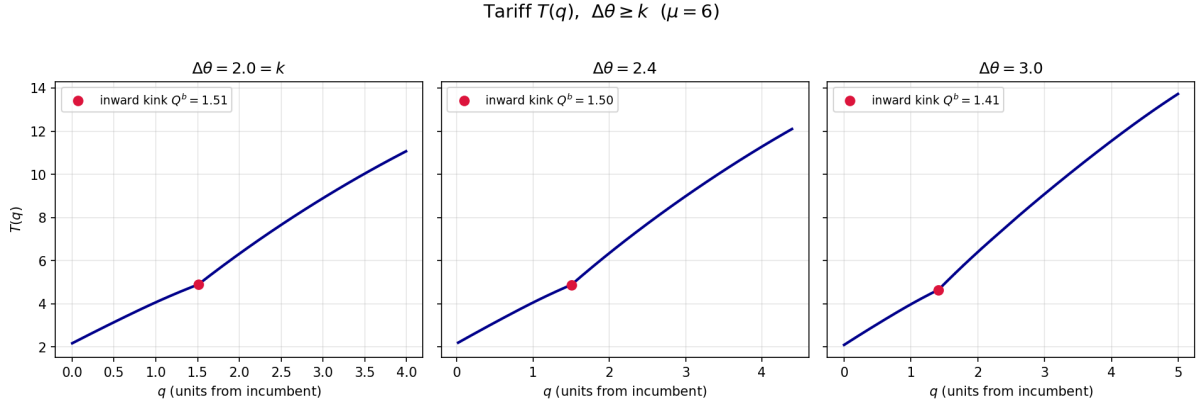


Figure OA.21: Reconstructed tariff $T(Q)$ for $\Delta\theta \geq k$ ($\mu = 6$, $s = 1$). The inward kink at Q^b persists across the overlap regime.

OA.8.3. Numerical robustness of the welfare and antitrust conclusions

The figures above show that the qualitative geometry of the non-conditional tariff (the inward kink at Q^b , read as a minimum-purchase requirement) survives away from $\Delta\theta = k$. We now check that the conclusions of Section 6 are equally robust: the welfare ranking of the optimal contract against the market-share and myopic benchmarks, and the verdicts of the antitrust tests.

Design. We solve the three contracts (NC, MS, and the myopic monopoly benchmark) on a grid in $(\Delta\theta, \Delta\theta/k, \mu)$ with $\Delta\theta \in [0.4, 1.2]$ (step 0.1), $\Delta\theta/k \in [0.5, 1.5]$ (step 0.1), $\mu \in [0.1, 1.9]$ (step 0.2) and $s = 1$, retaining the 970 couples that meet the essentiality bound $k \leq \underline{\theta} - c_0 = 2$ (the 20 excluded couples all lie at the smallest ratio). The grid is indexed by the ratio $\Delta\theta/k$ rather than by $\Delta\theta$ and k varied independently, for one structural and one numerical reason. Structurally, under the NC contract the single tariff ties the two branches through the delay identity $Q_1(\theta) = Q_0(\theta - k) + k$: the profile Q_0 lives on the extended domain $\Theta_{-k} \cup \Theta$, whose configuration (overlapping for $\Delta\theta > k$, joining at $\Delta\theta = k$, disjoint for $\Delta\theta < k$) depends on the parameters only through $\Delta\theta/k$. Numerically, the solver evaluates the shift $\theta \mapsto \theta - k$ exactly on the nodes of the type grid, without interpolation: with step $h = k/m$, this requires $\Delta\theta/k$ to be a multiple of $1/m$, a condition the ratio grid satisfies by construction (non-commensurate couples are rejected rather than interpolated). To rank welfare on a common basis, the incumbent's profit is recomputed for all three from the same type-by-type integral, normalized by $\Delta\theta$.

Continuation, not re-computation. At $\Delta\theta = k$ the solver reproduces the canonical solution of Section 6: on the 90 shared (k, μ) couples it matches every antitrust verdict (strict and in expectation, 0 discordant) with rebate differences at most 1.3×10^{-3} , and all aggregate quantities vary continuously in $\Delta\theta/k$ with no jump at the canonical ratio. The results below are thus a continuation of the main analysis, not an independent computation.

Regime covered. On this grid the cross-gap condition (CG) fails at every couple with $\Delta\theta < k$ (0 of 430); the grid's smallest ratio $\Delta\theta/k = 0.5$ is exactly the threshold below which Case 1 becomes possible, so for every finite μ the grid lies in Case 2. The optimal NC contract carries an active plateau Q^b throughout. The Case-1 equivalence $\mathcal{V}_I^{\text{NC}} = \mathcal{V}_I^{\text{MS}}$ needs entry rare enough for (CG) to be slack, reached only at larger μ (Figure OA.18, drawn at $\mu = 6$) or smaller $\Delta\theta/k$ than this grid spans.

Welfare ranking. The strict welfare dominance of the optimal contract over the MS contract documented at $\Delta\theta = k$ (Figure OA.13) extends to all 970 off-canonical ratio couples, with a median advantage $\mathcal{W}_{\text{NC}} - \mathcal{W}_{\text{MS}}$ of 4 to 6% of first-best welfare (+0.13 to +0.25 in model units) across $\Delta\theta/k$. This is the welfare face of the Case-1/Case-2 dichotomy: the grid lies in Case 2, where the NC contract carries its plateau and the MS contract (more flexible, foreclosing more) destroys more welfare, so $\mathcal{W}_{\text{NC}} > \mathcal{W}_{\text{MS}}$. The equality $\mathcal{W}_{\text{NC}} = \mathcal{W}_{\text{MS}}$ belongs to Case 1 (slack (CG)), which this grid does not reach; consistently, the gap closes nowhere here.

Strategic reaction versus the myopic benchmark. Section OA.7 gauges the strategic conduct by the gap $\Delta\mathcal{W} = \mathcal{W}_{\text{NC}} - \mathcal{W}_{\text{myopic}} = \Delta\mathcal{U} + (\mathcal{V}_{\text{real}} - \mathcal{V}_{\text{myopic}})$ relative to the myopic monopoly Q_m . Away from $\Delta\theta = k$ the reaction still raises total welfare at every interior couple ($k < \underline{\theta} - c_0$), by a median of 8 to 16% of first-best welfare across $\Delta\theta/k$, largest near $\Delta\theta/k \approx 1.1$ at the edge of the entry-barrier regime. The gain is the correction of the myopic monopoly's over-entry; the incumbent's profit is above the myopic one throughout ($\mathcal{V}_{\text{real}} \geq \mathcal{V}_{\text{myopic}}$, as revealed preference requires). The ranking inverts only at the exact essentiality boundary $k = \underline{\theta} - c_0$, and only marginally.⁶⁷ The MS contract also improves on the myopic benchmark, but not universally: it does so on 93% of couples, falling below the passive incumbent only in the deepest under-entry region ($\Delta\theta/k \leq 0.7$), where it forecloses aggressively. The optimal contract dominates the myopic benchmark more often (98%) and by more (a median of 8 to 16% of first-best welfare against 2 to 10% for the MS contract), so that $\mathcal{W}_{\text{NC}} > \mathcal{W}_{\text{MS}} > \mathcal{W}_{\text{myopic}}$ over most of the grid.

Antitrust verdicts. The diagnostic established at $\Delta\theta = k$, that the AEC test condemns the NC contract where it raises welfare (Table OA.22), is itself a feature of the entry-barrier regime, and the ratio $\Delta\theta/k$ makes this explicit. Under the MS contract $\omega(\theta) < c_0 k$ for every type throughout, so the test fails for all types under any reading, strict or in expectation, and independently of $\Delta\theta/k$ as well as of s : the market-share verdict is unconditional. Under the NC contract Table OA.23 reports, by $\Delta\theta/k$, the share of couples that improve on the myopic benchmark and the share failing the test in expectation. Welfare improves on every couple but the $k = \underline{\theta} - c_0$ boundary cases of the footnote above, while the expectation test fails on a fraction that declines monotonically with $\Delta\theta/k$: from the whole grid in the under-entry region ($\Delta\theta/k < 1$, where $k > \Delta\theta$ and entry is foreclosed) to none of it once $\Delta\theta/k$ is large. At the canonical ratio $\Delta\theta/k = 1$ the failure rate is 70%, consistent with the $s = 1$ row of Table OA.22.

The inversion is therefore concentrated in the entry-barrier regime and vanishes under over-entry. This is the $\Delta\theta/k$ counterpart of the under-/over-entry reversal of Section OA.7: where $k > \Delta\theta$ the incumbent forecloses an efficient entrant, so the buyer-oriented test condemns conduct that nonetheless raises welfare;

⁶⁷ At $k = \underline{\theta} - c_0 = 2$ (16 couples) $\Delta\mathcal{W} \in [-0.09, 0]$: the over-entering myopic monopoly ($\omega_m \in [5.75, 5.84] > c_0 k = 4$) confers a large buyer surplus that the non-conditional restriction removes, so $\Delta\mathcal{U} \approx -3$ and $\mathcal{V}_{\text{real}} - \mathcal{V}_{\text{myopic}} \approx +3$ nearly cancel and the balance tips slightly negative. Profit dominance $\mathcal{V}_{\text{real}} \geq \mathcal{V}_{\text{myopic}}$ holds here too, ruling out a numerical artifact.

as $\Delta\theta/k$ rises the contract tips into over-entry, the test ceases to condemn, and the inversion disappears. The off-canonical ratio grid thus locates the inversion precisely where the NC contract acts as a barrier to entry.

$\Delta\theta/k$	n	$\Delta\mathcal{W} > 0$	AEC fail	AEC fail $\Delta\mathcal{W} > 0$
0.6	90	93.3%	100.0%	100.0%
0.8	90	100.0%	94.4%	94.4%
1.0	90	100.0%	70.0%	70.0%
1.2	90	100.0%	28.9%	28.9%
1.4	90	100.0%	3.3%	3.3%

Table OA.23: NC contract at $s = 1$, robustness of the diagnostic inversion in the ratio $\Delta\theta/k$. For each ratio: share of admissible $(\Delta\theta, \mu)$ couples with $\Delta\mathcal{W} = \mathcal{W}_{\text{real}} - \mathcal{W}_{\text{myopic}} > 0$; share failing the AEC test in expectation ($\mathbb{E}_\theta[\omega] < c_0 k$); and the latter restricted to welfare-improving couples. $\Delta\theta/k < 1$ is the under-entry (entry-barrier) regime; the canonical ratio $\Delta\theta/k = 1$ reproduces the $s = 1$ row of Table OA.22. 90 couples per ratio.

OA.8.4. Economic discussion

The ratio $\Delta\theta/k$, measuring the dispersion of buyer types relative to the entrant's capacity, partitions the analysis into three regimes.

- $\Delta\theta < k$ with (CG) slack: the non-conditional restriction does not bind. The optimal contract replicates the market-share allocation, its two consumption ranges occupying disjoint portions of the tariff. Prohibiting conditional contracts would not eliminate their effect.
- $\Delta\theta < k$ with (CG) binding: a bunching plateau $\mathcal{Q}^b$ forms, the level identity generates a positive wedge γ in the sub-martingale property, and the features of Section 6 reappear. The wedge vanishes at the slack boundary and increases as $\Delta\theta \rightarrow k$.
- $\Delta\theta > k$: a new phenomenon arises on the overlap, where a single tariff quantity serves at once a low-demand exclusive buyer (type θ) and a high-demand split buyer (type $\theta + k$). The clean separation between branches breaks down and no closed-form characterization is available.

This structure clarifies the status of the restriction $\Delta\theta = k$ maintained in the main analysis. It is not a loss of generality that hides the phenomena of interest, but the point at which they are strongest. Two findings support this. First, when k is large relative to $\Delta\theta$ and (CG) is slack, the non-conditional tariff exactly replicates the market-share allocation. Second, and conversely, the inward kink does not require $\Delta\theta = k$: it persists throughout the bunching regime and across the entire overlap regime $\Delta\theta > k$ (Figure OA.21), where it coexists with the dual use of quantities rather than being replaced by it.

Between these two findings, the features documented in Section 6 (minimum purchase requirement, inward kink, below-cost pricing, sub-martingale property) are properties of the bunching regime: they are absent only when (CG) is slack. Their intensity is governed by the wedge γ , which is monotone in $\Delta\theta/k$ and peaks at $\Delta\theta = k$. That value is simultaneously the right boundary of the bunching regime and the left boundary of the overlap regime, and the only point at which the optimal NC contract admits a complete closed-form characterization while these features are at full strength. Studying the model at $\Delta\theta = k$ therefore captures the phenomena of interest in their strongest and most tractable form; the neighboring regimes, characterized above and confirmed numerically, attenuate them rather than introduce qualitatively new ones.

OA.9. NON-ESSENTIAL TRADING PARTNER

The main analysis assumes that the incumbent is an essential trading partner (ETP): the buyer cannot purchase from the entrant without first contracting with the incumbent. The buyer's participation constraint then reduces to

$$(IR-A) \quad \mathcal{U}_0(\theta) \geq 0 \quad \forall \theta \in \Theta,$$

binding only at $\underline{\theta}$ at the optimum, and the entry-branch constraint $\mathcal{U}_1(\theta) - c \geq 0$ is implied, since entry occurs if and only if $c \leq \mathcal{U}_1(\theta) - \mathcal{U}_0(\theta)$. This assumption is the empirically relevant one for the antitrust questions we study: exclusion concerns arise when the incumbent is hard to bypass, so its contract, rather than the buyer's outside access to the entrant, governs entry. We nonetheless show in this section that our main qualitative findings hold when the assumption is dropped. All numerical comparisons in this section pair each couple's ETP and NETP solutions and use a coarser grid than the rest of the appendix, (k, μ) at step 0.10 rather than 0.05; the rates and magnitudes reported below therefore differ marginally from their full-resolution counterparts.⁶⁸

When the incumbent is no longer essential (NETP), the buyer can purchase the k units from the entrant in isolation, with outside option in the entry branch $\mathcal{U}_{out}(\theta) = \theta k - \frac{1}{2}k^2 - c$. The entry-branch participation constraint becomes $\mathcal{U}_1(\theta) - c \geq \mathcal{U}_{out}(\theta)$, i.e.

$$(IR-B) \quad \mathcal{U}_1(\theta) \geq \theta k - \frac{1}{2}k^2 \quad \forall \theta \in \Theta.$$

By the envelope theorem $\dot{\mathcal{U}}_1(\theta) = Q_1(\theta) = \tilde{Q}_0(\theta) + k$ in both NC and MS, hence $\frac{d}{d\theta}(\mathcal{U}_1 - \mathcal{U}_{out}) = \tilde{Q}_0(\theta) \geq 0$ and (IR-B) reduces to a single inequality at the lowest type, $\mathcal{U}_1(\underline{\theta}) \geq \underline{\theta}k - \frac{1}{2}k^2$.⁶⁹

OA.9.1. NC & NETP

In the NC formulation $\tilde{\mathcal{U}}_0(\theta)$ is the utility of the fictitious type $\theta - k$, and $\mathcal{U}_1(\theta) - \tilde{\mathcal{U}}_0(\theta) = \theta k - \frac{1}{2}k^2$. Constraint (IR-B) is therefore equivalent to $\tilde{\mathcal{U}}_0(\theta) \geq 0$ on Θ and, by the same monotonicity argument, to

$$(OA.9.1) \quad \tilde{\mathcal{U}}_0(\underline{\theta}) \geq 0.$$

A single tariff serves both branches, imposing the consistency condition $\tilde{\mathcal{U}}_0(\bar{\theta}) = \mathcal{U}_0(\underline{\theta})$. The two participation constraints cannot both bind at $\underline{\theta}$: if $\tilde{\mathcal{U}}_0(\underline{\theta}) = \mathcal{U}_0(\underline{\theta}) = 0$, monotonicity gives $\tilde{\mathcal{U}}_0(\bar{\theta}) = \int_{\underline{\theta}}^{\bar{\theta}} \tilde{Q}_0$ while consistency forces it to zero, whence $\tilde{Q}_0 \equiv 0$ (degenerate). At any non-degenerate optimum it is the entry constraint that binds, $\tilde{\mathcal{U}}_0(\underline{\theta}) = 0$, and (IR-A) is slack. Since $\mathcal{U}_0$ enters the objective additively with coefficient -1 , the incumbent minimizes the common boundary level subject to (IR-B), giving

$$(OA.9.2) \quad \eta^* = \mathcal{U}_0(\underline{\theta}) = \tilde{\mathcal{U}}_0(\bar{\theta}) = \int_{\underline{\theta}}^{\bar{\theta}} \tilde{Q}_0(\theta) d\theta > 0.$$

The rent conceded in the entry branch propagates through the single tariff and leaves the buyer a strictly positive rent even without entry.

⁶⁸For instance the strict AEC failure rate under NC and ETP is 90.5% here, against 91.5% at full resolution (Table OA.21, $s = 1$); the gap is one of grid resolution, not of method.

⁶⁹In our simulations $\underline{\theta} = 4$ and $k \leq 2$, so that $\underline{\theta}k - \frac{1}{2}k^2 > c_0k$ throughout; equivalently $k < 2(\underline{\theta} - c_0)$.

Optimality conditions. Relaxing essentiality leaves I 's problem structurally unchanged except for the constraint that closes the system. The Hamiltonian, the adjoint equations (A.44)–(A.45), (A.49), (A.50), and the identity $\lambda_0(\theta) + \tilde{\lambda}_0(\theta) = \theta - \hat{k}$ all carry over; only the boundary conditions and the closing constant $\hat{k}$ change. As in ETP there is no end-point constraint on $\mathcal{U}_0$ at $\bar{\theta}$, so $\lambda_0(\bar{\theta}) = 0$. The only difference is at $\underline{\theta}$, where the free end-point moves from $\tilde{\mathcal{U}}_0(\underline{\theta})$ (free in ETP, giving $\tilde{\lambda}_0(\underline{\theta}) = 0$) to $\mathcal{U}_0(\underline{\theta})$ (free in NETP, since (IR-A) is slack). Through the identity $\tilde{\mathcal{U}}_0(\bar{\theta}) = \mathcal{U}_0(\underline{\theta})$, the boundary term of the variation yields $\tilde{\lambda}_0(\bar{\theta}) = \lambda_0(\underline{\theta})$, hence $\tilde{\lambda}_0(\bar{\theta}) = \bar{\theta} - \hat{k}$ as before, while

$$(OA.9.3) \quad \tilde{\lambda}_0(\underline{\theta}) = \underline{\theta} - \bar{\theta} = -k$$

is the multiplier of the now-binding constraint (OA.9.1). The separating-arc expressions (A.52)–(A.53) and the change of variable (A.54) involve only $\tilde{\lambda}_0$, $\hat{k}$ and $G(\omega)$, never the closing condition, so the optimal quantities admit exactly the same representation as under ETP,

$$Q_0 = Q_m + \bar{\theta} - \hat{k} - \frac{\mu}{1 - G(\omega)}, \quad Q_1 = Q_m + \bar{\theta} - \hat{k} + \frac{\mu}{G(\omega)},$$

with μ as in (A.54); only $\hat{k}$ and the boundary behaviour of $\tilde{\lambda}_0$ differ. Relative to ETP, the single change is $\tilde{\lambda}_0(\underline{\theta}) : 0 \mapsto -k$. This pulls the $\tilde{\lambda}_0$ profile downward and lowers $\tilde{Q}_0$: with the entry rent no longer extractable, I distorts the entry-state quantity downward less.

OA.9.2. MS & NETP

Under NETP the buyer faces two participation constraints, (IR-A) and (IR-B). Since $\mathcal{U}_0(\underline{\theta})$ need no longer be zero, the substitution leading to (A.2) is unavailable and the problem cannot be reduced to the single state ω of the ETP analysis. We therefore carry $\mathcal{U}_0$ and ω as two states, with adjoints ψ and λ . Up to the density $1/\Delta\theta$,

$$(OA.9.4) \quad \mathcal{H} = G(\omega) \left(c_0 k + [S(\theta, q)]_{Q_0}^{Q_1} - \omega \right) + S(\theta, Q_0) - \mathcal{U}_0 + \psi Q_0 + \lambda(Q_1 - Q_0).$$

The adjoint equation for ω coincides with the ETP case (A.3); that for $\mathcal{U}_0$ is $\dot{\psi}(\theta) = 1$. Let $\alpha \geq 0$ multiply $\mathcal{U}_0(\underline{\theta}) \geq 0$ and $\beta \geq 0$ multiply $\mathcal{U}_1(\underline{\theta}) \geq \underline{\theta}k - \frac{1}{2}k^2$. With $\mathcal{U}_0(\bar{\theta})$ and $\omega(\bar{\theta})$ free, the boundary term of the variation gives

$$(OA.9.5) \quad \lambda(\bar{\theta}) = \psi(\bar{\theta}) = 0, \quad \lambda(\underline{\theta}) = -\beta, \quad \psi(\underline{\theta}) = -(\alpha + \beta).$$

The conditions in (Q_0, Q_1) are formally those of (A.5)–(A.6), since (OA.9.4) differs from the ETP Hamiltonian only through additive adjoint terms. Summing them yields the martingale property (5.6), and with μ as in (A.7) the optimal quantities again take the form (A.10)–(A.11),

$$(OA.9.6) \quad Q_0 = Q_m - \frac{\mu}{1 - G(\omega)}, \quad Q_1 = Q_m + \frac{\mu}{G(\omega)}.$$

Consumption is structurally invariant; only the boundary behaviour of μ changes, with $\mu(\bar{\theta}) = 0$ and $\mu(\underline{\theta}) = \Delta\theta G(\omega(\underline{\theta})) - \beta$, subtracting β from the ETP value (A.9).

Integrating $\dot{\psi} = 1$ with $\psi(\bar{\theta}) = 0$ gives $\psi(\underline{\theta}) = -\Delta\theta$, which combined with (OA.9.5) yields the identity

$$(OA.9.7) \quad \alpha + \beta = \Delta\theta.$$

The entry constraint always binds ($\beta > 0$): were it slack, the solution would coincide with the ETP one, for which ω increases with $\omega(\bar{\theta}) < c_0 k$, so $\omega(\underline{\theta}) < c_0 k < \underline{\theta}k - \frac{1}{2}k^2$ under the maintained assumption $k < 2(\underline{\theta} - c_0)$, a contradiction. The no-entry constraint binds ($\alpha > 0$, $\mathcal{U}_0(\underline{\theta}) = 0$) or is slack ($\alpha = 0$, $\beta = \Delta\theta$). The slack case is an entry-suppression regime: the incumbent raises $\mathcal{U}_0(\underline{\theta})$ above zero to lower $\omega(\underline{\theta})$ and dampen entry. On the grid the identity (OA.9.7) holds at every point, complementarity is never violated, and the slack regime occupies an intermediate- k band.

Under both contracts (IR-A) may be slack, but for different reasons: under NC the single tariff forces it through the consistency condition, mechanically; under MS the incumbent chooses it, and only where suppressing entry is worthwhile. In both cases, however, the optimal quantities keep the ETP representation (OA.9.6), the only change being in the multipliers and the closing constant. The architecture of the optimal contract is thus unchanged by essentiality.

OA.9.3. Numerical comparisons

We now compare the NETP and ETP optima at fixed contract form, on the grid $(k, \mu) \in [0.1, 2]^2$ at $s = 1$.

Figure OA.22 reports the buyer's no-entry rent at the lowest type, $\mathcal{U}_0(\underline{\theta})$, which is positive exactly when (IR-A) is slack. Under NC it is positive almost everywhere: the consistency condition forces $\mathcal{U}_0(\underline{\theta}) = \int_{\underline{\theta}} \tilde{Q}_0 > 0$, so the relaxation of (IR-A) is mechanical. Under MS it is positive only on an intermediate- k band, where $\mathcal{U}_0(\underline{\theta})$ is a free instrument the incumbent raises to deter entry; outside the band (notably at large k , small μ , where entry is cheap) the incumbent accommodates entry and (IR-A) binds. The same observed feature thus arises from two distinct mechanisms.

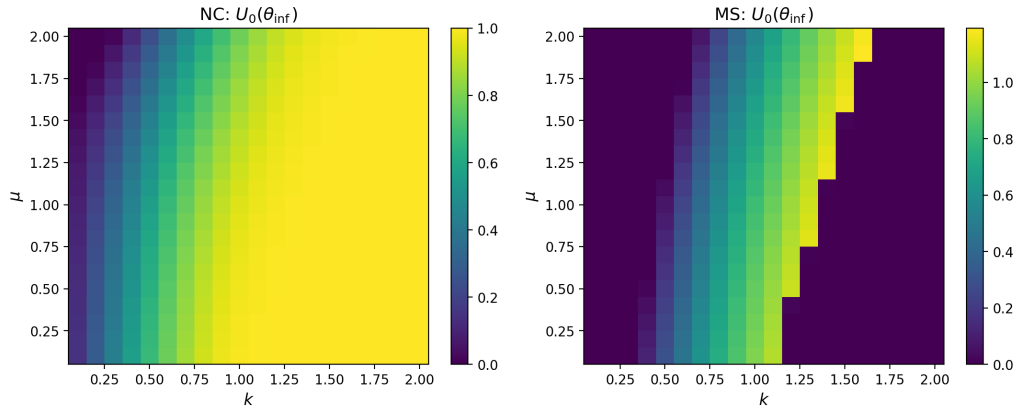


Figure OA.22: Buyer's no-entry rent at the lowest type, $\mathcal{U}_0(\underline{\theta})$, under NETP, across the grid (k, μ) at $s = 1$.

Table OA.24 reports the average effect on the main aggregates. In both contracts the incumbent's profit falls (about -58% under NC, -62% under MS) and is transferred to the buyer; entry rises, the more so under MS. The welfare effect, however, differs in sign: -2.3% under NC, $+4.0\%$ under MS. The channel is the level of entry relative to its first-best benchmark $G(c_0 k)$. Under ETP entry is insufficient: strongly

so under MS, mildly under NC. Relaxing essentiality raises the entry-branch rent and pushes entry above the first-best level on almost the whole grid. Whether this helps depends on the starting point: under MS, where entry was far too low, the correction is beneficial; under NC, where entry was already near-efficient, the same push overshoots and reduces welfare. The welfare loss concentrates at small and intermediate k (Figure OA.23), while large k , where the entrant supplies a large volume, yields a gain in both contracts. The cross-couple correlation between excess entry and the welfare change is strongly negative (-0.83 under NC, -0.84 under MS), confirming excessive entry as the channel of welfare deterioration.

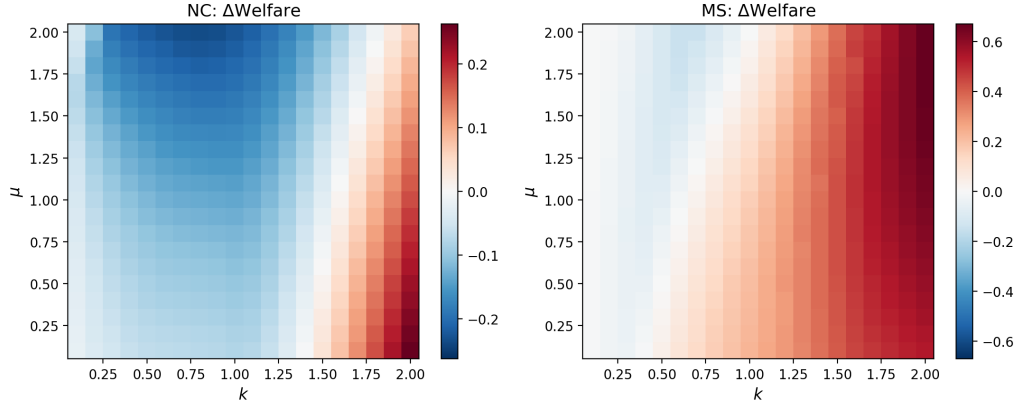


Figure OA.23: Change in expected welfare, $\Delta\text{Welfare}$ (NETP – ETP), across the grid (k, μ) at $s = 1$. Red indicates a welfare gain, blue a loss.

	Mean	Median
NC		
Profit $\mathcal{V}_I$	−58.4%	−67.2%
Buyer surplus $\mathcal{U}$	+67.0%	+63.2%
Welfare $\mathcal{W}$	−2.3%	−2.4%
Entry probability	+18.3%	+14.4%
MS		
Profit $\mathcal{V}_I$	−61.9%	−74.5%
Buyer surplus $\mathcal{U}$	+198.2%	+223.6%
Welfare $\mathcal{W}$	+4.0%	+4.9%
Entry probability	+135.9%	+70.8%

Table OA.24: Average effect of relaxing essentiality (NETP relative to ETP) at $s = 1$. Profit, buyer surplus, welfare and entry probability are expectations over types; figures are relative to the ETP level.

Table OA.25 reports the share of couples failing the AEC test, $\omega(\theta) \geq c_0 k$ for all θ . Under ETP the test fails on almost the entire grid (90.5% under NC, all couples under MS); under NETP the failure rate drops to 6.8% and 0%. The test is satisfied when ω is large, and NETP raises ω above $c_0 k$ for nearly all types, so AEC indirectly captures the incumbent's market power over the buyer, which NETP curtails. It remains an imperfect efficiency proxy, however: it tracks only the buyer's rent, not total surplus, so the clearance it grants under NETP coincides with excessive (not efficient) entry, and under NC with a welfare loss. Test and welfare criterion diverge where entry overshoots its first-best level. The residual 6.8% of NC failures are the couples where $\omega(\bar{\theta}) < c_0 k$: since ω is decreasing under NETP, it can dip below $c_0 k$ near $\bar{\theta}$, leaving entry insufficient at the top types, which the all-or-nothing criterion detects (Table OA.26).

At the lowest type, $\omega(\underline{\theta}) > c_0 k$ on the entire grid for both contracts: the binding entry constraint gives

	ETP	NETP
NC	90.5%	6.8%
MS	100.0%	0.0%

Table OA.25: Share of couples failing the AEC test ($\omega(\theta) \geq c_0 k$ for all θ), by contract and regime, at $s = 1$.

	(+, +) both excessive	(+, -) low exc., high insuf.	(-, +) —	(-, -) —
NC	93%	7%	0%	0%
MS	100%	0%	0%	0%

Table OA.26: Sign of $\omega(\theta) - c_0 k$ at the two ends of the type space under NETP, at $s = 1$. A + denotes excessive entry ($\omega > c_0 k$), a – insufficient entry.

$\omega(\underline{\theta}) = \underline{\theta}k - \frac{1}{2}k^2 - \mathcal{U}_0(\underline{\theta})$, and $\mathcal{U}_0(\underline{\theta}) < (\underline{\theta} - c_0)k - \frac{1}{2}k^2$ throughout under our maintained assumption. Entry is therefore excessive for the lowest types in every case (Table OA.26). At the highest type the picture is contract-dependent: excessive everywhere under MS, but insufficient on the 7% of NC couples noted above.

Two conclusions follow. First, the analysis is robust to essentiality: the same mechanics govern both regimes (the martingale structure, the representation of quantities through μ , and the role of the entry margin), so relaxing essentiality changes the boundary conditions and the binding constraint but not the qualitative features of the optimal contract. This robustness, together with the fact that the ETP case is the one of antitrust interest (exclusion matters when the incumbent is hard to bypass), supports the maintained assumption of the main analysis. Second, the common architecture still delivers quantitatively distinct outcomes, since NETP forces the incumbent to concede more surplus to the buyer: profit falls, buyer surplus and entry rise. The sign of the welfare effect then depends on whether the induced entry corrects a pre-existing deficit (MS) or overshoots an already near-efficient level (NC). More entry is therefore not equivalent to higher welfare, and form-based screens based on the buyer's rent may clear contracts where entry has become excessive.

OA.10. OTHER EXTENSIONS

Sensitivity to G . The truncated normal distribution adopted in the main analysis is a baseline; we also performed a sweep with a logistic distribution and recovered qualitatively identical conclusions on rankings and antitrust correlations.

Sensitivity to parameters. We have run our simulations with other values for $(\underline{\theta}, c_0)$ with no qualitative changes to our analysis.